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Series Editors: Jagdish Chand Bansal · Kusum Deep · Atulya K. Nagar

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Artificial Intelligent Algorithms for Image Dehazing and Non-Uniform Illumination Enhancement

Springer

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Series Editors

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Teena Sharma · Nishchal K. Verma

Artificial Intelligent Algorithms for Image Dehazing and Non-Uniform Illumination Enhancement

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Preface

Artificial Intelligence (AI) has introduced a range of effective image processing tools that play a pivotal role in fortifying vision-based applications. The success of these applications hinges on the quality of input images, particularly those containing intricate textural details. The merit of an image is typically gauged through visible aspects such as illumination, color, and shapes, all of which can be adversely impacted by atmospheric factors like haze and uneven illumination.

When haze particles are present, they impede the transmission of reflected light rays to the observer, resulting in an elevation of airlight and causing the image to exhibit a hazy appearance. Conversely, non-uniform illumination conditions contribute to a darkened depiction of the captured scene, leading to the obscuring of visible features. Consequently, image enhancement emerges as a crucial preliminary step, aiming to revive textural intricacies and refine image details. This enhancement process proves indispensable in bolstering the resilience of vision-based applications, ensuring their optimal performance even in challenging environmental circumstances.

This book provides a detailed explanation of AI algorithms for image dehazing and non-uniform illumination enhancement. It outlines unsupervised and supervised methods with their advantages and limitations. The structure of this book will be helpful for a reader first to understand the unsupervised methods considering the scenario when it is impractical to have a large set of datasets for training an AI algorithm for image enhancement. Since there have been extensive advancements in deep learning methods and the availability of resources, this book provides a detailed insight into the run-time and accuracy adjustments of deep learning-based AI algorithms to learn the pattern better while performing enhancement. It also introduces the fuzzy set theory-based decision-making tools to deal with uncertainties

present in a digital image. The content presented in this book will open research pathways to increase the robustness of various vision-based applications like object detection, vehicle tracking, security, medical diagnostics, etc.

Guwahati, India — Teena Sharma
Kanpur, India — Nishchal K. Verma
2024

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Acronyms

AHE	Adaptive Histogram Equalization
AI	Artificial Intelligence
AIPNet	Atmospheric Illumination Prior Network
AIT2FF	Adaptive Interval Type-2 Fuzzy Filter
AOD-Net	All-in-One Dehazing Network
AUC	Area Under Curve
BCCR	Boundary Constraint and Contextual Regularization
BM3D	Block Matching and 3D
CAP	Color Attenuation Prior
CLAHE	Contrast Limited Adaptive Histogram Equalization
CNN	Convolutional Neural Network
CSIDNet	Compact Single Image Dehazing Network
CVC	Contextual and Variational Contrast
DCP	Dark Channel Prior
DICM	DIgital CaMeras
ENR	Efficient Naturalness Restoration
FFA-Net	Feature Fusion Attention Network
FL	Fuzzy Logic
FST	Fuzzy Set Theory
GFN	Gated Fusion Network
GMAN	Generic Model-Agnostic Network
GMF	Gaussian Membership Function
GT2	General Type-2
HDR	High Dynamic Range
HE	Histogram Equalization
HSTS	Hybrid Subjective Testing Set
IT2FL	Interval Type-2 Fuzzy Logic
LDR	Layered Difference Representation
LECARM	Low-light image Enhancement CAmera Response Model
LIME	Low-light IMage Enhancement
LMF	Lower Membership Function

LOE	Lightness Order Error
MAE	Mean Absolute Error
MF	Membership Function
MSCNN	Multi-Scale Convolutional Neural Network
MSE	Mean Square Error
MSR	Multi-Scale Retinex
NIQE	Naturalness Image Quality Evaluator
OTS	Outdoor Training Set
PSNR	Peak Signal-to-Noise Ratio
ReLU	Rectified Linear Unit
RESIDE	REalistic Single Image DEhazing
ROC	Receiver Operating Characteristic
ROI	Region Of Interest
RTTS	Real-world Task-driven Testing Set
SCN	Supply Chain Network
SIFT	Scale Invariant Feature Transform
SOTS	Synthetic Objective Testing Set
SSIM	Structural SIMilarity
SSR	Single Scale Retinex
SURF	Speeded Up Robust Features
T1FF	Type-1 Fuzzy Filter
T2FF	Type-2 Fuzzy Filter
UMF	Upper Membership Function
Zero-DCE	Zero-reference Deep Curve Estimation
ZSWFM	Z-Score Weighting Function Map

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Chapter 1
Introduction



1.1 Image Enhancement

Image quality enhancement is a pre-processing step that aims to improve the information content in an image for vision-based decision-making applications [1–14]. The information content in an image can be interpreted in terms of the local features like edges, color, contrast, etc. [15]. In the literature, several feature detectors have been introduced to detect and describe the local features in an image. Mikolajczyk et al. [16] compared various feature detectors based on scale, rotation, and illumination invariance. In order to perform the feature matching between two images, the information of local features is encoded by feature descriptors [17, 18] in terms of a series of numbers. Mikolajczyk and Schmid [19] compared several feature descriptors based on their distinctiveness, repeatability, and computational complexity. Even in the presence of the robustness of existing feature detectors and descriptors, several factors affect the quality of an image, such as noise, motion blur, illumination, haze, etc. [20–22]. These factors often degrade the performance of algorithms relying on the information content of the image. Hence, image quality becomes an indispensable requirement for vision-based decision-making systems. In computer vision, the information content described by local features of an image gets affected due to various factors. This book mainly focuses on the following:

(i) **Haze:** The recognition of objects in any image under hazy conditions is a bottleneck for vision-based decision-making algorithms, as poor visibility strongly influences the accuracy and general functionality of such algorithms. In Fig. 1.1, it is clearly seen that the image visibility is obstructed as the haze increases from slight to moderate to heavy.
(ii) **Non-Uniform Illumination:** The recognition of objects in any image under uneven illumination results in wrong highlights and tucks away the image features, which results in an unrealistic captured scene image. Figure 1.2 shows a few example images of non-uniform illumination.

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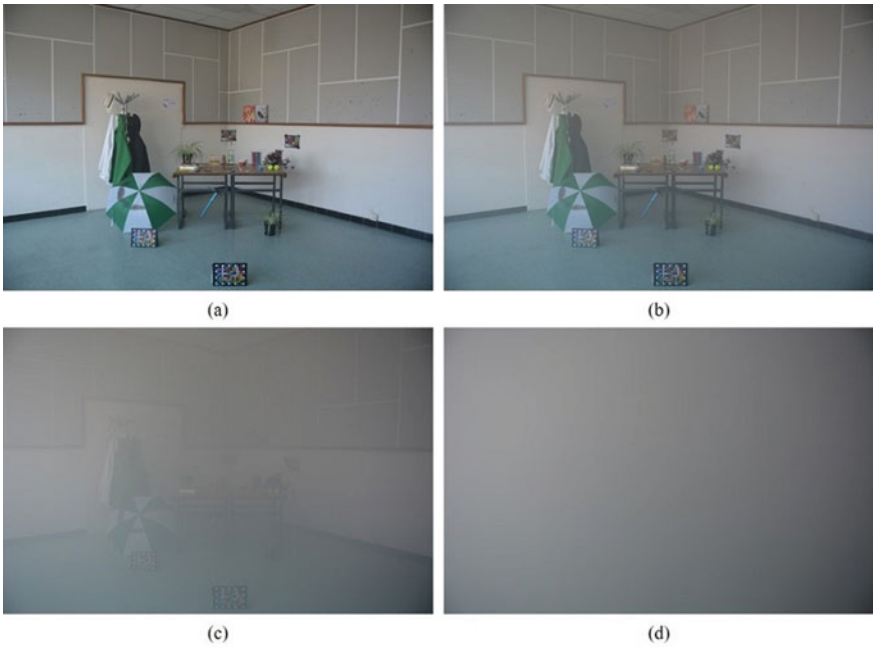


Fig. 1.1 Examples of hazy images [23]: A sample image with **a** No haze present, with **b** Slight, **c** Moderate, and **d** Heavy presence of haze

This book introduces the methods to address these major factors affecting image quality using artificial intelligence-based algorithms for image dehazing and non-uniform illumination enhancement. The literature survey of image matching using local features, followed by the literature survey of image dehazing and non-uniform illumination enhancement, have been discussed in the following subsections.

1.2 Image Matching

Image matching [25] aims to find a particular pattern for correspondence between two images of the same scene. The pattern within an image comprises the local features described by pixel intensity values. Some of the initial attempts at feature-based image matching were based on image histograms. The histogram of an image refers to the frequency of occurrence of each pixel intensity value. Swain and Ballard [26] introduced a color histogram approach to finding the Region of Interest (ROI) for counting the occurrence of a predefined template image in a test scene. This was further improved in [27–29] to overcome the illumination variations using the fuzzy color histogram. Thereafter, correlation-based matching [30] was introduced to optimize the speed. In [31], various correlation-based matching algorithms have



Fig. 1.2 Examples of non-uniform illumination [24]. **Top row:** Some scene images captured with illumination distributed unevenly and dark appearance and **Bottom row:** Enhanced version of the corresponding images shown in the top row

been compared. Though correlation-based matching algorithms perform faster than histogram-based matching algorithms, these algorithms fail with rotation and scale variations. Thus, a combination of Hessian and Laplacian methods was proposed in [32] to deal with rotation and scale variations. In an attempt to deal with such variations, supervised [33] and unsupervised [34, 35] approaches using Speeded Up Robust Features (SURF) [36] have also been proposed in the literature. Since the aforementioned state-of-the-art image-matching algorithms depend upon the textural information extracted from the visual appearance of the scene under consideration, the obstruction in visibility due to haze and non-uniform illumination hinders the performance of the system.

1.3 Image Dehazing

The performance of computer vision-based decision-making algorithms [1–14] is dependent on the input images characteristics. These characteristics can be ascertained by the information contents in the digital images. The information content constitutes visible elements like color, shapes, illumination, etc. These visible elements get compromised because of noise, blur, etc. [37]. There are several particles present in the atmosphere. These particles cause smog, fog, haze, etc. When reflected

light rays from the object travel toward the observer, the visible elements of images get distorted. Images captured under hazy conditions appear whitish or brighter than their haze-free version. Minnaert [38], Henderson [39], McCartney [40], and Middleton [41] have studied and explored the concepts of atmospheric optics for the effect of weather conditions during image formation.

The objective of image dehazing is to further improve the information of visible elements in the image. This increases the robustness of the algorithms. In contemporary times, the pre-processing stage of input images holds paramount significance in real-time applications. These applications may be for multi-media [2, 3], remote-sensing [42, 43], multi-spectral-imagery [44, 45], etc. The degradation of visible elements in images is a prevalent concern, particularly due to the poor scenes caused by suspended air particles during adverse weather conditions. In tackling these challenges, the atmospheric scattering model [46, 47] has emerged as a benchmark in the literature, offering a crucial solution for enhancing the quality of hazy images.

1.3.1 Atmospheric Scattering Model

As per the findings outlined in reference [47], when light rays reflect from an object with a true image radiance $I_{\mathrm{RL}}(p, q)$ (here, (p, q) represents the pixel location) and proceed toward the observer with a transmission factor $T_{\mathrm{R}}(p, q)$, the presence of haze particles impedes this transmission through what is referred to as "Direct Attenuation". The scene transmission, influenced by the angular scattering coefficient β and the depth map $D_{\mathrm{Map}}(p, q)$ can be described as follows:

$$T_{\mathrm{R}}(p, q) = \exp\left(-\beta \times D_{\mathrm{Map}}(p, q)\right). \tag{1.1}$$

Furthermore, the cumulative effect of scattered light rays results in the manifestation of true radiance, giving the pixel an overall whitish appearance. This phenomenon is commonly referred to as "Airlight".

Hence, The hazy image $I_{\mathrm{Hazy}}(p, q)$ formation, with A_{G} as the "Global Atmospheric Light", can be interpreted as follows:

$$I_{\mathrm{Hazy}}(p, q) = I_{\mathrm{RL}}(p, q) \times T_{\mathrm{R}}(p, q) + A_{\mathrm{G}}\,(1 - T_{\mathrm{R}}(p, q)) \tag{1.2}$$

Figure 1.3 visualizes the phenomenon of the formation of a hazy scene in the presence of suspended air particles.

1.3.2 Related Work

Based on the input requirements of any image dehazing method, it can be broadly categorized into dehazing methods with multiple images as input and dehazing

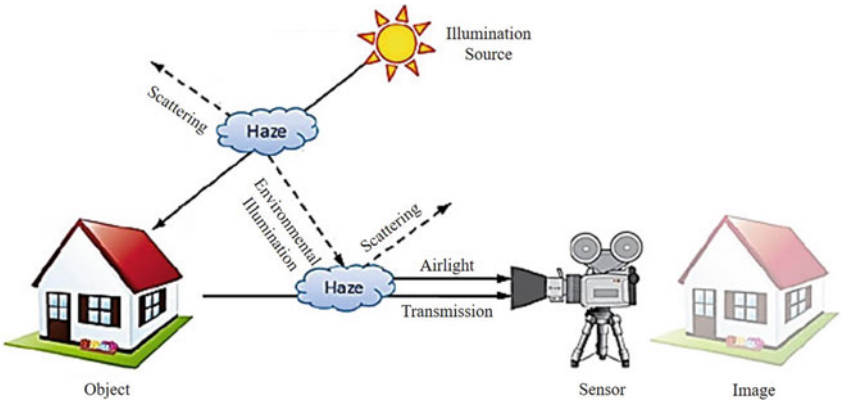


Fig. 1.3 Phenomenon of the formation of a hazy scene [47]

methods with a single image as input. In further sub-points, the state-of-the-art algorithms introduced under these two broader categories are discussed in detail.

(i) **Image Dehazing Methods with Multiple Images**: During initial research in this area, the image dehazing methods had to rely upon multiple images of the same scene for producing the corresponding haze-free image. One of the popular methods was based on the use of the polarizing filter. Schechner et al. [48, 49] used images captured from two perspectives using a polarizing filter. Thereafter, Narasimhan et al. [46, 50] rigorously studied the atmospheric optics for the development of physics-based atmospheric scattering model as explained in Sect. 1.3.1. The objective was to study and explore the image features that are affected by the presence of haze. One of those features is the depth map. The depth map, also termed scene depth, refers to the distance between the location of the observer view and the location of the information being captured. Kopf et al. [51] proposed a method that uses scene depth to quantify the thickness of the haze for obtaining a haze-free image. Despite the fact that these approaches have been proven effective in outputting the haze-free image, the supplementary multi-modal information, like multiple images of a similar scene, depth, etc., are not feasible to capture, thus decreasing the robustness and real-time performance of vision-based applications.

(ii) **Image Dehazing Methods with Single Image**: Image dehazing methods using a single image as an input [23, 52–61] is an ill-posed and arduous problem but practically desirable and advantageous for real-time applications. These methods rely upon haze-relevant features such as saturation [62], contrast [63, 64], histogram [65], etc., to measure the thickness of the haze. The sub-categorization of single image dehazing methods may include priors or assumptions-based methods and deep learning-based methods as follows:

(a) **Priors and Assumptions based Image Dehazing Methods**: These methods focus on estimating the scene parameters, i.e., depth map, transmission, and global atmospheric light, mentioned in the atmospheric scattering model by Narasimhan et al. [47]. Initially, Fattal [66] proposed a method to remove the haze by segmenting constant albedos for surface shading and locally statistically uncorrelated transmission function. Though the proposed method was able to produce a reliable transmission, it is time-consuming for real-time applications. Tan [52] developed an optimization approach for contrast maximization using a Markov random fields-based cost function. The cost function includes the data cost as well as the smoothness cost, which relies on the observations that the contrast of a haze-free image is better than the hazy one, and the smoothness increases with the increase in the distance of the object from the observer's view, respectively. Similarly, a Boundary Constraint and Contextual Regularization (BCCR) method was proposed by Meng et al. [67] using contrast-based assumption, but regions with poor contrast generate discontinuities in the textural information. These discontinuities were further tackled by Ancuti et al. [25] by restoring the edge information in such poor contrast regions. On the basis of contrast-based assumption, the dehazing methods somehow restrict themselves from achieving a uniform pixel intensity distribution, which results in an unrealistic restored image. Thus, researchers came up with various priors and assumptions. One of the popular benchmarked methods is Dark Channel Prior (DCP) proposed by He et al. [23]. DCP is based on the observation that some pixel intensities tend to have low gray values in one or more color channels of an RGB image. This assumption fails in the sky region, and haze overestimation takes place, which limits the real-time use of this method. Apart from this limitation, DCP becomes highly time-consuming due to soft matting [68]. To deal with the computational efforts, He et al. further introduced median of median filter [69], fast matting [70], and guided filter [71]. Being the priors and assumptions-based dehazing method, DCP became quite popular as a baseline for the development of single image dehazing methods like Long et al. [42]. It estimates an atmospheric veil for the dehazing of images in remote-sensing applications. In continuation of DCP as prior and assumption, Tang et al. [72] proposed to estimate the thickness of haze in terms of several features like hue disparity, maximum saturation, and illumination channel. Apart from these, several other modified versions of single image dehazing algorithms are also introduced in the literature [73–75].

(b) **Deep Learning based Image Dehazing Methods**: With the emergence of human brain-inspired Convolutional Neural Network (CNN) models in various domains like detection [76], segmentation [77], colorization [78], feature fusion [79], disease identification [80], speech enhancement [81], etc., several researchers attempted to develop various dehazing models designed using CNN-based architecture. These models focus on restoring the scene depth, transmission, or directly obtaining the haze-free image. Among the popular ones, initially, the Color Attenuation Prior (CAP) model was intro-

duced by Zhu et al. [54] for scene depth estimation. The accuracy of CAP gets affected during airlight calculation. Cai et al. introduced an end-to-end dehazing network named DehazeNet [55] for the estimation of scene transmission but left with some haze in order to sustain the textural information. Thus, the combination of fine and coarse scale architectures was proposed named Multi-Scale Convolutional Neural Network (MSCNN) by Ren et al. [56] for preserving textural information along with haze removal in the obtained scene transmission. Thereafter, the atmospheric scattering model is further utilized to obtain the hare-free output. As of now, the underlying assumption of the atmospheric scattering model limits the performance of CNN-based dehazing networks. Hence, state-of-the-art methods for the development of trained networks have been taken into consideration for the development of dehazing networks. Li et al. developed All-in-One Dehazing Network (AOD-Net) [58] and Ren et al. developed contrast relevant features based [82, 83] Gated Fusion Network (GFN) [59]. Though the performance has increased with respect to the aforementioned state-of-the-art methods, the addition of priors and assumptions with trained dehazing networks have also been experimented with by Wang et al. in Atmospheric Illumination Prior Network (AIPNet) [60]. The assumption was based on the Least Mean Square Value (MSE) obtained for illumination channels of hazy and haze-free images in comparison to chrominance channels. Furthermore, the efforts were also made by Yang et al. [84] to enhance the textural details in the obtained scene transmission. The priors and assumptions based on the atmospheric scattering model somehow have become saturated to have experimented with natural hazy scenes. Thus, Liu et al. proposed Generic Model-Agnostic Network (GMAN) [61] irrespective of the atmospheric scattering model-based assumption. Recently, with the combination of pixel and channel attention, Qin et al. proposed Feature Fusion Attention Network (FFA-Net) [85] for restoring the image details. Following the inspiration and observations using biologically inspired deep learning methods, including Dark Channel Network (DCNet) [86], image dehazing using Faster RCNN [87], and Faster RCNN based haze detection network (FRHDNet) [88] are also introduced. In [89], a review of image dehazing algorithms has been presented.

It is evident from the state-of-the-art works done for image enhancement strategies under hazy conditions that image dehazing is a quite popular area of research to achieve robustness in vision-based decision-making applications. The oversaturation in the darker regions due to lack of illumination emerges as the major field of research. In the next subsection, the description and state-of-the-art non-uniform illumination enhancement strategies have been detailed.

1.4 Non-uniform Illumination Enhancement

Image enhancement under non-uniform illumination is a critical task for various applications such as object detection. The performance of these applications undoubtedly depends on the visual quality of the captured scene. The quality of a scene image is vastly degraded by illumination variations. This results in non-uniformly illuminated scene images and dark scene appearances. This also causes the loss of visible scene features. Therefore, non-uniform illumination enhancement is essential to increase the robustness of such applications. Various image enhancement algorithms have been proposed for this purpose in the literature. In the further subsections, the state-of-the-art works done in this field have been discussed.

1.4.1 Retinex Theory

The modeling of human vision for non-uniformly illuminated images was first introduced by Land and McMann in Retinex theory [91]. As per this theory, a non-uniformly illuminated image denoted as $I_{\text{NUI}}(p, q)$ can be expressed through the decomposition into its reflectance $R(p, q)$ and illumination $I_l(p, q)$ components at each pixel location (p, q). This decomposition is formulated as

$$I_{\text{NUI}}(p, q) = R(p, q) \times I_l(p, q). \tag{1.3}$$

Figure 1.4 visualizes the illumination and reflectance components of a non-uniformly illuminated image.



Fig. 1.4 Non-uniform illumination [90]: **a** Non-uniformly illuminated input image, **b** Illumination component, and **c** Reflectance component

1.4.2 Related Work

In the realm of image enhancement, numerous algorithms have been proposed, with histogram-based methods emerging as the simplest and most widely utilized approaches.

One fundamental technique is Histogram Equalization (HE) [92], which uniformly adjusts the histogram of an input image across its dynamic range. However, this method tends to cause over-enhancement in low-contrast areas, leading to a degradation of sharpness at boundaries due to uncertainties.

The introduction of Adaptive Histogram Equalization (AHE) [93] has significantly enhanced performance by considering the histogram of distinct regions as a transformation function. This approach effectively addresses halo artifacts, particularly in low-contrast regions, while preserving brightness. However, AHE encounters challenges when applied to non-uniformly illuminated scene images. Building upon this, the performance of Histogram Equalization (HE) and AHE has been further elevated through the incorporation of Contextual and Variational Contrast (CVC) enhancement [94] and the Layered Difference Representation (LDR) of 2D histograms method [95]. Despite their success in mitigating edge artifacts, these methods still exhibit visual inconsistency in the enhanced scene image.

An alternative category of enhancement algorithms draws inspiration from the Retinex theory [91].

Retinex-based algorithms [96–98], such as those proposed by Jobson et al. [96], leverage human vision models for image enhancement. The widely adopted Single Scale Retinex (SSR) introduced by Jobson et al. [96] is akin to the difference of the Gaussian function and is extensively utilized in computer vision for reflectance estimation. Similarly, the Multi-Scale Retinex (MSR) method [97] was proposed for reflectance estimation, combining weighted sums of different SSR outputs. However, images enhanced using SSR and MSR often exhibit unnatural appearances due to color distortions, primarily caused by the amplification of uncertainties.

To address these issues, a modified MSR approach [98] has been developed, incorporating a color restoration function in chromaticity space. While this modification helps handle uncertainties, the resulting images may still lack naturalness. Another approach, the bright pass filter by Wang et al. [99], estimates illumination by applying a mean filter to pixel intensities higher than the center. However, this method may introduce unnatural halo artifacts near edges, detracting from the overall scene's realism.

Fu et al. introduced a novel approach for simultaneous estimation of reflectance and illumination, as detailed in [100]. Their method utilizes a weighted variational model, effectively handling uncertainties in reflectance while preserving intricate image details.

Differing from the traditional Retinex theory-based image decomposition, Dong et al. approached the enhancement of inverted low-light images by treating them as hazy ones, as described in [101]. Similarly, the image decomposition technique employed in [102] aimed at preserving the naturalness of enhanced images.

A recent advancement in non-uniformly illuminated scene image enhancement is presented in [90], demonstrating promising results. However, this method tends to introduce blurriness near edges due to the elimination of uncertainties. Gao et al. addressed this issue in [103] by incorporating a reflectance estimation approach that maintains the visibility of edges, resulting in improved clarity.

To address issues related to color and illumination distortions, Ren et al. introduced the Low-light image Enhancement CAmera Response Model (LECARM) [104]. Conversely, Guo et al. proposed a lightweight Zero-Reference Deep Curve Estimation (Zero-DCE) network [105]. Zero-DCE was trained to learn the mapping of input images and produce enhanced scene images with uniform illumination. When enhancing hazy or unevenly illuminated images, a primary challenge is to preserve the image's natural appearance, ensuring that the enhanced result is visually appealing to the human eye with minimal error. To tackle this, a fuzzy transform-based approach for illumination enhancement [24, 106], leveraging the smoothing property of fuzzy transform, has been employed.

1.5 Organization of the Book

Following the introduction and previous work done for dealing with hazy and non-uniform illumination conditions, this book will further give a deeper explanation and insights into the novel artificial intelligence methods to address these issues. The chapters in this book are organized as follows:

- **Chapter** 2 presents a computationally efficient transmission map estimation function designed for single image dehazing. It effectively addresses challenges related to over-saturation in darker regions resulting from excessive dehazing.
- **Chapter** 3 introduces the "Compact Single Image Dehazing Network (CSIDNet)", providing insights into its architectural design, network parameters, training process, and a comprehensive experimental analysis. The chapter establishes the superior performance of CSIDNet compared to other deep learning-based dehazing networks.
- **Chapter** 4 details an enhancement approach using the Z-score for hazy and non-uniformly illuminated scene images. This chapter presents a unified enhancement framework utilizing the Z-score weighting function map to enhance the minimum and maximum channels under hazy and non-uniform illumination conditions, respectively.
- **Chapter** 5 introduces a Type-2 fuzzy approach for estimating scene depth and global atmospheric light, utilizing a single similarity function matrix for single image dehazing.
- **Chapter** 6 presents an innovative AI agent in the form of an adaptive interval Type-2 fuzzy filter. This filter effectively handles uncertainties in digital images, preserving image naturalness, and excels in enhancing non-uniformly illuminated scene images while maintaining visual realism and textural information.

- **Chapter** 7 summarizes the methods discussed in this book for single image dehazing and non-uniform illumination enhancement. It also outlines potential future research directions to guide further exploration in these areas.

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Chapter 2
Modified Transmission Map Estimation Function



2.1 Introduction

Inclement weather conditions often introduce various atmospheric particulate matter, including dust, water drops, and aerosols, which significantly impact environmental clarity. Haze, a prevalent phenomenon in such conditions, contributes to the obscurity of the surroundings. This atmospheric effect entails the absorption and scattering of reflected light, resulting in diminished brightness and potential loss of color information in captured images.

Numerous methodologies have been proposed in the literature to address challenging atmospheric conditions, including dehazing techniques [1–6]. A pivotal model that has significantly influenced the development of dehazing methodologies is the "atmospheric scattering model" [7], which is mathematically represented as

$$I_{\text{Hazy}}(p,q) = I_{\text{RL}}(p,q) \times T_{\text{R}}(p,q) + A_{\text{G}}\,(1 - T_{\text{R}}(p,q)) \tag{2.1}$$

where,

- I_{Hazy} is the observed hazy image,
- I_{RL} is the true scene radiance,
- (p, q) is the pixel location,
- A_G is the global atmospheric light, and
- T_{R} represents the transmission map, characterizing the proportion of light that reaches the camera. This is defined as

$$T_{\text{R}}(p,q) = \exp\left(-\beta D_{\text{Map}}(p,q)\right). \tag{2.2}$$

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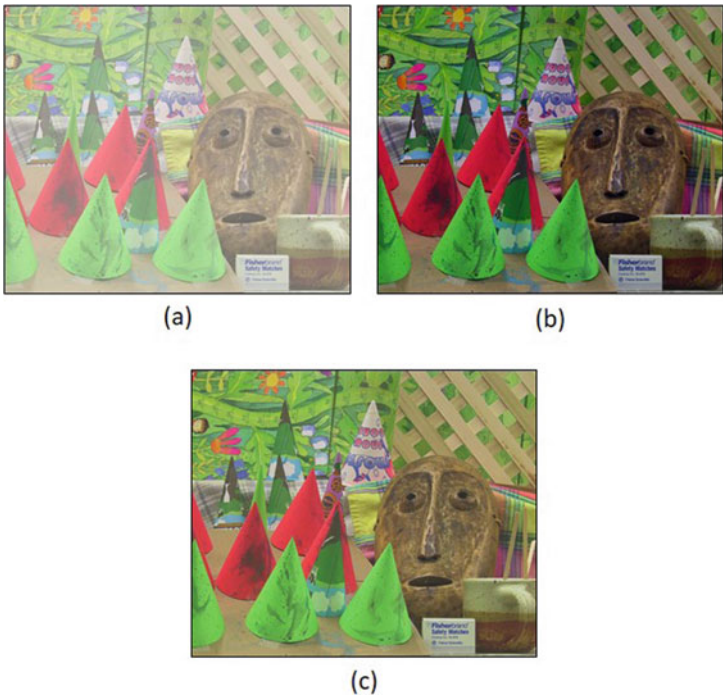


Fig. 2.1 Example of haze overestimation on "Cones" image from Middlebury Stereo dataset [10–12]: **a** Hazy image, **b** Dehazed image using DCP [8] with over-saturation, and **c** Dehazed version of hazy image shown in **a** using unsupervised method presented, i.e., modified transmission map estimation function based method [23]

where

– β is the angular scattering coefficient and
– $D_{\text{Map}}(p, q)$ is the depth map at pixel location (p, q).

Several conventional methods utilize the atmospheric scattering model to generate a clear, haze-free image by computing scene transmission and global atmospheric light. However, certain dehazing techniques, particularly those employing dark channels [8], may encounter the challenge of overestimating airlight in the atmospheric scattering model. This overestimation can result in the excessive intensification of color channels.

Figure 2.1 illustrates an instance of the phenomenon of haze overestimation. The issue arises because transmission maps generated using dark channels exhibit limited variation with corresponding scene depth, leading to the overestimation of haze thickness even in regions with subtle haze. Beyond scene depth, other image channels, such as saturation and value channels, are also affected by haze [9].

In this chapter, a transmission map estimation function is introduced for unsupervised single image dehazing, specifically designed to address the issue of over-

saturation. The function incorporates the utilization of saturation and value channels in addition to the dark channel derived from the input image. This estimation function aims to approximate a haze-free image without encountering the problem of over-intensification in the color channels. Subsequent sections provide a detailed explanation of the methodology employed in this context.

2.2 Methodology

Within this section, a comprehensive exploration of the methodology is presented, elucidating the utilization of a modified transmission map estimation function to mitigate over-saturation in the context of single image dehazing.

Figure 2.2 illustrates the block diagram with the key components of the methodology. Subsequent subsections detail the discussions of each fundamental building block.

2.2.1 Dark Channel

The method extracts the dark channel from an input hazy image for pixel location (p, q) as

$$I_{\text{dark}}(p, q) = \min_{(i,j)\in N_k} \left(\min_{\tau \in \{\text{R, G, B}\}} \left(I^{\tau}_{\text{Hazy}}(i, j) \right) \right) \tag{2.3}$$

where

- N_k is a local patch of size $\sqrt{k} \times \sqrt{k}$ with k elements centered at pixel location (i, j), $\forall 1 \leq (i, j) \leq \sqrt{k}$,
- and τ represents R, G, and B color channels of the input hazy image.

The foundation of the dark channel stems from the observation that the intensity of dark or low pixels is predominantly influenced by the presence of airlight. The airlights get added to the corresponding pixel locations within the captured scene. However, relying solely on this observation to derive transmission results in inaccuracies in measuring the depth map of the captured scene image. To address this limitation, the presented method incorporates the saturation and value channels from the HSV color domain. This aims to provide a more accurate estimation of the haze, enhancing the overall transmission map.

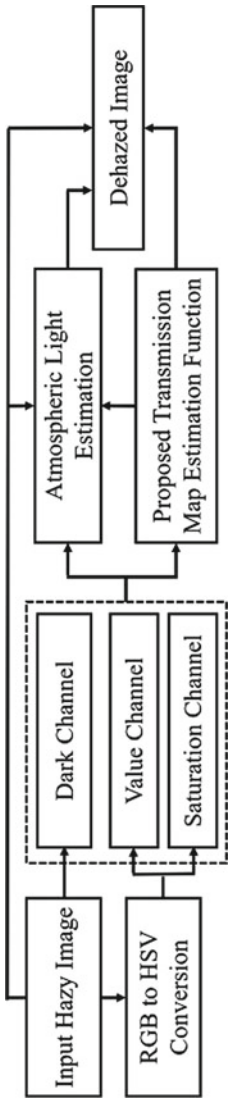


Fig. 2.2 Visual representation of dehazing process of an image for preventing over-saturation during single image dehazing using modified transmission map estimation function based method [23]

2.2.2 RGB to HSV Conversion

Consider an input RGB hazy image $I^{r}_{\text{Hazy}}(p, q)$ having the pixel intensity values for color channels {R, G, B} in the range of $[0, C_{\alpha_0}]$, where C_{α_0} is the maximum possible pixel intensity value.

The value channel $I^{\text{V}}_{\text{Hazy}}(p, q)$ and saturation channel $I^{\text{S}}_{\text{Hazy}}(p, q)$ at pixel location (p, q) in the corresponding HSV color space [13] is given by

$$I^{\text{V}}_{\text{Hazy}}(p, q) = \frac{C_{\alpha_1}(p, q)}{C_{\alpha_0}} \tag{2.4}$$

and

$$I^{\text{S}}_{\text{Hazy}}(p, q) = \begin{cases} \dfrac{C_r(p, q)}{C_{\alpha_1}(p, q)} & C_{\alpha_1}(p, q) > 0, \\ 0 & \text{otherwise} \end{cases} \tag{2.5}$$

where

$$C_{\alpha_1}(p, q) = \max(\text{R, G, B}) \tag{2.6}$$

and

$$C_r(p, q) = C_{\alpha_1}(p, q) - C_{\alpha_2}(p, q) \tag{2.7}$$

for

$$C_{\alpha_2}(p, q) = \min(\text{R, G, B}). \tag{2.8}$$

The value channel serves as an indicator of the amount of light rays that an object reflects. In contrast, saturation gauges the purity of a color, diminishing when additional pigments are introduced to the original hue. Atmospheric haze can detrimentally impact color purity in observed scenes, leading to a reduction in the saturation of the captured image.

In Fig. 2.3, the histogram visualization illustrates the impact on saturation and value channels in a selection of sample hazy images [14]. To enhance comprehension, Fig. 2.4 depicts the dark channel and saturation channel of a hazy image.

The thickness of haze in an image varies with scene depth, being minimal near the leaves within the red rectangle and maximal toward the center marked by a yellow rectangle, as depicted in Fig. 2.4. Notably, in the region enclosed by the red rectangle, where haze is minimal, the dark channel's intensity is close to zero, as seen in Fig. 2.4d. This is attributed to the fact that areas with little or no haze exhibit a minimum intensity value of zero. Conversely, the saturation channel's intensity is close to one, indicating higher purity in the absence of significant haze, as illustrated in Fig. 2.4g.

Similarly, within the yellow rectangle, the dark channel appears close to white due to higher minimum intensity values, as shown in Fig. 2.4f. In contrast, the saturation

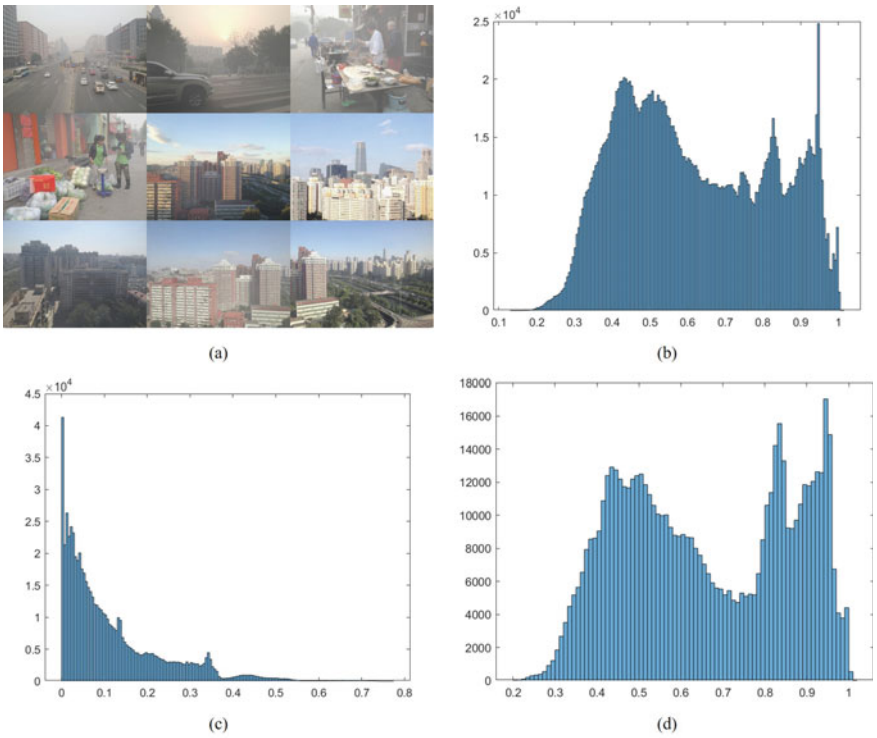


Fig. 2.3 Effect of haze on image channels [23]: **a** Sample hazy images [14], **b** Occurrence of gray level pixel intensities values for the images shown in **a**, **c** Occurrence of pixel intensities values of image saturation for the images shown in **a**, i.e., Saturation channel, and **d** Occurrence of pixel intensities values of image illumination for the images shown in **a**, i.e., Value/Illumination channel

channel is close to zero, signifying a lack of purity in this region with considerable haze, as depicted in Fig. 2.4i.

Consequently, it can be deduced that with an increase in haze, the intensity of the dark channel transitions from zero to one, while saturation decreases from one to zero. Regions with intermediate haze, denoted by the white rectangles in both the dark channel and saturation channel, exhibit a mid-range gray value due to the presence of moderate haze, as depicted in Figs. 2.4e, 2.4h. Consequently, a more comprehensive understanding of haze thickness estimation can be achieved by considering both channels concurrently.

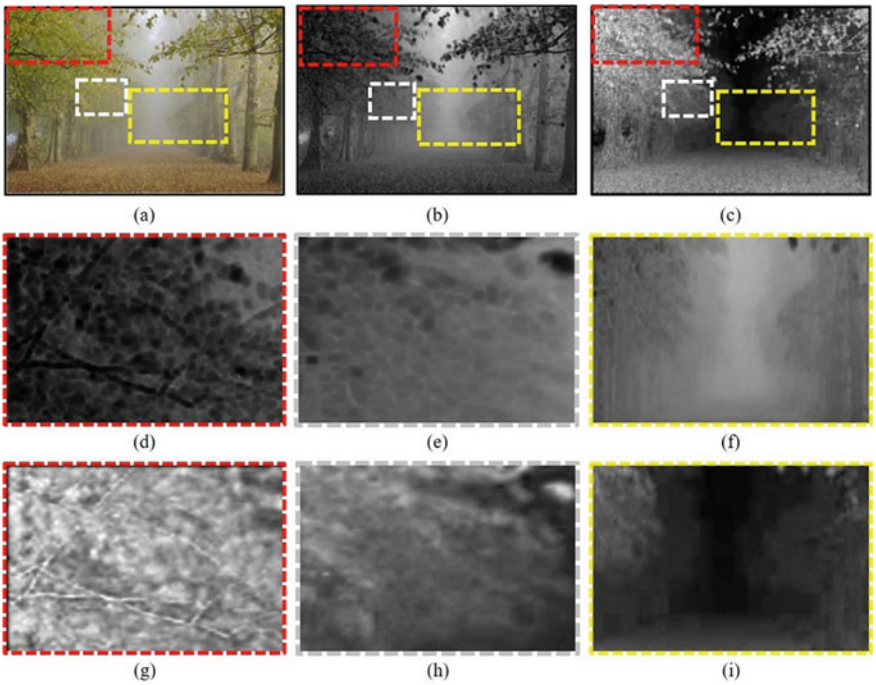


Fig. 2.4 Illustration of over-saturation effects on a natural hazy image of a "tree" [8][23]: **a** Observed hazy image, **b** Dark channel using DCP method for the image shown in **a**, **c** Saturation channel for the image shown in **a**, Enlarged version of image regions marked with **d** Red square, **e** White square, **f** Yellow square for the image shown in **b**, Enlarged version of image regions marked with **g** Red square, **h** White square, and **i** Yellow square for the image shown in **c**. The colored squares shown using red, white, and yellow are represented as left, center, and right patches in the images shown in **a**—**c**

2.2.3 Modified Transmission Map Estimation Function

The transmission map function $T_{\mathrm{R}}(p, q)$ according to the conventional atmospheric scattering model [7] is given by

$$T_{\mathrm{R}}(p, q) = \exp\left(-\beta D_{\mathrm{Map}}(p, q)\right). \tag{2.9}$$

where

- β is the angular scattering coefficient and
- $D_{\mathrm{Map}}(p, q)$ is the depth map at pixel location (p, q).

The method exclusively relies on dark channels for haze estimation, producing a transmission map with minimal variation corresponding to scene depth. This characteristic leads to an overestimation of haze thickness. Additionally, the approach

introduces a heuristic parameter, the angular scattering coefficient β, necessitating manual adjustment for different images. Consequently, the reliance on this parameter adds a need for supplementary information in the process of estimating haze thickness.

Inspired by DCP [8], AIPNet [15], and CAP [16] methods, the modified transmission map function considers saturation and value channels of input hazy image, as they are the most affected by the haze in comparison to the hue channel.

The modified transmission map estimation function is given by

$$T'_{\mathrm{R}}(p,q) = \exp\left(-\frac{I_{\mathrm{dark}}(p,q)}{\exp\left(\left(I^{\mathrm{S}}_{\mathrm{Hazy}}(p,q)\right)^{4} \times \left(I^{\mathrm{V}}_{\mathrm{Hazy}}(p,q) + I^{\mathrm{S}}_{\mathrm{Hazy}}(p,q)\right)^{0.01}\right)}\right) \tag{2.10}$$

where

– $T'_{\mathrm{R}}(p,q)$ is the modified transmission map function.

Inference drawn from Sect. 2.2.2 suggests that saturation decreases in the presence of haze. Consequently, the following conclusions can be derived:

(i) Saturation in regions with thick haze tends to approach zero. Consequently, in such cases, the denominator of (2.10) will be close to one, emphasizing the reliance on the dark channel for scene transmission.
(ii) Conversely, regions with fine haze exhibit higher saturation. In these instances, the transmission map becomes more sensitive to the value channel in the denominator of (2.10). This sensitivity serves to prevent over-enhancement of the image, ensuring that color fidelity and contrast information are preserved by the value channel, in addition to preventing over-saturation.

2.2.4 Atmospheric Light Estimation

The transmission map generated using the modified estimation function as in (2.10) is further used to calculate the global atmospheric light A_G as

$$A_{\mathrm{G}} = \max_{(i,j)\in\{(p,q)|T'_{\mathrm{R}}(p,q)<T_0\}} \left(I_{\mathrm{Hazy}}(i,j)\right) \tag{2.11}$$

where

– T_0 is the threshold for medium transmission.

In this chapter, the value of T_0 is considered as 0.1.

2.2.5 *Image Dehazing*

Finally, a haze-free output image $I_{\text{enh}}(p, q)$ is obtained by rearranging the atmospheric scattering model (1.2) as

$$I_{\text{enh}}(p, q) = \frac{I_{\text{Hazy}}(p, q) - A_{\text{G}}}{T'_{\text{R}}(p, q)} + A_{\text{G}}. \tag{2.12}$$

Another merit of using (2.10) is that it depends only on I_{Hazy}, and does not require any manual adjustment of parameters, i.e., angular scattering coefficient β.

2.3 Results and Discussions

In this section, the experimental results generated using the methodology introduced in this chapter have been presented and compared with several popular state-of-the-art dehazing methods in terms of qualitative and quantitative analysis. The comparison has been presented with the following state-of-the-art dehazing methods: Contrast Limited Adaptive Histogram Equalization (CLAHE) [17], DCP [8], BCCR [18], CAP [16], DehazeNet [19], and GMAN [20]. For quantitative comparison, Peak Signal to Noise Ratio (PSNR) [21] and Structural SIMilarity (SSIM) index [22] scores have been used as performance measures. In further subsections, the qualitative and quantitative analyses have been discussed in detail.

2.3.1 *Quantitative Comparison*

For quantitative analysis, Table 2.1 tabulates the PSNR and SSIM index values for images from Middlebury Stereo dataset [10–12] as shown in Fig. 2.5.

The visual comparison of these images is shown in Figs. 2.6, 2.7, 2.8, 2.9, and 2.10. It can be seen that the modified transmission map estimation function-based method significantly outperforms nearly all the considered state-of-the-art dehazing methods in terms of PSNR, and gives a comparable performance in terms of the SSIM index.

The method demonstrates higher PSNR and SSIM values compared to DCP, indicating the efficient estimation of haze thickness through the use of additional channels. Notably, it eliminates the need for soft matting, a requirement in DCP, suggesting that the modified transmission map estimation function directly yields a haze-free image without additional operations. This implies a potential speed advantage over DCP, as the soft matting employed by DCP is often time-consuming. While CAP shows slightly higher results, it's important to acknowledge that CAP is a deep

Table 2.1 Comparative performance assessment of methodology using modified transmission map on images subset from the Middlebury Stereo dataset [10–12]. These images are shown in Fig. 2.5

Dataset	Metric	CLAHE [17]	DCP [8]	BCCR [18]	CAP [16]	DehazeNet [19]	GMAN [20]	Modified function [23]
Computer [10–12]	**PSNR** [21]	27.428	27.648	27.928	27.548	27.565	28.232	**29.962**
	SSIM [22]	0.907	0.922	0.793	0.962	0.948	0.835	**0.958**
Cones [10–12]	**PSNR** [21]	27.558	27.751	28.939	30.123	27.630	28.726	**30.817**
	SSIM [22]	0.858	0.956	0.936	0.976	0.952	0.886	**0.964**
Laundry [10–12]	**PSNR** [21]	27.436	27.502	28.634	29.304	28.164	28.128	**29.215**
	SSIM [22]	0.867	0.890	0.922	0.975	0.957	0.830	**0.957**
Moebius [10–12]	**PSNR** [21]	27.623	27.648	28.500	28.541	29.140	28.436	**29.221**
	SSIM [22]	0.850	0.901	0.928	0.957	0.949	0.861	**0.952**
Reindeer [10–12]	**PSNR** [21]	27.545	28.673	27.986	27.725	27.335	29.909	**27.912**
	SSIM [22]	0.652	0.884	0.764	0.831	0.790	0.863	**0.798**



Fig. 2.5 **Top row:** Images with some haze added to it synthetically. **Bottom row:** Image with no haze, i.e., ground truth version, of corresponding images shown in the top row. The images shown are named as (read from left to right) "Computer", "Cones", "Laundry", "Moebius", and "Reindeer" from the Middlebury Stereo dataset [10–12]

learning-based dehazing model trained on millions of images. In contrast, the introduced method in this chapter is an unsupervised model that requires no training. Considering these factors, the modified transmission map estimation function-based method yields remarkable results.



Fig. 2.6 Perceptual analysis on an image named "Computer" from the Middlebury Stereo dataset [10–12]: **a** Image with no haze (Ground truth), **b** Image with some haze present, Dehazed version of image shown in **b** using **c** CLAHE [17], **d** DCP [8], **e** BCCR [18], **f** CAP [16], **g** DehazeNet [19], **h** GMAN [20], and **i** Modified transmission map estimation function based method [23]

2.3.2 Qualitative Comparison

The comparison of experimental results, on some natural and synthetic hazy images commonly used in literature [10–12], are shown in Figs. 2.11, 2.12, 2.6, 2.7, 2.8, 2.9, and 2.10 for "tree", "city", "Computer", "Cones", "Laundry", "Moebius", and "Reindeer" images, respectively.

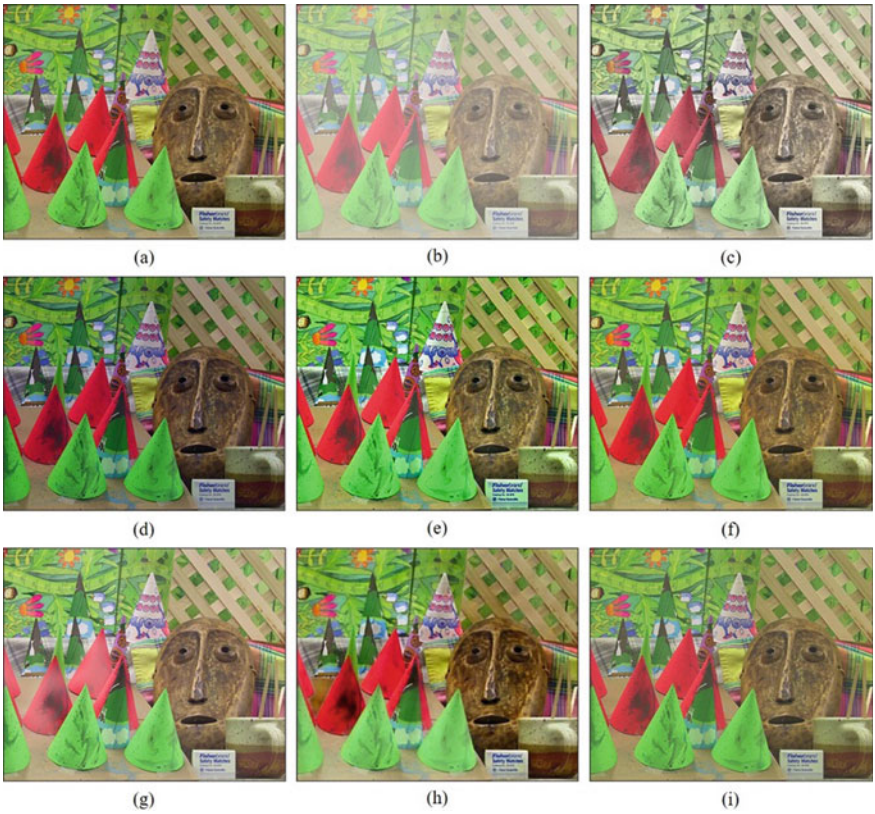


Fig. 2.7 Perceptual analysis on an image named "Cones" from the Middlebury Stereo dataset [10–12]: **a** Image with no haze (Ground truth), **b** Image with some haze present, Dehazed version of image shown in **b** using **c** CLAHE [17], **d** DCP [8], **e** BCCR [18], **f** CAP [16], **g** DehazeNet [19], **h** GMAN [20], and **i** Modified transmission map estimation function based method [23]

The synthetic hazy images are generated from images of Middlebury Stereo dataset [10–12]. These synthetic hazy images have been generated using the conventional atmospheric scattering model [7] by considering different transmission values for different haze densities.

The visual comparison presented in these figures illustrates the efficacy of the modified transmission map function by producing visually appealing results without any over-saturation effects.

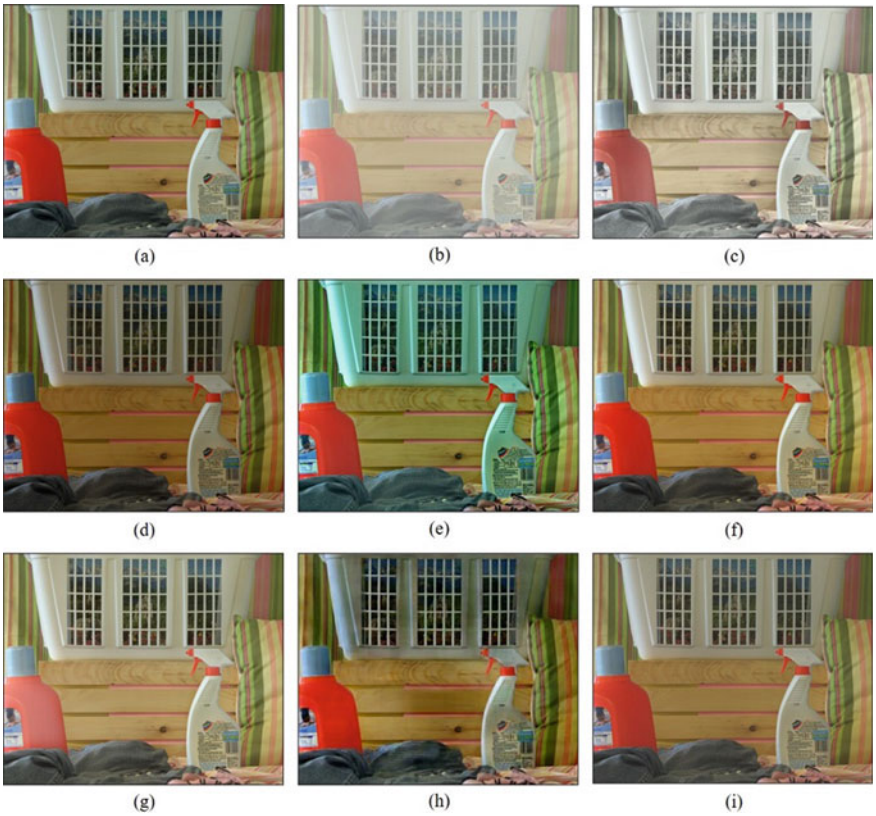


Fig. 2.8 Perceptual analysis on an image named "Laundry" from the Middlebury Stereo dataset [10–12]: **a** Image with no haze (Ground truth), **b** Image with some haze present, Dehazed version of image shown in **b** using **c** CLAHE [17], **d** DCP [8], **e** BCCR [18], **f** CAP [16], **g** DehazeNet [19], **h** GMAN [20], and **i** Modified transmission map estimation function based method [23]

For both natural and synthetic hazy images, it is evident that CLAHE falls short of completely removing haze and tends to lose color information. Similarly, both BCCR and GMAN methods exhibit noticeable artifacts. DCP, although effective to some extent, introduces noticeable artifacts, particularly in the sky regions of Figs. 2.11 and 2.12. Moreover, it tends to overestimate haze, resulting in an over-intensification of the color channels. In contrast, the presented method exhibits no such distortions. The absence of artifacts suggests that the modified transmission map estimation function successfully and accurately estimates the haze thickness.



Fig. 2.9 Perceptual analysis on an image named "Moebius" from the Middlebury Stereo dataset [10–12]: **a** Image with no haze (Ground truth), **b** Image with some haze present, Dehazed version of image shown in **b** using **c** CLAHE [17], **d** DCP [8], **e** BCCR [18], **f** CAP [16], **g** DehazeNet [19], **h** GMAN [20], and **i** Modified transmission map estimation function based method [23]

2.4 Summary

This chapter has presented a comprehensive overview of a modified transmission map estimation function-based method for single image dehazing. The approach incorporates the value and saturation channels alongside the dark channel of the input hazy image, effectively eliminating haze without inducing over-saturation or over-intensification effects. This addresses the limitations associated with conventional transmission estimation functions. Notably, the method does not necessitate

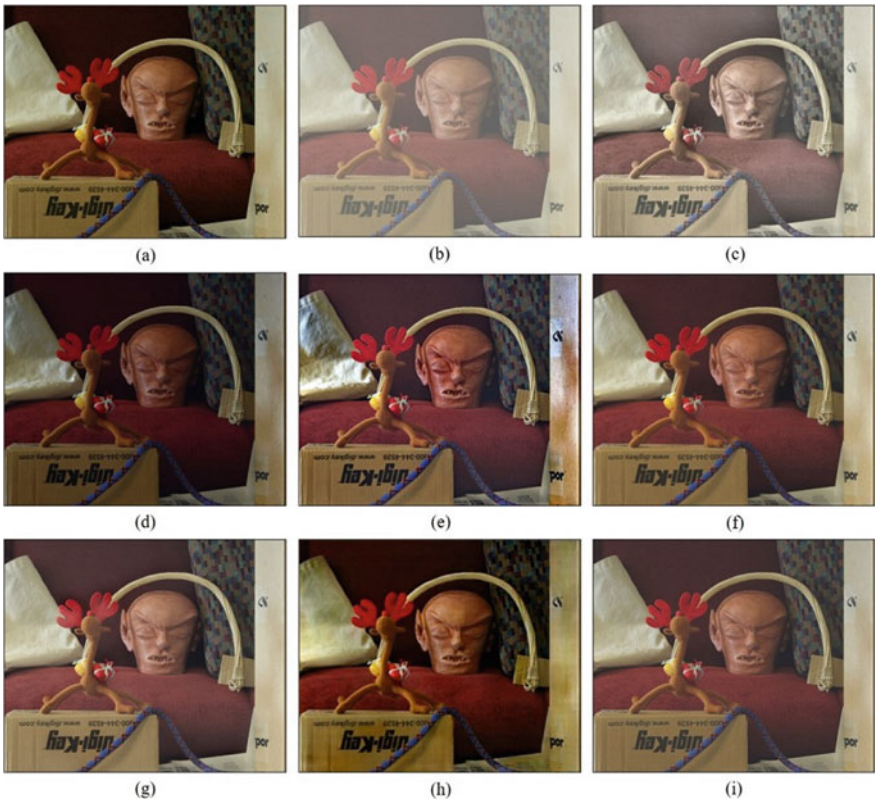


Fig. 2.10 Perceptual analysis on an image named "Reindeer" from the Middlebury Stereo dataset [10–12]: **a** Image with no haze (Ground truth), **b** Image with some haze present, Dehazed version of image shown in **b** using **c** CLAHE [17], **d** DCP [8], **e** BCCR [18], **f** CAP [16], **g** DehazeNet [19], **h** GMAN [20], and **i** Modified transmission map estimation function based method [23]

additional operations like filtering or soft matting. While the results of deep learning-based models may marginally surpass its performance, it is essential to highlight that the introduced method is unsupervised and requires no training. These characteristics underscore its suitability and efficiency for real-time vision-based applications. Future enhancements could involve considering additional color domains and exploring other image attributes affected by the haze phenomenon to further refine the method.



Fig. 2.11 Perceptual analysis of a naturally hazy image of "tree" [8]: **a** Image captured under haze, Dehazed versions of image shown in **a** using **b** CLAHE [17], **c** DCP [8], **d** BCCR [18], **e** CAP [16], **f** DehazeNet [19], **g** GMAN [20], and **h** Modified transmission map estimation function method [23]

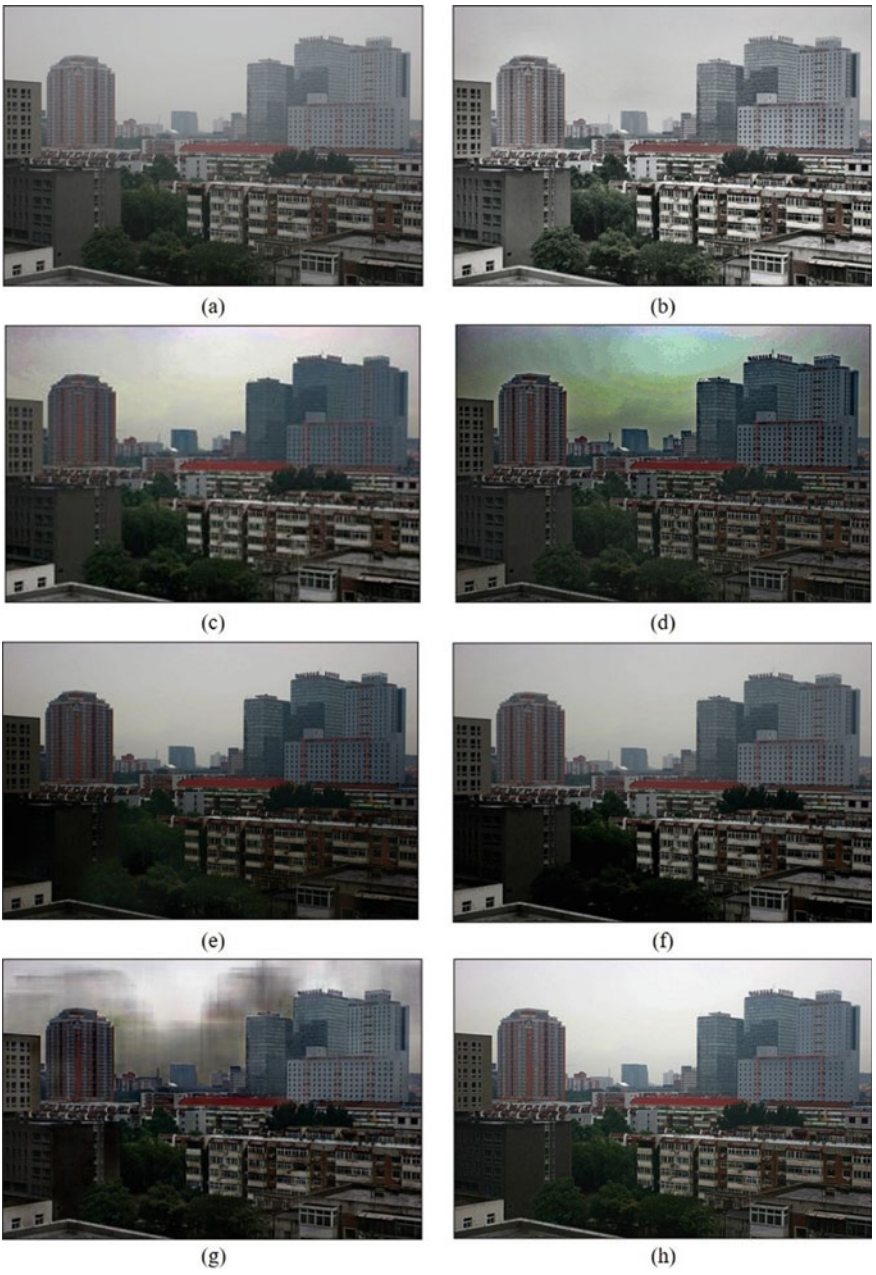


Fig. 2.12 Perceptual analysis of a naturally hazy image of "city" [8]: **a** Image captured under haze, Dehazed versions of image shown in **a** using **b** CLAHE [17], **c** DCP [8], **d** BCCR [18], **e** CAP [16], **f** DehazeNet [19], **g** GMAN [20], and **h** Modified transmission map estimation function method [23]

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Chapter 3
Compact Single Image Dehazing Network



3.1 Introduction

Haze is characterized as signal-dependent non-linear noise, causing attenuation in an image with an increase in scene depth [1]. Consequently, different pixel locations within a scene image experience varying degrees of degradation. In recent times, single image dehazing has gained greater popularity compared to methods that require additional data, such as multiple images [2] or different degrees of polarization [3]. The preference for single image dehazing is attributed to its practicality for real-time applications where acquiring additional information is often infeasible [4–15]. Consequently, the challenge of single image dehazing is intensified [16].

A prominent challenge in dehazing research arises from the scarcity of datasets containing pairs of natural images with and without haze. Atmospheric conditions are unlikely to remain consistent between hazy and clear days, making it difficult to obtain matching pairs. Consequently, researchers often resort to using synthetic hazy images for training purposes and subsequently evaluate dehazing methods on natural hazy scene images. Despite considerable advancements in removing haze through deep learning approaches [17–22], challenges persist, particularly in dealing with the intricate architecture of deep learning models and the necessity for extensive training.

This chapter introduces the "Compact Single Image Dehazing Network (CSIDNet)", a deep learning-based model designed for enhancing outdoor scenes through single image dehazing. The key features of CSIDNet can be summarized as follows:

(i) CSIDNet lives up to its name by being a more compact network compared to existing deep learning-based dehazing models, consisting of only three convolutional layers.
(ii) The network is trained on a smaller dataset with a reduced number of images, without compromising its performance. This characteristic facilitates faster training and makes CSIDNet more accessible for real-time applications.

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https://doi.org/10.1007/978-981-97-2011-8_3

(iii) The dehazed images produced by CSIDNet exhibit visual appeal and surpass the performance of benchmarked deep learning-based dehazing models, as measured by PSNR and SSIM index evaluations.

To the best of our knowledge, many existing dehazing models tend to be computationally expensive, resulting in longer run-times, or demand substantial resources for implementation. In contrast, CSIDNet has been purposefully crafted with a reduced number of layers and trained on a notably smaller dataset. Despite these streamlined attributes, CSIDNet demonstrates superior performance compared to state-of-the-art dehazing methods. Subsequent sections provide a detailed explanation of the architectural design of CSIDNet.

3.2 Compact Single Image Dehazing Network

This section provides an overview of the dehazing network, with Fig. 3.1 illustrating the architecture. Notably, unlike other deep learning-based dehazing networks, CSIDNet is distinctive for its utilization of only three convolutional layers. Further elucidation on the architectural design of CSIDNet is provided in subsequent sections.

3.2.1 Hazy Input Features

Inspired by DCP [23] and AIPNet [24], CSIDNet extracts dark channels and illumination channels from input hazy images to learn the pattern of haze.

3.2.1.1 Dark Channel Prior

DCP [23] operates under the assumption that pixels in a local patch for at least one color channel consistently exhibit low pixel intensity values in the non-sky region. Consequently, the dark channel is computed using

$$I_{\text{dark}}(p, q) = \min_{(i,j)\in N_k} \left(\min_{\tau \in \{\text{R, G, B}\}} \left(I^{\tau}_{\text{Hazy}}(i, j) \right) \right) \tag{3.1}$$

where

- N_k is a local patch of size $\sqrt{k} \times \sqrt{k}$ with k elements centered at pixel location (i, j), $\forall 1 \leq (i, j) \leq \sqrt{k}$,
- and τ represents R, G, and B color channels of the input hazy image.

Equation (3.1) suggests that the minimum intensity at each pixel location across all color channels tends to have a very low value in a haze-free image. This is

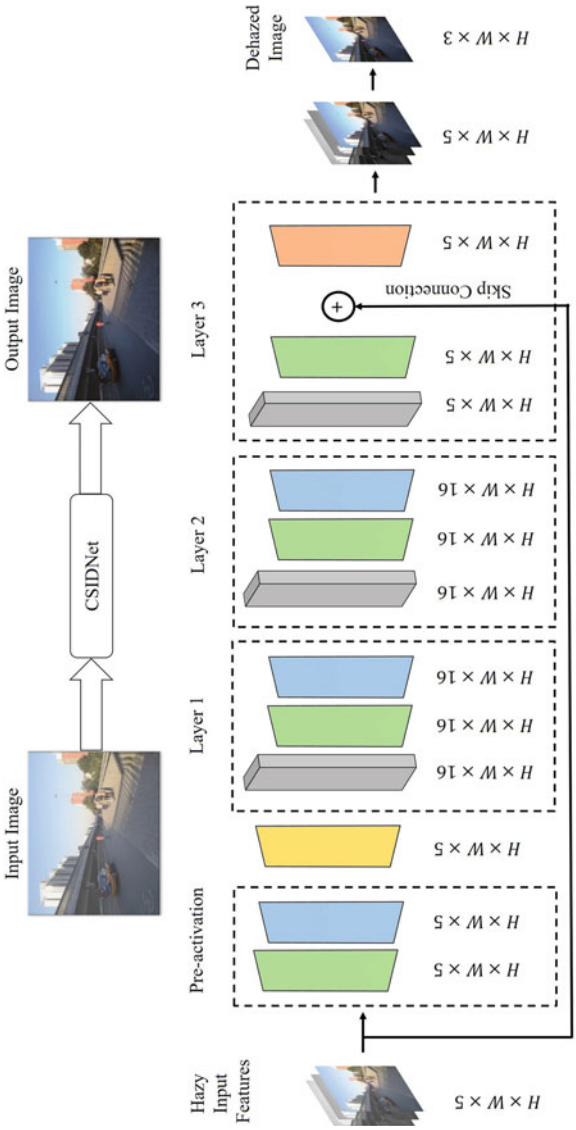


Fig. 3.1 Visual representation of the architectural design of the presented network, i.e., "CSIDNet" [39]. In this design shown, the input layer takes image size as $H \times W$ with H height (rows) and W width (columns). The input layer comprises five input features which include R, G, and B color channels of the image with haze (I^{τ}_{Hazy}), dark channel (I_{dark}), and value/illumination channel (I^{V}_{Hazy}). Here, $\tau \in \{\text{R, G, B}\}$

primarily attributed to the dominance of one color channel in colorful objects or the inherently low intensity of dark objects such as tree trunks, shadows, etc. Throughout this chapter, experimentation has been conducted using the dark channel with a patch size of 1×1.

3.2.1.2 Illumination Channel

For a hazy input image $I_{\text{Hazy}}(p, q)$, the illumination channel or Y channel is obtained by YCbCr color domain [25]. The RGB color channels are converted to YCbCr color channels using

$$\begin{bmatrix} I^{\text{Y}}_{\text{Hazy}}(p,q) \\ I^{\text{Cb}}_{\text{Hazy}}(p,q) \\ I^{\text{Cr}}_{\text{Hazy}}(p,q) \end{bmatrix} = \begin{bmatrix} 0.299 & 0.587 & 0.144 \\ -0.169 & -0.331 & 0.500 \\ 0.500 & -0.419 & -0.081 \end{bmatrix} \begin{bmatrix} I^{\text{R}}_{\text{Hazy}}(p,q) \\ I^{\text{G}}_{\text{Hazy}}(p,q) \\ I^{\text{B}}_{\text{Hazy}}(p,q) \end{bmatrix} \tag{3.2}$$

where

- (p, q) is the pixel location,
- $I^{\text{Y}}_{\text{Hazy}}$ is the illumination channel,
- $I^{\text{Cb}}_{\text{Hazy}}$ and $I^{\text{Cr}}_{\text{Hazy}}$ are the corresponding chrominance channels,
- and $I^{\text{R}}_{\text{Hazy}}$, $I^{\text{G}}_{\text{Hazy}}$, and $I^{\text{B}}_{\text{Hazy}}$ are the red, blue, and green color channels of the input image I_{Hazy}, respectively.

Figure 3.2 shows the histogram visualization of the effect on dark and illumination channels on a few sample hazy images [26, 27].

The R, G, and B color channels, dark channel, and illumination channel are then concatenated to form the hazy input features for the network as

$$I^{\text{Input}}_{\text{Hazy}}(p,q) \xleftarrow{\text{Concatenate}} \left(I^{\text{R}}_{\text{Hazy}}(p,q), I^{\text{G}}_{\text{Hazy}}(p,q), I^{\text{B}}_{\text{Hazy}}(p,q), I_{\text{dark}}(p,q), I^{\text{Y}}_{\text{Hazy}}(p,q) \right). \tag{3.3}$$

3.2.2 Pre-activation

The hazy input features obtained by (3.3) are fully pre-activated [28] with batch normalization (Ψ) and leaky ReLU activation function (Φ) as

$$I_{\text{Normalized}}(p,q) = \Phi\left(\Psi\left(I^{\text{Input}}_{\text{Hazy}}(p,q)\right)\right). \tag{3.4}$$

A dropout layer (Ω) has also been included after the pre-activation stage to avoid any overfitting as

$$I_{\text{Dropout}}(p,q) = \Omega\left(I_{\text{Normalized}}(p,q)\right). \tag{3.5}$$

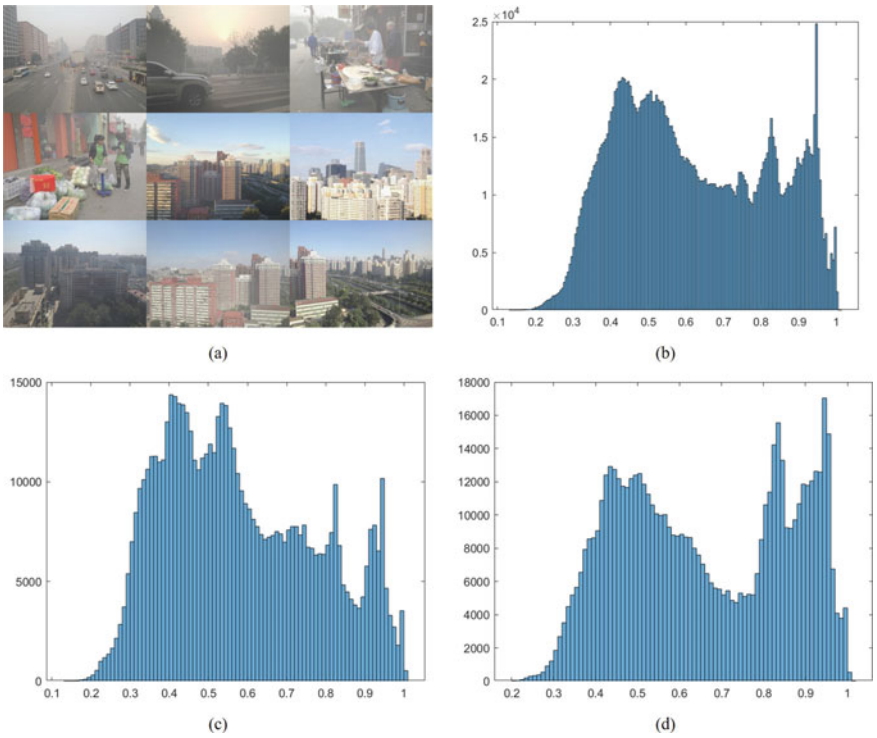


Fig. 3.2 Effect of haze on hazy input features [39]: **a** Sample hazy images [26], **b** Occurrence of gray level pixel intensities values for the images shown in **a**, **c** Occurrence of pixel intensities values of image saturation for the images shown in **a**, i.e., Saturation channel, and **d** Occurrence of pixel intensities values of image illumination for the images shown in **a**, i.e., Value/Illumination channel

3.2.3 Convolutional Layers

The output of the dropout layer obtained by (3.5) is passed through three consecutive combinations of the following: convolutional layer, batch normalization layer, and activation layer, i.e., Layers 1, 2, and 3 of Fig. 3.1, as

$$I_{\mathrm{L}+1}(p,q) = \Phi\left(\Psi\left(W_{\mathrm{L}} * I_{\mathrm{L}}(p,q)\right)\right); \quad \forall\, \mathrm{L} = 1, 2, 3 \tag{3.6}$$

where W is the kernel weight matrix between two layers, $I_{\mathrm{L}}(p,q)$ and $I_{\mathrm{L}+1}(p,q)$, which act as input and output of Lth layer, respectively. The input for the first layer is

$$I_1(p,q) = I_{\mathrm{Dropout}}(p,q). \tag{3.7}$$

3.2.3.1 Skip Connection

Furthermore, to compensate for any possible information loss, the network contains one global skip connection [29]. The output of this skip connection connects hazy input features $I_{\text{Hazy}}^{\text{Input}}(p, q)$ and output of 3rd convolutional layer after batch normalization as

$$I_{\text{Skip}}(p, q) = \Psi\left(W_3 * I_3(p, q)\right) + I_{\text{Hazy}}^{\text{Input}}(p, q). \tag{3.8}$$

3.2.3.2 Sigmoid Activation

The output of global skip connection I_{Skip} is then passed through Sigmoid activation (f_σ) to constraint the output in the range of 0 to 1 as

$$I_{\text{Sigmoid}}(p, q) = f_\sigma\left(I_{\text{Skip}}(p, q)\right). \tag{3.9}$$

3.2.4 Dehazed Image

The output $I_{\text{Sigmoid}}(p, q)$ comprises five channels. Among these, the first three channels correspond to the desired enhanced RGB output image, wherein

$$I_{\text{enh}}(p, q) \xleftarrow{\text{Extract first three channels}} I_{\text{Sigmoid}}(p, q). \tag{3.10}$$

Hence, CSIDNet directly produces an enhanced, haze-free, or dehazed image $I_{\text{enh}}(p, q)$ interactively, without the need for intermediate results such as a transmission map or atmospheric light.

3.3 Results and Discussions

This segment provides a comprehensive comparison, both qualitatively and quantitatively, of the outcomes achieved through the utilization of CSIDNet. The performance metrics are compared with those from existing studies, including BCCR [30], DCP [23], CAP [31], DehazeNet [32], MSCNN [33], AOD-Net [34], and GMAN [35]. The evaluation is conducted based on measures such as PSNR [36] and the SSIM index [37].

While many deep learning-based dehazing models undergo training on datasets comprising thousands to millions of images, CSIDNet distinguishes itself by achieving high performance with a modest training set of only 200 images. These images were randomly selected from the Outdoor Training Set (OTS) within the REalis-

tic Single Image DEhazing (RESIDE) dataset [26]. Notably, CSIDNet maintains its performance despite this comparatively smaller training data size.

The training process for CSIDNet is remarkably efficient, requiring approximately 30 min on hardware equipped with an Nvidia GeForce 940MX 2GB Graphics card. This stands in contrast to other models, which typically demand significantly longer training times, often ranging from 12 to 36 h. The training sessions for CSIDNet were conducted on an Intel Core i5-7200U 7th generation processor boasting a 2.5GHz clock speed and 8 GB DDR4 RAM.

The subsequent sections delve into specific aspects of CSIDNet, providing detailed explanations of datasets used, network parameters, loss functions employed, and both quantitative and qualitative comparisons. These discussions contribute to a comprehensive understanding of CSIDNet's architecture, training methodology, and comparative performance metrics.

3.3.1 Datasets

The RESIDE dataset [26] comprises a substantial collection of synthetic hazy images for training and testing purposes. Specifically, it includes 72,135 synthetic hazy images in the OTS used for training. For testing, the dataset incorporates 500 synthetic hazy images in the Synthetic Objective Testing Set (SOTS) and an additional 10 synthetic hazy images in the Hybrid Subjective Testing Set (HSTS), each accompanied by their respective ground truths.

In the training phase of CSIDNet, a subset of 200 hazy images has been randomly chosen from the OTS. These selected images, along with their corresponding ground truths, form one set of training data. This process is repeated to create a total of five distinct sets, labeled as Set 1, Set 2, Set 3, Set 4, and Set 5. Each set consists of 200 hazy images and their respective ground truths, offering diversity and variability in the training data for robust model learning.

3.3.2 Network Parameters and Loss Function

CSIDNet incorporates convolutional layers with depths of 16 for the first layer, 16 for the second layer, and 5 for the third layer. The network initially trained on 224×224 images, is versatile enough to be tested on images of varying resolutions. Using a 3×3 kernel size, the convolutional layers employ the Leaky ReLU activation function with a slope of 0.2. A dropout rate of 0.2 is applied for improved model robustness.

The convolutional layer kernel weights are initialized using the He Uniform initializer, as suggested by [38]. Training involves 100 iterations with the Adam optimizer, incorporating momentum values of $\beta_1 = 0.9$ and $\beta_2 = 0.999$. The training process is guided by the Mean Square Error (MSE) loss function:

$$(\text{Loss})_{\text{MSE}} = \frac{1}{\tau \times H \times W} \sum_{i=1}^{\tau} \sum_{j=1}^{H} \sum_{k=1}^{W} \left(I^i_{\text{enh}}(j,k) - I^i_{\text{GT}}(j,k) \right)^2 \tag{3.11}$$

where I_{enh} is the enhanced or dehazed image, I_{GT} is the ground truth image, H, and W represents the height and width of the image, respectively, and τ denotes the R, G, B color channels.

For further analysis, the CSIDNet has also been trained with Mean Absolute Error (MAE) as a loss function. It is also termed as $L1$ loss and can be defined by

$$(\text{Loss})_{L1} = \frac{1}{\tau \times H \times W} \sum_{i=1}^{\tau} \sum_{j=1}^{H} \sum_{k=1}^{W} \left| I^i_{\text{enh}}(j,k) - I^i_{\text{GT}}(j,k) \right| \tag{3.12}$$

The number of epochs used for training with $L1$ loss function are 100, 150, and 200 in order to find the best possible hyper-parameters.

3.3.3 Quantitative Comparison

The performance evaluation of dehazed images generated by CSIDNet is presented in Table 3.1, with a focus on average PSNR and SSIM measures for images sourced from the SOTS and the HSTS within the RESIDE dataset [26].

Notably, while GMAN exhibits a slightly higher PSNR than CSIDNet for the HSTS dataset, it concurrently demonstrates a lower SSIM index. The lower SSIM suggests a greater presence of distortions in GMAN's output. Similarly, AOD-Net

Table 3.1 Performance assessment of CSIDNet. Here, PSNR and SSIM values are the averages of PSNR and SSIM values obtained for all images present in the dataset. (∗ First, † Second, and ††Third: Top 3 performance metrics values)

Methods	Test set			
	SOTS [26]		HSTS [26]	
	PSNR [36]	SSIM [37]	PSNR [36]	SSIM [37]
BCCR [30]	28.147	0.820	28.187	0.791
DCP [23]	16.620	0.818	14.840	0.761
CAP [31]	28.009	0.793	28.333	$0.858^{\dagger\dagger}$
DehazeNet [32]	$28.613^{\dagger\dagger}$	$0.864^{\dagger\dagger}$	$29.091*$	$0.890^{\dagger}$
MSCNN [33]	28.216	0.840	28.287	0.807
AOD-Net [34]	28.234	$0.902*$	28.254	$0.891*$
GMAN [35]	$28.836^{\dagger}$	0.848	$29.052^{\dagger}$	0.846
CSIDNet [39]	$28.995*$	$0.883^{\dagger}$	$28.828^{\dagger\dagger}$	0.848

Table 3.2 Run-time comparison of CSIDNet. Here, the run-time mentioned is the average run-time for all images present in the dataset. The run-time is measured in seconds

Methods	Test set		
	SOTS [26]	HSTS [26]	Natural
BCCR [30]	1.174	1.002	1.604
DCP [23]	>50	>50	>50
CAP [31]	0.327	0.289	0.497
DehazeNet [32]	1.951	1.660	3.014
MSCNN [33]	0.954	0.742	1.689
AOD-Net [34]	0.349	0.297	0.562
GMAN [35]	1.441	1.373	2.036
CSIDNet [39]	**0.326**	**0.282**	**0.435**

outperforms CSIDNet in terms of SSIM for the SOTS dataset. However, CSIDNet showcases superior run-time efficiency compared to AOD-Net.

Further details on the comparative run-time performance across SOTS, HSTS, and natural hazy images are presented in Table 3.2. This comprehensive analysis provides insights into both the quantitative performance metrics and the efficiency of CSIDNet in comparison to other dehazing models.

An extended run-time can result in a lag in the processing pipeline, adversely affecting real-time performance. Timely availability of output is crucial for interactive applications. Examining Table 3.2, it is evident that DehazeNet exhibits one of the lengthier run-times among all models, with GMAN closely following suit. This extended processing time makes both models impractical for real-time applications. In contrast, CSIDNet stands out as the fastest among the compared models.

The efficiency of CSIDNet in terms of run-time, coupled with its competitive performance in terms of PSNR and SSIM values, positions it as a compelling choice for dehazing tasks. The combination of robust performance and swift processing makes CSIDNet well-suited for applications where real-time output is paramount.

3.3.4 Qualitative Comparison

The visual representation of dehazed images from different datasets is illustrated in Figs. 3.3, 3.4, and 3.5. CSIDNet demonstrates its capability to produce dehazed images without introducing any discernible visual artifacts.

In contrast, BCCR and MSCNN exhibit color alterations in the sky region, primarily focusing on contrast enhancement. DCP introduces halo artifacts near edges and struggles to address the sky regions. CAP tends to produce over-saturated dehazed images with color information alterations. DehazeNet, while generally visually appealing, retains some residual haze. GMAN, however, produces dehazed images



Fig. 3.3 Perceptual analysis of an image belongs to SOTS [26]: **a** Image with no haze (Ground truth), **b** Image with some haze present, Dehazed version of image shown in **b** using **c** BCCR [30], **d** DCP [23], **e** CAP [31], **f** DehazeNet [32], **g** MSCNN [33], **h** AOD-Net [34], **i** GMAN [35], **j** CSIDNet [39], **k** Enlarged version of an image region shown in **a**, and **l** in **j**

with noticeable distortions. AOD-Net performs well compared to other methods, but it introduces blur near edges, impacting textural information.

CSIDNet, in this visual comparison, stands out by generating aesthetically pleasing results without visible artifacts. Most other methods exhibit distortions, especially in the sky region, possibly attributed to excessive dehazing in areas with light or fine haze. CSIDNet's ability to maintain visual quality while effectively reducing haze contributes to its appeal among state-of-the-art dehazing methods.

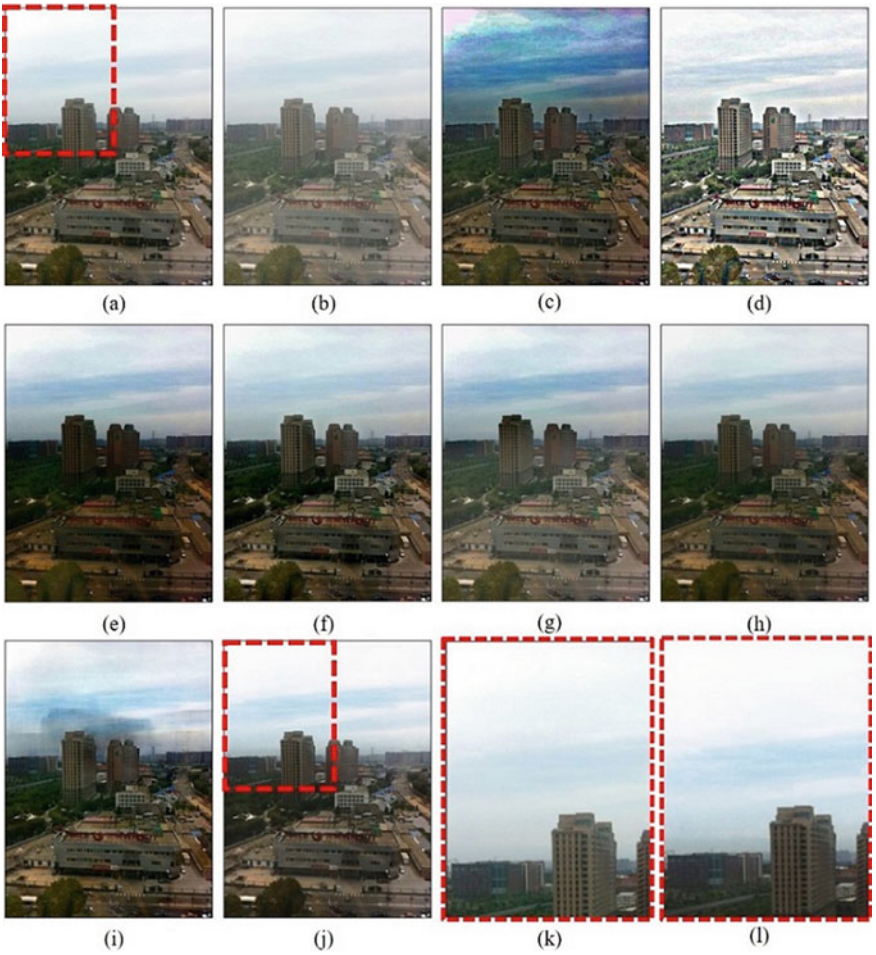


Fig. 3.4 Perceptual analysis of an image belongs to HSTS [26]: **a** Image with no haze (Ground truth), **b** Image with some haze present, Dehazed version of image shown in **b** using **c** BCCR [30], **d** DCP [23], **e** CAP [31], **f** DehazeNet [32], **g** MSCNN [33], **h** AOD-Net [34], **i** GMAN [35], **j** CSIDNet [39], **k** Enlarged version of an image region shown in **a**, and **l** in **j**

3.3.5 Discussions

The impact of network architecture variations on performance has been explored by experimenting with different numbers of layers and filters/depths in CSIDNet. Table 3.3 presents the corresponding values of PSNR and SSIM index obtained using trained models with varying numbers of layers. Notably, the performance on test datasets appears to be optimized with a configuration of three layers.

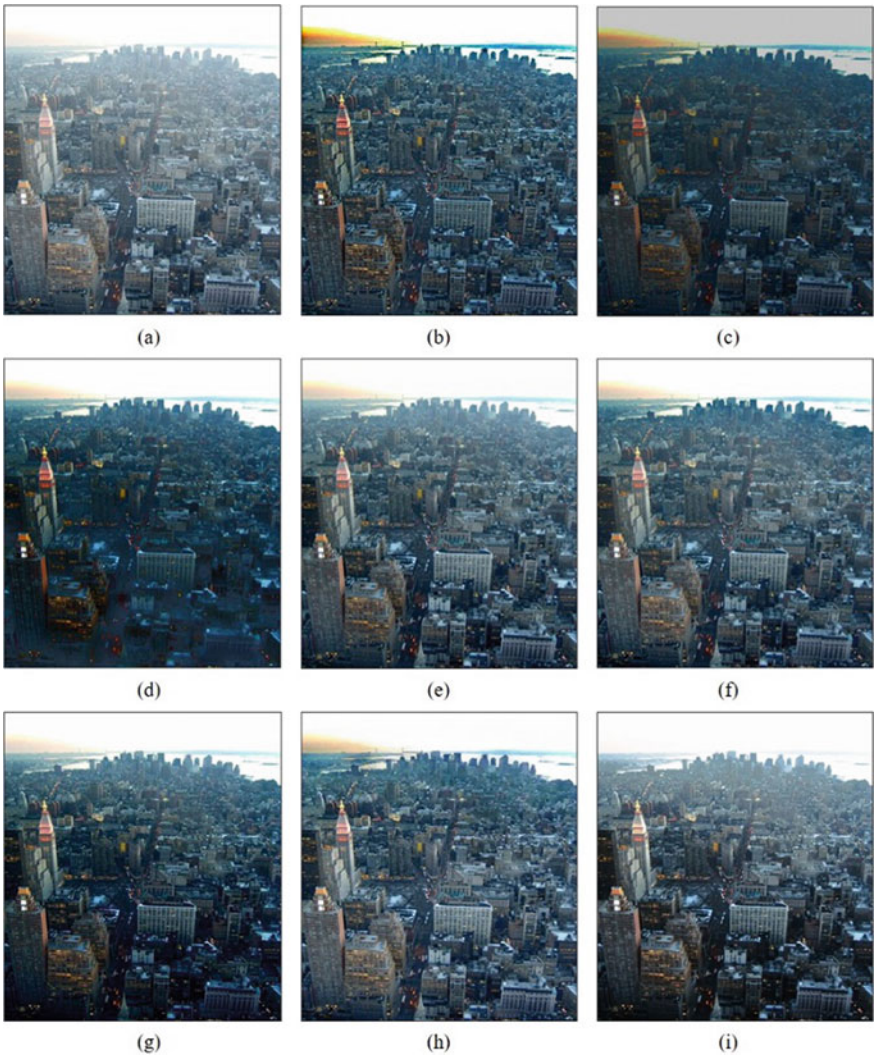


Fig. 3.5 Perceptual analysis of a naturally hazy image [23]: **a** Image captured under haze, Dehazed version of image shown in **a** using **b** BCCR [30], **c** DCP [23], **d** CAP [31], **e** DehazeNet [32], **f** MSCNN [33], **g** AOD-Net [34], **h** GMAN [35], and **i** CSIDNet [39]

Similarly, Table 3.4 provides an overview of the performance metrics, specifically PSNR and SSIM index values, obtained from trained models with different numbers of filters. This analysis sheds light on the influence of filter count on overall performance.

These experiments contribute insights into the architectural choices within CSID-Net, showcasing the optimal configurations for achieving superior performance on

Table 3.3 Performance assessment of CSIDNet with variation in number of layers

Number of layers	Test set			
	SOTS [26]		HSTS [26]	
	PSNR [36]	SSIM [37]	PSNR [36]	SSIM [37]
2	29.049	0.883	28.856	0.849
3	**28.995**	**0.887**	**28.975**	**0.859**
4	29.008	0.884	28.764	0.854
5	28.850	0.862	28.660	0.844
6	28.770	0.853	28.472	0.818
7	28.551	0.841	28.295	0.801
8	28.606	0.834	28.328	0.792

Table 3.4 Performance assessment of CSIDNet with variation in number of filters

Number of filters	Test set			
	SOTS [26]		HSTS [26]	
	PSNR [36]	SSIM [37]	PSNR [36]	SSIM [37]
12	28.999	0.885	28.765	0.853
16	**29.045**	**0.887**	**28.975**	**0.856**
24	29.056	0.888	28.912	0.857
32	29.033	0.887	28.892	0.855

test datasets. Adjustments in the number of layers and filters impact the model's ability to effectively dehaze images, as evidenced by the quantitative metrics presented in the respective tables.

The training progress in terms of MSE loss for different numbers of layers and filters is visualized in Figs. 3.6 and 3.7, respectively. These plots provide a graphical representation of how the loss function evolves during the training process, offering insights into the convergence and stability of the model with varying architectural parameters.

Following these experiments, a decision was made to adopt 16 filters for CSIDNet. This choice was informed by considerations such as memory usage and calculation time. The use of 16 filters strikes a balance, ensuring that the model remains compact while maintaining accuracy. Importantly, this configuration mitigates the risk of increased memory requirements, contributing to the overall efficiency of CSIDNet. The decision to opt for a more compact structure is aligned with the goal of achieving optimal performance with practical resource considerations.

CSIDNet has undergone training on five distinct sets, each comprising 200 images for training: Set 1, Set 2, Set 3, Set 4, and Set 5. The quantitative evaluation involved testing images from the SOTS and the HSTS using PSNR and SSIM index measures. The performance metrics obtained using the MSE loss function are detailed in

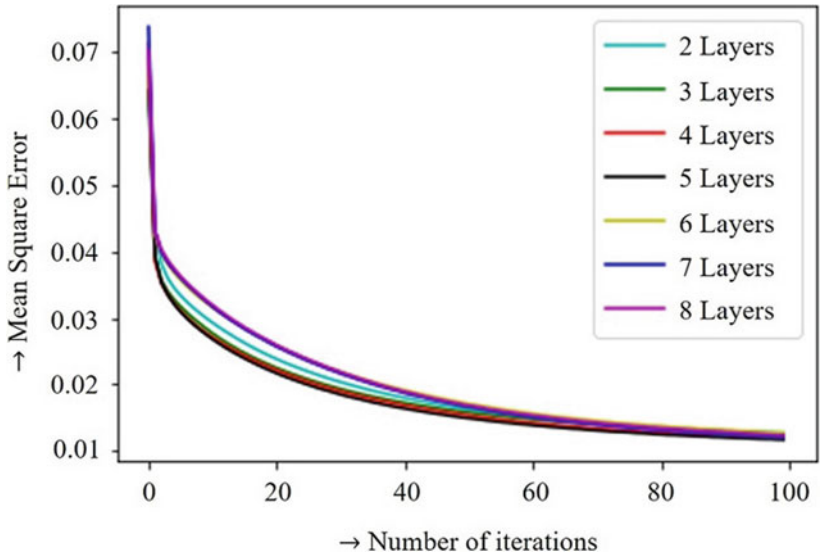


Fig. 3.6 Plots illustrating the mean square error obtained for CSIDNet with varying numbers of layers [39]

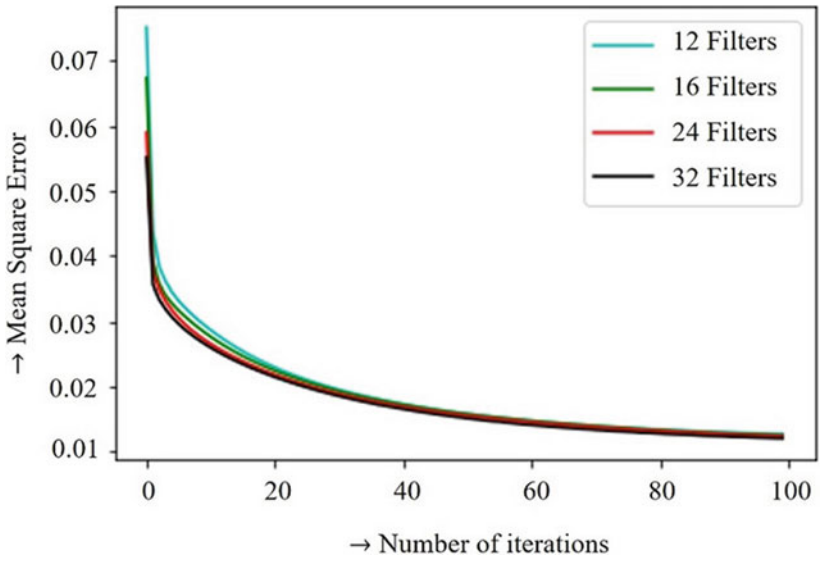


Fig. 3.7 Plots illustrating the mean square error obtained for CSIDNet with varying numbers of filters [39]

Table 3.5 Performance assessment of CSIDNet using MSE as a loss function

Training set	Test set			
	SOTS [26]		HSTS [26]	
	PSNR [36]	SSIM [37]	PSNR [36]	SSIM [37]
Set 1	29.045	0.887	28.975	0.856
Set 2	29.108	0.891	28.895	0.856
Set 3	28.982	0.880	28.838	0.844
Set 4	28.908	0.880	28.640	0.844
Set 5	28.932	0.878	28.791	0.842
Average	28.995	0.883	28.828	0.848

Table 3.6 Performance assessment of CSIDNet using $L1$ loss function: 100 epochs

Training set	Test set			
	SOTS [26]		HSTS [26]	
	PSNR [36]	SSIM [37]	PSNR [36]	SSIM [37]
Set 1	28.952	0.878	28.742	0.844
Set 2	28.960	0.878	28.727	0.842
Set 3	29.019	0.880	28.872	0.842
Set 4	28.791	0.878	28.600	0.843
Set 5	28.836	0.872	28.545	0.831
Average	28.912	0.877	28.697	0.840

Table 3.7 Performance assessment of CSIDNet using $L1$ loss function: 150 epochs

Training set	Test set			
	SOTS [26]		HSTS [26]	
	PSNR [36]	SSIM [37]	PSNR [36]	SSIM [37]
Set 1	29.041	0.876	28.910	0.852
Set 2	29.067	0.885	28.937	0.852
Set 3	28.911	0.871	28.814	0.828
Set 4	28.805	0.878	28.577	0.840
Set 5	28.963	0.869	28.793	0.835
Average	28.957	0.876	28.806	0.841

Table 3.5. This table provides a comprehensive overview of the model's performance based on the training sets.

Additionally, Tables 3.6, 3.7, and 3.8 present performance measures obtained using the $L1$ loss function for 100, 150, and 200 epochs, respectively. These tables provide further insights into the impact of training duration on the model's quantitative performance.

Table 3.8 Performance assessment of CSIDNet using $L1$ loss function: 200 epochs

Training set	Test set			
	SOTS [26]		HSTS [26]	
	PSNR [36]	SSIM [37]	PSNR [36]	SSIM [37]
Set 1	28.966	0.871	28.874	0.844
Set 2	29.062	0.885	28.889	0.855
Set 3	28.932	0.873	28.725	0.834
Set 4	28.959	0.880	28.751	0.847
Set 5	28.951	0.865	28.743	0.832
Average	28.974	0.875	28.796	0.842

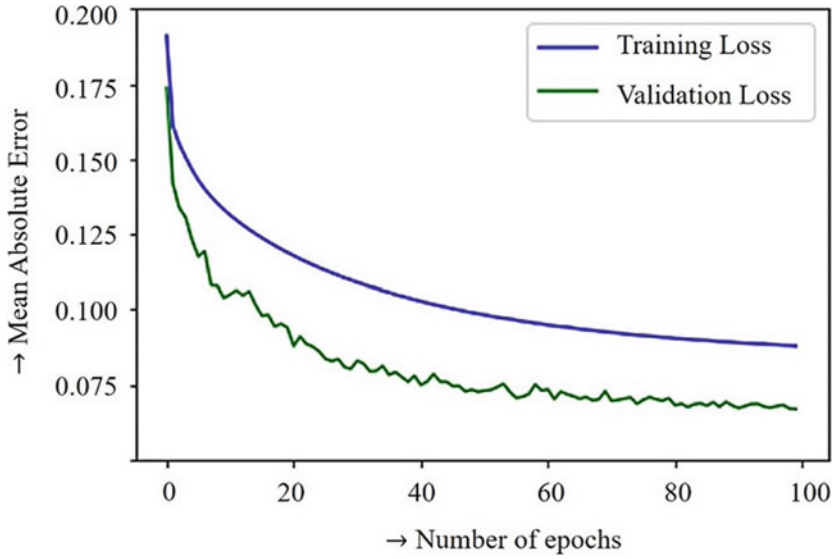


Fig. 3.8 Plots illustrating the mean absolute error/$L1$ loss obtained for CSIDNet [39]: 100 epochs

The choice of using MSE as the loss function during training, leading to improved PSNR and SSIM index measures, can be explained by the robustness of MSE in penalizing differences between predicted and ground truth values. The squaring of errors in MSE makes it more sensitive to larger discrepancies, which is advantageous in tasks such as image processing where accurate representation of pixel values is crucial.

In contrast, the $L1$ loss function, relying on absolute differences, tends to be less sensitive to outliers and offers a more resilient model. However, it may not capture fine details as effectively as MSE.

The visual representation of training progress in Figs. 3.6, 3.7, 3.8, 3.9, and 3.10 underscores the dynamics of model learning with different loss functions, reinforcing the observed quantitative differences in performance.

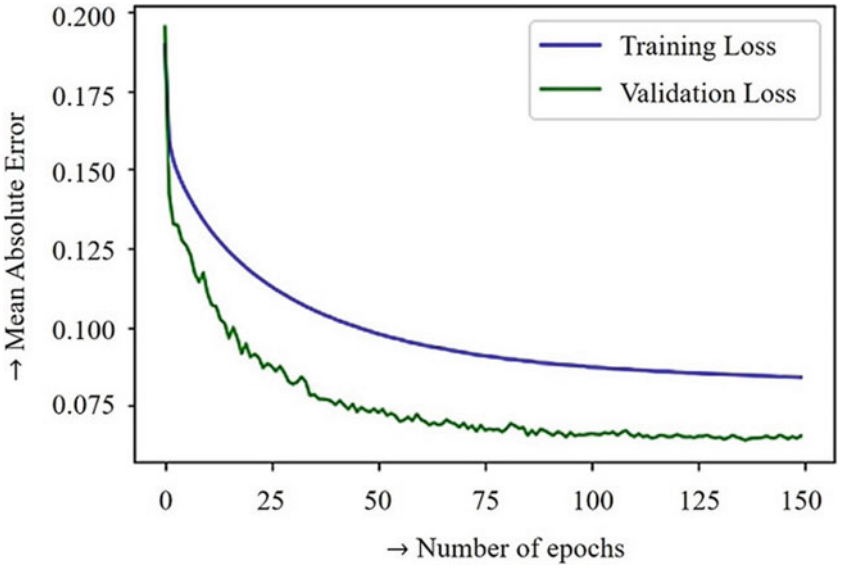


Fig. 3.9 Plots illustrating the mean absolute error/$L1$ loss obtained for CSIDNet [39]: 150 epochs

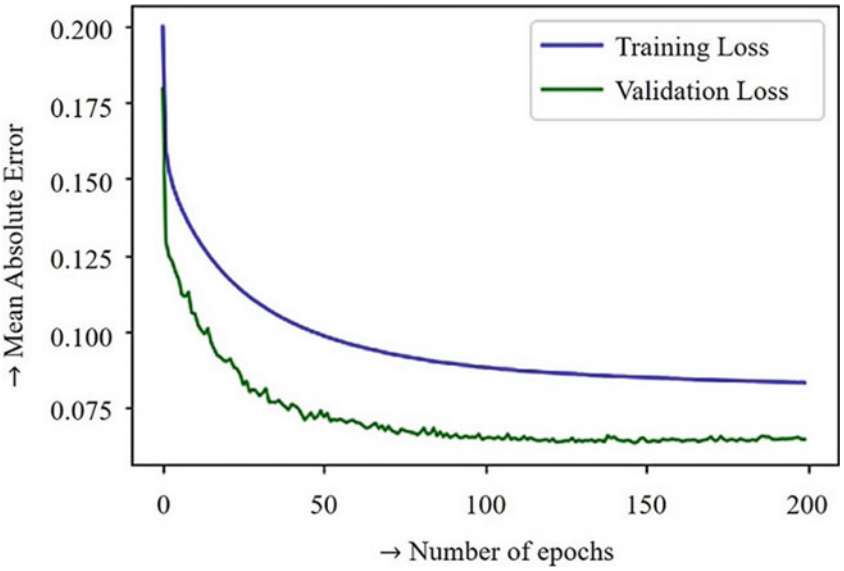


Fig. 3.10 Plots illustrating the mean absolute error/$L1$ loss obtained for CSIDNet [39]: 200 epochs

3.4 Summary

In this chapter, an in-depth exploration of the *Compact Single Image Dehazing Network (CSIDNet)* is provided for the enhancement of outdoor scenes through deep learning-based single image dehazing. CSIDNet not only surpasses several state-of-the-art dehazing models but also distinguishes itself as a compact model with minimal resource requirements. The resulting enhanced scene images achieved through CSIDNet are capable of maintaining a balance between speed and accuracy while performing enhancement. Through comparative analysis, it is evident that CSIDNet exhibits faster training times and reduced run-time while maintaining quantitative and visual performance robustness.

The network's noteworthy results suggest its potential application in critical and real-time scenarios. Furthermore, the chapter outlines potential future scopes, particularly the prospect of developing an end-to-end network designed for simultaneous image dehazing and denoising under non-uniform illumination conditions, without relying on priors and assumptions. This signifies a promising direction for advancing the capabilities of image enhancement technologies.

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Chapter 4
Z-Score Method



4.1 Introduction

This chapter delves into the intricate process of refining perceptible visual information through single image enhancement. The focus is on elucidating approaches for single image dehazing and non-uniform illumination enhancement using the Z-score [1]. The overarching goal is to elevate the visual quality of images intended for diverse applications [2–20]. Specifically, this approach enhances the minimum channel for hazy images and the illumination channel for non-uniformly illuminated images. The experimental validation of this approach, tailored for two distinct scenarios, is conducted on various benchmark datasets. The results indicate its efficiency compared to state-of-the-art dehazing methods in existing literature. The subsequent sections elaborate on the preliminaries, observations, and the detailed image enhancement approach for both single image dehazing and non-uniform illumination enhancement.

4.2 *Z*-Score

In the context of a digital image, the Z-score characterizes the relationship between a pixel intensity and the mean of a set of pixel intensities. The Z-score is quantified in terms of standard deviation from the mean [1].

Consider a group of pixel intensities $p_{i,j}$ in an image, where $1 \leq i \leq H$ and $1 \leq j \leq W$, with H and W representing the height and width of the image, respectively.

The Z-score for each pixel intensity is defined as follows:

$$Z^{\text{Score}}\left(p_{i,j}\right) = \frac{p_{i,j} - \mu}{\sigma} \tag{4.1}$$

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where
– μ and σ are the mean and standard deviation of the image, respectively.

The probability value for a normal distribution using Z-score value is defined as

$$Z^f\left(p_{i,j}\right) = 0.5 \times erfc\left(-\frac{Z^{\text{Score}}\left(p_{i,j}\right)}{\sqrt{2}}\right). \tag{4.2}$$

4.3 Observations

In literature, the Z-score has found widespread applications in diverse fields such as medical analysis [21], image processing [22], and more. The Z-score can take positive or negative values, where a positive value indicates that the Z-score surpasses the mean, and a negative value signifies that the Z-score is below the mean.

The Z-score approach presented for enhancing hazy and non-uniformly illuminated images is grounded in the following observations:

(i) For hazy images, the pixel intensity histogram tends to be skewed toward the brighter side, as depicted in Fig. 4.1. Consequently, the means for local patches in a hazy image are positioned toward the brighter side of the histogram.
(ii) Conversely, for non-uniformly illuminated images, the pixel intensity histogram is inclined toward the darker side, as illustrated in Fig. 4.2. This results in the means for all local patches in a non-uniformly illuminated image lying toward the darker side of the histogram.

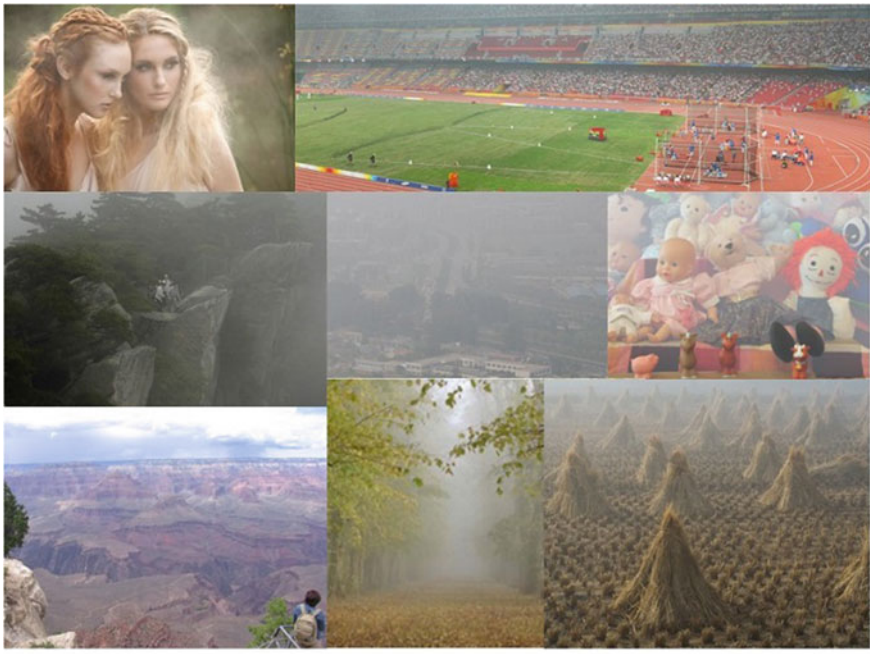
According to the assumption, an enhanced image under hazy or non-uniformly illuminated conditions should exhibit an equalized histogram for improved visibility. Hence, the following statements can be made:

(i) Given that the minimum channel shifts from the darker side to the brighter side for hazy images, the Z-score function endeavors to adjust the minimum channel to enhance contrast and achieve a clear, haze-free output image.
(ii) Conversely, for non-uniform illumination, the Z-score function strives to adjust the pixel intensities of the illumination channel, aiming to produce an enhanced output image with uniform illumination.

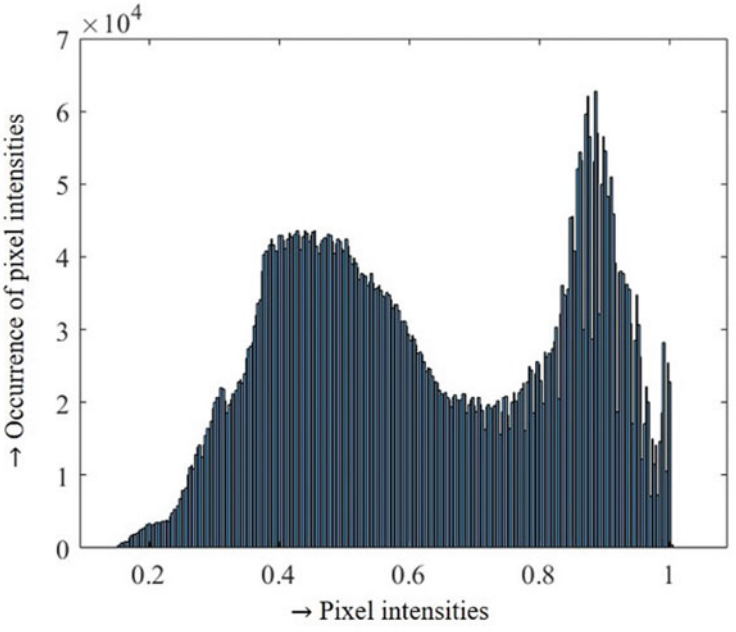
In the upcoming section, a detailed explanation of single image dehazing and non-uniform illumination enhancement using the Z-score approach will be provided.

4.4 Methodology

This section provides a detailed explanation of single image dehazing and illumination enhancement using the Z-score approach for images captured under hazy and non-uniform illumination conditions, respectively.



(a)



(b)

Fig. 4.1 Visual analysis of images with haze [45]: **a** Some images for visualization [23] and **b** Histogram of **a**

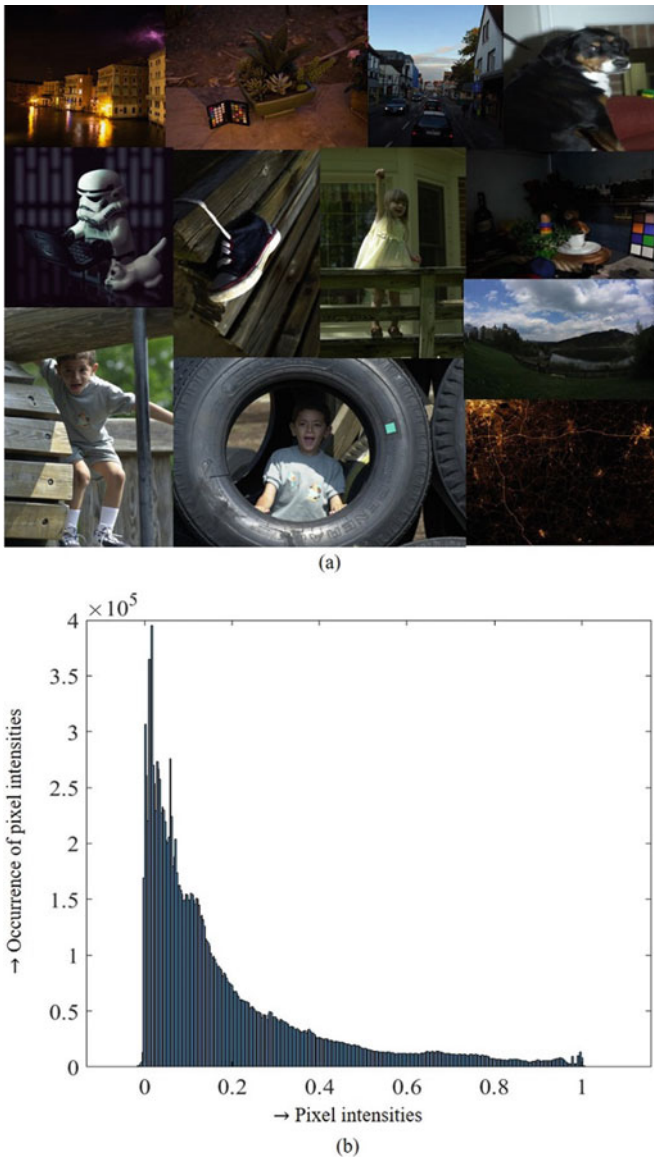


Fig. 4.2 Visual analysis of images with non-uniform illumination [45]: **a** Some images for visualization [24] and **b** Histogram of **a**

4.4.1 Single Image Dehazing

Figure 4.3 shows the block diagram of single image dehazing using the Z-score approach. Firstly, the Z-score weighting function map is estimated for the minimum channel to obtain the depth map and scene transmission. The minimum channel of the hazy input scene image is obtained as

$$I_{\min}(p,q) = \min_{\tau_1 \in \{\mathrm{R,G,B}\}} \left(I_{\mathrm{Hazy}}^{\tau_1}(p,q) \right). \tag{4.3}$$

Then, the global atmospheric light is calculated using Quad tree subdivision [25, 26] based on the Z-score approach for the Value/ Illumination channel of the image in the HSV color domain. Finally, the enhanced image is obtained by atmospheric scattering model [27] using

$$I_{\mathrm{enh}}(p,q) = \frac{I_{\mathrm{Hazy}}(p,q) - A_{\mathrm{G}}}{\max\left(T_{\mathrm{R}}(p,q), T_0\right)} + A_{\mathrm{G}} \tag{4.4}$$

where
– T_0 is a lower bound non-zero value of the scene transmission.

The value of T_0 is adjusted as 0.1 to avoid indefinite values at pixel locations with zero transmission.

Furthermore, the estimation of scene transmission and global atmospheric light have been explained in detail to recover the haze-free clear image.

4.4.1.1 Scene Transmission

The scene transmission estimation using the Z-score approach comprises the estimation of Z-Score Weighting Function Map (ZSWFM) for the minimum channel followed by depth map estimation. For a local patch of size $(2F_1 + 1) \times (2F_1 + 1)$ centered at (i_1, j_1) in a hazy input image I_{Hazy} of height H_1 and width W_1, the neighborhood pixel set N' with k_1 number of pixels is extracted using

$$N'_{k_1} = \{I_{\min}(i_1 + u_1, j_1 + v_1)\} \; \forall \; (u_1, v_1) \in [-F_1, F_1] \tag{4.5}$$

where
– $\forall\ 1 \leq i_1 \leq H_1$ and
– $\forall\ 1 \leq j_1 \leq W_1$.

For the pixel intensities of neighborhood pixel set N' with mean μ_1 and standard deviation σ_1, the Z-score is evaluated by

$$Z_{k_1}^{\mathrm{Score}}(p,q) = \frac{N'_i(p,q) - \mu_1(p,q)}{\sigma_1(p,q) + \epsilon_1} \quad \forall\ 1 \leq i \leq k_1 \tag{4.6}$$

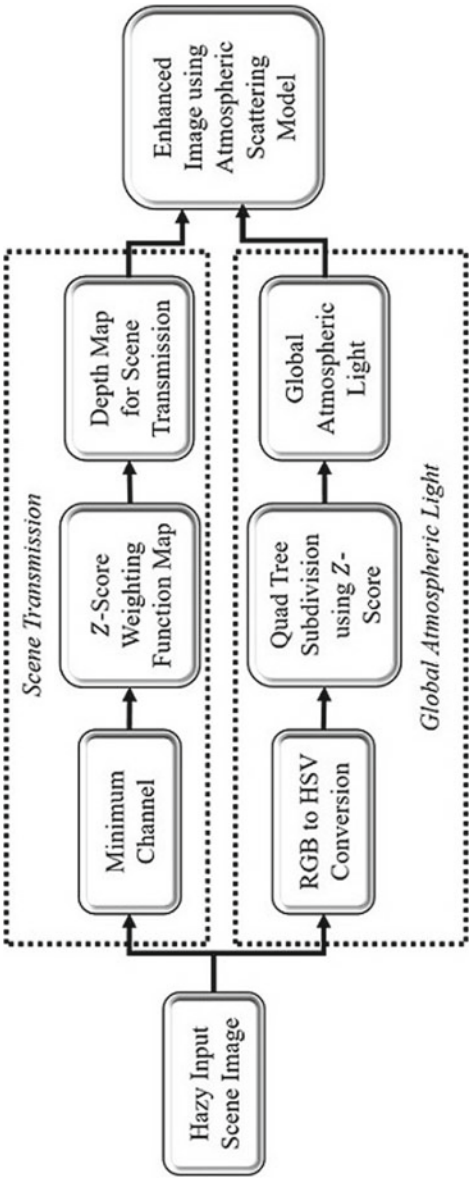


Fig. 4.3 Visual representation of dehazing process of an image using Z-score approach [45]

where
– ϵ_1 is an empirical parameter.

Thereafter, the ZSWFM is obtained by

$$Z_{k_1}^{f}(p,q) = 0.5 \times erfc\left(-\frac{Z_i^{\text{Score}}(p,q)}{\sqrt{2}}\right) \ \forall\, 1 \le i \le k_1. \tag{4.7}$$

Now, the depth map is estimated using the weighted average of ZSWFM and the neighborhood pixel set as

$$D_{\text{Map}}(p,q) = \frac{\sum_{i=1}^{k_1} Z_i^f(p,q) \times N_i'(p,q)}{\sum_{i=1}^{k_1} Z_i^f(p,q)}. \tag{4.8}$$

Therefore, the scene transmission is obtained by

$$T_{\text{R}}(p,q) = \exp\left(-\beta \times D_{\text{Map}}(p,q)\right) \tag{4.9}$$

where
– β is the angular scattering coefficient of the atmosphere, and the value has been set as 1 for experimentation.

Remark 4.1 If standard deviation σ_1 in (4.6) is 0, then this leads to the condition of infinite value of $Z_{k_1}^{\text{Score}}$. Thus, a lower bound of standard deviation, i.e., ϵ_1 has been set experimentally as 0.01.

4.4.1.2 Global Atmospheric Light

The global atmospheric light in the Z-score approach is estimated by Quad tree subdivision [25, 26] based on Z-score. The input RGB image is converted into the HSV image domain to obtain the illumination channel $I_{\text{Hazy}}^{\text{V}}$ as

$$I_{\text{Hazy}}^{\tau_1} \xrightarrow{\text{RGB to HSV Conversion}} I_{\text{Hazy}}^{\tau_2} \tag{4.10}$$

where
– $\tau_1 \in$ {R,G,B} and
– $\tau_2 \in$ {H,S,V}.

Then, the illumination channel $I_{\text{Hazy}}^{\text{V}}$ is divided into d equal regions represented by I_d^l for $d \in [1, 2, 3, 4]$ under l^{th} iteration. For each subdivision, the ZSWFM value as defined in (4.7), is obtained by

$$A_d^l = Z^f\left(I_d^l\right). \tag{4.11}$$

Then, the maximum mean of ZSWFM value for each subdivision is evaluated by

$$A^l_{\max} = \max_d \left(\text{mean}\left(A^l_d\right)\right). \tag{4.12}$$

Thereafter, the subdivision with a maximum ZSWFM value is considered for further subdivisions. The Quad tree subdivision is followed until the following criteria are satisfied,

$$\min |A^l_{\max} - A^{l-1}_{\max}| \leq \epsilon_{\mathrm{T}} \tag{4.13}$$

where

– ϵ_{T} is a predefined threshold and has been set experimentally.

Hence, the global atmospheric light A_{G} is obtained using

$$A_{\mathrm{G}} = A^l_{\max}. \tag{4.14}$$

Finally, the enhanced dehazed image is obtained using (4.4).

Remark 4.2 For experiments, the value of ϵ_{T} has been set as 0.1. The decrease in the value of ϵ_{T} leads to more accurate but increased run-time for estimation of the global atmospheric light, whereas the increase in the value of ϵ_{T} leads to less accurate estimation of the global atmospheric light with faster run-time.

4.4.2 Non-uniform Illumination Enhancement

Figure 4.4 shows the block diagram of the Z-score approach for the enhancement of non-uniformly illuminated images. Initially, the illumination channel $I^{\mathrm{V}}_{\mathrm{NUI}}$ of non-uniformly illuminated input image $I^{\tau_3}_{\mathrm{NUI}}$ for $\tau_3 \in \{\mathrm{R}, \mathrm{G}, \mathrm{B}\}$ is obtained by converting it to HSV image domain i.e., $I^{\tau_4}_{\mathrm{NUI}}$ for $\tau_4 \in \{\mathrm{H}, \mathrm{S}, \mathrm{V}\}$. Then, the illumination channel is processed with ZSWFM to obtain the coarse illumination, followed by applying adaptive gamma correction. Thereafter, the scene reflectance is estimated. Finally, the enhanced image is composed using (1.3).

Further, the coarse illumination estimation, obtaining the scene reflectance, and adaptive gamma correction have been explained in detail to restore the enhanced output image.

4.4.2.1 Coarse Illumination

For a local patch of size $(2F_2 + 1) \times (2F_2 + 1)$ centered at (i_2, j_2) in the illumination channel $I^{\mathrm{V}}_{\mathrm{NUI}}$ of non-uniformly illuminated image I_{NUI} of height H_2 and width W_2, the neighborhood pixel set N'' with k_2 number of pixels is extracted using

$$N''_{k_2} = \{I^{\mathrm{V}}_{\mathrm{NUI}}(i_2 + u_2, j_2 + v_2)\} \; \forall \; (u_2, v_2) \in [-F_2, F_2] \tag{4.15}$$

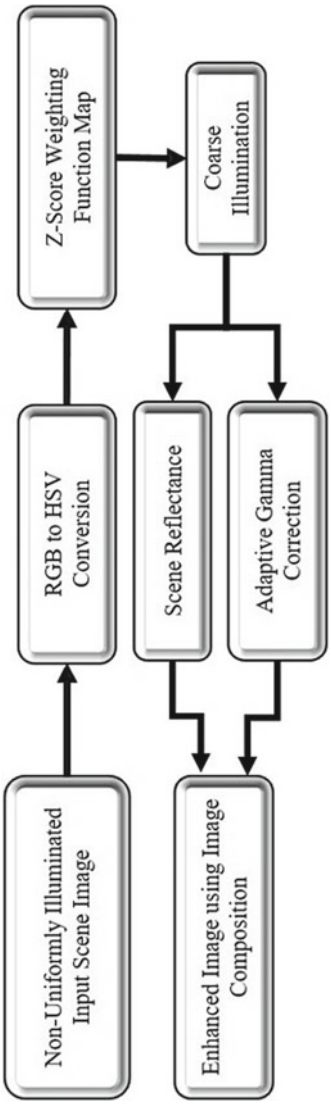


Fig. 4.4 Visual representation of enhancement process using Z-Score approach for an image with non-uniform/uneven illumination component [45]

where
– $\forall\ 1 \leq i_2 \leq H_2$ and
– $\forall\ 1 \leq j_2 \leq W_2$.

The Z-score for neighborhood pixel set N'' is evaluated by

$$Z_{k_2}^{\text{Score}}(p,q) = \frac{N_i''(p,q) - \mu_2(p,q)}{\sigma_2(p,q) + \epsilon_2} \quad \forall\ 1 \leq i \leq k_2 \tag{4.16}$$

where
– ϵ_2 is an empirical parameter adjusted to 0.01 for avoiding the infinite value of $Z_{k_2}^{\text{Score}}$,
– μ_2 and σ_2 are the mean and standard deviation, respectively for N''.

Then, the ZSWFM is obtained by

$$Z_{k_2}^{f}(p,q) = 0.5 \times erfc\left(-\frac{Z_i^{\text{Score}}(p,q)}{\sqrt{2}}\right) \quad \forall\ 1 \leq i \leq k_2. \tag{4.17}$$

Thus, the coarse illumination is estimated using the weighted average of ZSWFM and the neighborhood pixel set as

$$I_l(p,q) = \frac{\sum_{i=1}^{k_2} Z_i^f(p,q) \times N_i''(p,q)}{\sum_{i=1}^{k_2} Z_i^f(p,q)}. \tag{4.18}$$

4.4.2.2 Scene Reflectance

The scene reflectance for non-uniformly illuminated input image is estimated as

$$R^{\tau_1}(p,q) = \frac{I_{\text{NUI}}^{\tau_1}(p,q)}{I_l(p,q)}. \tag{4.19}$$

4.4.2.3 Adaptive Gamma Correction

The coarse illumination obtained by (4.18) is processed with adaptive gamma correction [24] to obtain the refined coarse illumination as

$$I_l^{\text{AGC}}(p,q) = I_l(p,q)^{\gamma(p,q)} \tag{4.20}$$

where

$$\gamma(p,q) = \frac{I_l(p,q) + \alpha}{(1+\alpha)^{\alpha}} \tag{4.21}$$

and

$$\alpha = 1 + \text{mean}\left(I_{\text{NUI}}^{\text{V}}(p,q)\right). \tag{4.22}$$

Thus, the final enhanced output image is recomposed with obtained scene reflectance and refined coarse illumination as

$$I_{\text{enh}}^{\tau_1}(p,q) = R^{\tau_1}(p,q) \times I_l^{\text{AGC}}(p,q). \tag{4.23}$$

4.5 Results and Discussions

This section showcases the experimental outcomes of the presented approach, providing a comprehensive evaluation through both quantitative and qualitative analyses against established methods in the field. The efficacy of the image dehazing technique is gauged against a range of benchmark methods, including Contrast Limited Adaptive Histogram Equalization (CLAHE) [28], DCP [23], CAP [29], MSCNN [30], DehazeNet [31], AOD-Net [32], and GMAN [33].

To quantify the performance, benchmarked performance metrics are employed such as PSNR [34], SSIM index [35], Lightness Order Error (LOE) [36], and Naturalness Image Quality Evaluator (NIQE) [37]. The assessment encompasses natural hazy images sourced from existing literature and those drawn from the RESIDE dataset [38], which encompasses 500 indoor SOTS, 500 outdoor SOTS, and 10 HSTS images.

In contrast, the evaluation of enhanced results for non-uniformly illuminated images involves a comparison with established methods, including CLAHE [28], CVC [39], LDR [40], Efficient Naturalness Restoration (ENR) [24], Low-light IMage Enhancement (LIME) [41], LECARM [42], and Zero-DCE [43]. The quantitative assessment relies on LOE and NIQE scores, and experiments are conducted on naturally non-uniformly illuminated images from existing literature. Additionally, the evaluation extends to images sourced from the High Dynamic Range (HDR) dataset [44], comprising 8 scene images. For each scene image, the reconstructed image derived from input images with diverse views and exposures serves as the ground truth for benchmarking.

4.5.1 Quantitative Comparison

Tables 4.1, 4.2, and 4.3 provide a comprehensive performance analysis comparing the Z-score approach with other methods. The evaluation is based on PSNR, SSIM, LOE, and NIQE metrics, applied to images from the RESIDE dataset's indoor SOTS, outdoor SOTS, and synthetic HSTS categories. Higher PSNR and SSIM values signify enhanced fidelity to the ground truth, while lower LOE and NIQE values indicate improved naturalness aligned with human perception.

Table 4.1 Performance assessment of Z-score approach on images of indoor SOTS subset from the RESIDE dataset [38]

Methods → Performance metric ↓	CLAHE [28]	DCP [23]	CAP [29]	MSCNN [30]	DehazeNet [31]	AOD-Net [32]	GMAN [33]	Z-score approach [45]
PSNR [34]	13.1359	11.8304	18.9634	17.1292	21.1671	18.7337	28.0794	**28.7952**
SSIM [35]	0.7140	0.6346	0.8524	0.8069	0.8741	0.8456	0.9664	**0.9695**
LOE [36]	1.326	1.485	0.941	0.982	0.839	1.012	0.947	**0.822**
NIQE [37]	3.128	2.947	3.159	3.178	3.12	3.224	3.753	**2.871**

Table 4.2 Performance assessment of Z-score approach on images of outdoor SOTS subset from the RESIDE dataset [38]

Methods → Performance metric ↓	CLAHE [28]	DCP [23]	CAP [29]	MSCNN [30]	DehazeNet [31]	AOD-Net [32]	GMAN [33]	Z-score approach [45]
PSNR [34]	16.4549	15.0527	18.1236	19.4735	22.2760	19.4997	18.8861	**22.5978**
SSIM [35]	0.7921	0.6993	0.7622	0.8334	0.8570	0.8606	0.8548	**0.8679**
LOE [36]	0.872	1.039	0.549	0.730	0.547	0.607	0.939	**0.530**
NIQE [37]	2.799	3.097	2.826	2.745	2.746	2.837	2.903	**2.601**

Table 4.3 Performance assessment of Z-score approach on images of synthetic HSTS subset from the RESIDE dataset [38]

Methods → Performance metric ↓	CLAHE [28]	DCP [23]	CAP [29]	MSCNN [30]	DehazeNet [31]	AOD-Net [32]	GMAN [33]	Z-score approach [45]
PSNR [34]	13.2242	18.9331	11.8863	16.3276	14.2192	13.7651	10.0533	**20.1251**
SSIM [35]	0.7117	0.6917	0.6126	0.8048	0.7334	0.8027	0.5353	**0.8219**
LOE [36]	0.905	1.151	0.487	0.696	0.495	0.375	0.932	**0.318**
NIQE [37]	2.75	3.28	2.73	2.73	2.76	3.43	2.67	**2.59**

Table 4.4 Performance assessment of Z-score approach on images from the HDR dataset [44] for enhancement of illumination

Methods → Performance metric ↓	CLAHE [28]	CVC [39]	LDR [40]	ENR [24]	LIME [41]	LECARM [42]	Zero-DCE [43]	Z-score approach [45]
LOE	2.611	3.609	3.596	3.576	2.573	3.026	2.723	**2.095**
NIQE	2.9859	3.1518	3.0575	2.8530	2.9719	2.8680	2.6035	**2.5506**

To emphasize the Z-score approach's focus on preserving naturalness, Tables 4.4 and 4.5 present comparisons of LOE and NIQE measures. Lower values in these tables indicate superior performance in maintaining naturalness, showcasing the Z-score approach's efficacy.

Table 4.5 Performance assessment of Z-score approach on images from the DICM dataset [40] for enhancement of illumination

Methods → Performance metric ↓	CLAHE [28]	CVC [39]	LDR [40]	ENR [24]	LIME [41]	LECARM [42]	Zero-DCE [43]	Z-score approach [45]
LOE	3.4025	1.377	0.961	1.875	3.810	1.989	0.916	**0.892**
NIQE	2.9582	2.8759	2.9080	3.5892	3.0569	3.7955	3.1546	**2.8051**

4.5.2 Qualitative Comparison

For a visual demonstration, Figs. 4.5 and 4.6 offer comparative visualizations of remote sensed hazy and non-uniformly illuminated outdoor scene images. These visuals underscore the Z-score approach's capability to outperform other methods in handling challenging scenarios while preserving visual fidelity and naturalness.

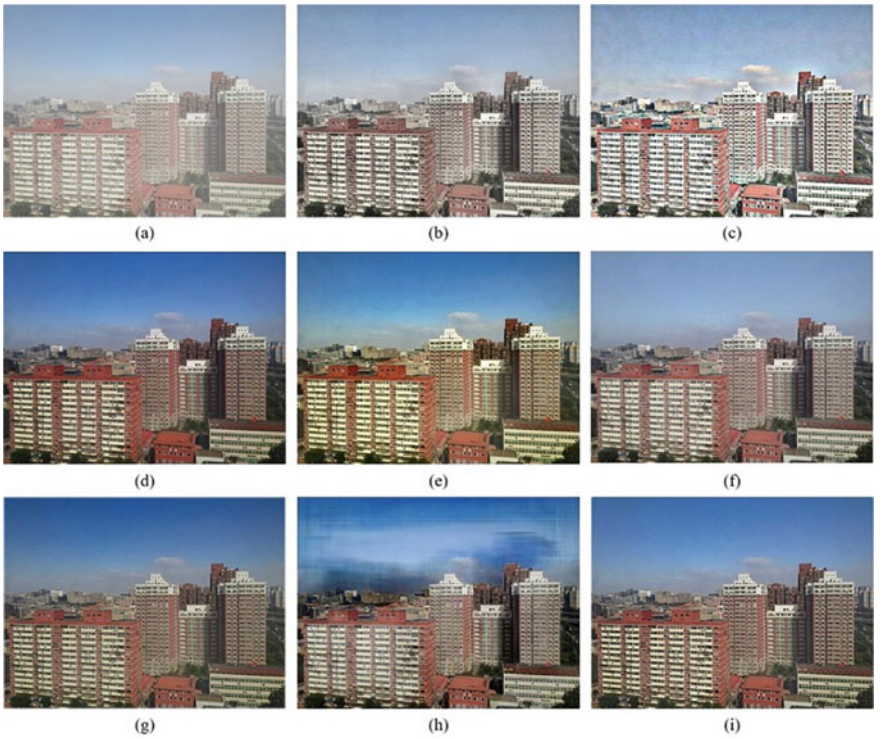


Fig. 4.5 Perceptual analysis of an outdoor image with presence of haze in it [38]: **a** Image captured under haze, Dehazed version of image shown in **a** using **b** CLAHE [28], **c** DCP [23], **d** CAP [29], **e** MSCNN [30], **f** DehazeNet [31], **g** AOD-Net [32], **h** GMAN [33], and **i** Z-Score approach [45]

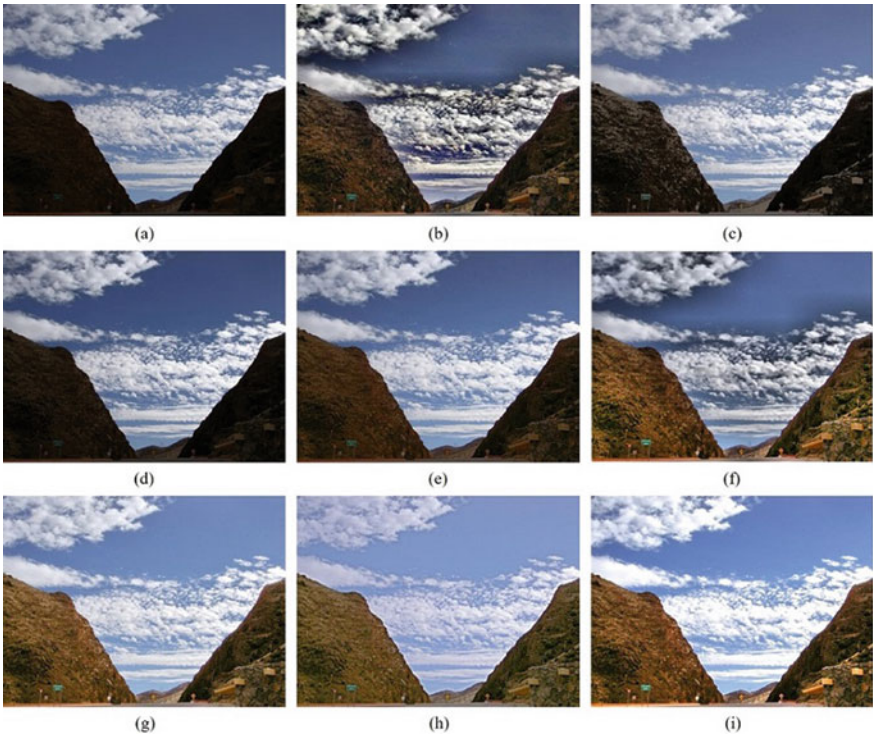


Fig. 4.6 Perceptual analysis of an outdoor image with presence of uneven illumination in it [40]: **a** Image captured under bad illumination, Enhanced version of image shown in **a** using **b** CLAHE [28], **c** CVC [39], **d** LDR [40], **e** ENR [24], **f** LIME [41], **g** LECARM [42], **h** Zero-DCE [43], and **i** Z-Score approach [45]

In Fig. 4.7, the dehazed images are visually compared, revealing distinct performance characteristics. Notably, CLAHE struggles with haze removal, introducing information distortions. DCP and CAP successfully eliminate haze but exhibit over-saturation. Meanwhile, MSCNN, DehazeNet, and GMAN prioritize naturalness, retaining some haze to avoid excessive contrast and saturation. AOD-Net maintains contrast but introduces edge-blurring, degrading textural details. In contrast, the Z-score approach strikes a balance, effectively restoring contrast while preserving scene texture.

Figure 4.8 illustrates the enhanced images for a non-uniformly illuminated scenario. The Z-score approach excels by preserving textural information during enhancement. LIME performs well, retaining image texture with minimal noise. However, CLAHE, CVC, LDR, ENR, LECARM, and Zero-DCE struggle to enhance visual information for optimal human perception.

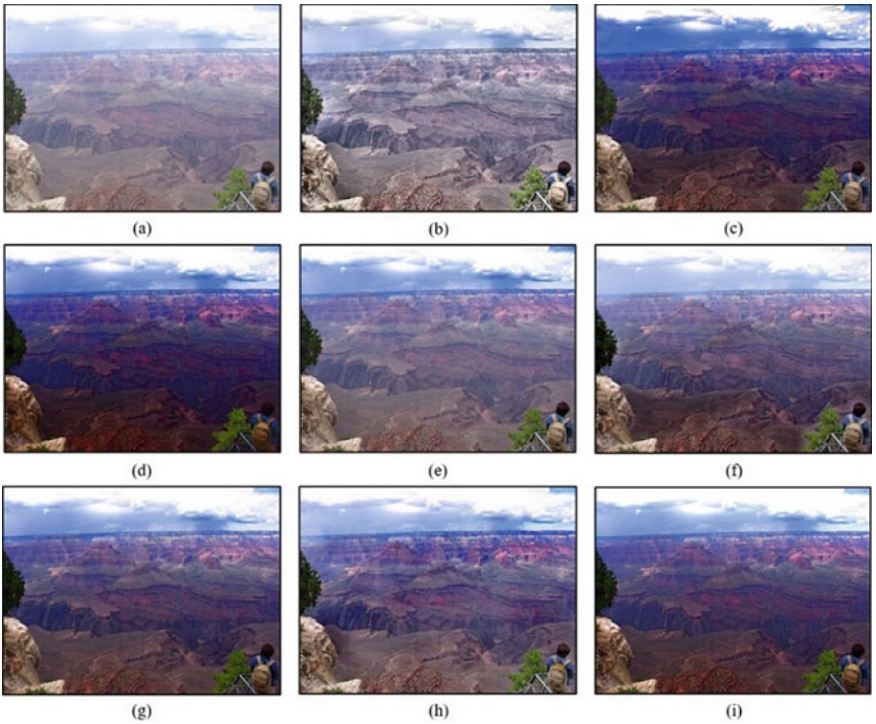


Fig. 4.7 Perceptual analysis of a naturally hazy image [23]: **a** Image captured under haze, Dehazed version of image shown in **a** using **b** CLAHE [28], **c** DCP [23], **d** CAP [29], **e** MSCNN [30], **f** DehazeNet [31], **g** AOD-Net [32], **h** GMAN [33], and **i** Z-Score approach [45]

Overall, the Z-score approach emerges as a robust performer, effectively addressing the challenges posed by haze and non-uniform illumination in image enhancement.

4.6 Summary

This chapter introduces an image enhancement approach for hazy and non-uniformly illuminated scene images using the Z-score. The effectiveness of the Z-score approach is demonstrated in image dehazing and non-uniform illumination enhancement, ensuring the absence of visual artifacts and the preservation of textural information.

However, there is still potential for future improvements, particularly in addressing the noise introduced during the enhancement process. Despite the presented approach outperforming others, mitigating noise could be considered as a pre-processing task to further enhance image textural information and improve overall visual appearance.

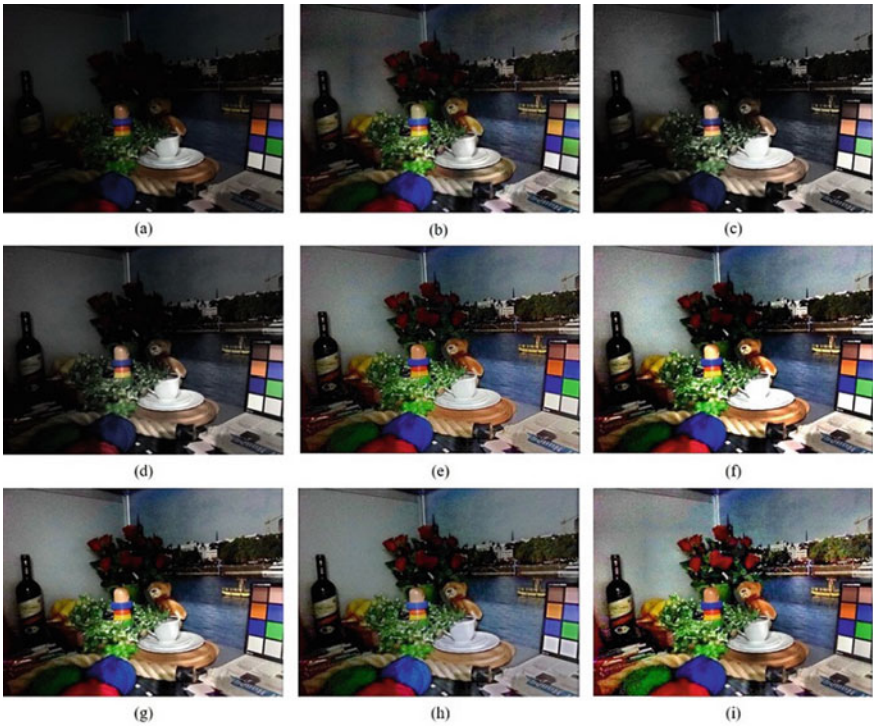


Fig. 4.8 Perceptual analysis of enhancement on an image with non-uniform/uneven illumination [24]: **a** Image captured under bad illumination, Enhanced version of image shown in **a** using **b** CLAHE [28], **c** CVC [39], **d** LDR [40], **e** ENR [24], **f** LIME [41], **g** LECARM [42], **h** Zero-DCE [43], and **i** *Z*-Score approach [45]

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Chapter 5
Image Dehazing Using Type-2 Fuzzy Approach



5.1 Introduction

Poor-quality input images significantly impact the performance of vision-based algorithms [1–20]. Therefore, image enhancement methods are crucial for images captured in hazy environments. The formation of an image depends on the light rays reflected and reaching the observer, known as transmission. In the presence of haze particles, the transmission to the observer is impeded, leading to an increase in airlight. The reduction in transmission, termed attenuation, is a consequence of the presence of haze particles. Airlight represents perceptible brightness due to environmental illumination by haze particles. As the distance between an image location and the observer increases, the transmission decreases due to attenuation, resulting in an increase in airlight.

As per the theory of atmospheric scattering model [21], the observed hazy image is formed with the combination of attenuation and airlight components as mentioned in (1.2). According to this model, the following statements can be made for the scene transmission with respect to the depth map:

- If the observer is very close to the object, then

$$\begin{aligned} D_{\text{Map}}(p,q) \to 0 \Rightarrow T_{\text{R}}(p,q) &\to 1 \\ I_{\text{Hazy}}(p,q) &\approx I_{\text{RL}}(p,q). \end{aligned} \tag{5.1}$$

- If the observer is very far from the object, then

$$\begin{aligned} D_{\text{Map}}(p,q) \to \infty \Rightarrow T_{\text{R}}(p,q) &\to 0 \\ I_{\text{Hazy}}(p,q) &\approx A_{\text{G}}. \end{aligned} \tag{5.2}$$

From (5.1) and (5.2), it can be stated that the depth map $D_{\text{Map}}(p,q)$ for scene transmission estimation plays a key role in single image dehazing.

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https://doi.org/10.1007/978-981-97-2011-8_5

Another unknown parameter crucial for solving the atmospheric scattering model to obtain a dehazed image is the global atmospheric light A_G. Increased haze in an image tends to introduce whiteness, affecting the overall perception. Existing literature primarily focuses on bright pixel intensities and their variations to estimate A_G. For instance, Tan et al. [22] considered the brightest pixel as A_G, while He et al. [23] opted for the top 0.1% of the brightest pixels. However, these methods encountered challenges in distinguishing white objects from the sky region. Kim et al. [24] introduced a hierarchical search method based on quad-tree subdivisions. To enhance position accuracy and avoid conflicts with the sky region in the presence of white objects, Wang et al. [25] built upon Kim et al.'s work by leveraging knowledge of sky region distribution.

As both the direct attenuation and airlight in the atmospheric scattering model are contingent on the depth map, this chapter introduces a single similarity function matrix. The purpose is to estimate the depth map, facilitating the determination of true scene transmission and global atmospheric light for image dehazing using (1.2).

Remark 5.1 Equation (1.2) implies that the attenuation and airlight are directly affected by the scene transmission.

The interpretation of image regions, involving edges and contrasts between the background and objects, is inherently vague as human understanding is never perfectly crisp. This introduces ambiguity into the final interpretation, and a hard decision-making system can often compromise naturalness. Therefore, the incorporation of a soft decision-making approach becomes crucial. The ambiguity in image regions is characterized by (i) Grayness ambiguity, representing uncertainty in deciding whether a pixel is white or black, and (ii) Spatial ambiguity, reflecting vagueness in defining object boundaries concerning the background.

In contrast to deep learning-based approaches in the literature, fuzzy logic proves beneficial for estimating the enhanced image while considering sparse and boundary constraints. The sparse constraint helps maintain the smoothness of homogeneous regions, while the boundary constraint preserves edges, addressing grayness and spatial ambiguities, respectively. The contents of this chapter can be summarized as follows.

- Following the principle stated in the Dark Channel Prior (DCP) method [23], which asserts the presence of a dark pixel in the neighborhood patch of the haze-free image, this chapter introduces a Type-2 Membership Function (MF) [26] based similarity function matrix. This matrix is designed to capture neighborhood information in the minimum channel of the observed hazy image.
- The Type-2 MF-based similarity function matrix is then utilized to estimate the depth map of the observed hazy scene. Subsequently, the estimated depth map is employed to compute the true scene transmission of the observed hazy image.
- Additionally, the similarity function matrix is leveraged to estimate the global atmospheric light for the observed hazy image.

Ultimately, the estimated scene transmission and global atmospheric light of the observed hazy image are applied to obtain the dehazed image using the atmospheric scattering model given by (1.2).

Furthermore, the methodology has been illustrated in detail and provides the experimental analysis.

5.2 Methodology

The methodology for single image dehazing using the Type-2 MF, also referred to as the Type-2 fuzzy approach, is elucidated in this section. Figure 5.1 illustrates the block diagram of the Type-2 fuzzy approach. The process begins by obtaining the minimum channel of the input hazy image. This minimum channel undergoes further processing with a Type-2 MF-based similarity function matrix to estimate the depth map for true scene transmission and global atmospheric light. Finally, a dehazed image is generated using the estimated transmission and global atmospheric light through the atmospheric scattering model. Subsequent subsections delve into a detailed discussion of each block.

5.2.1 Minimum Channel

As detailed in Sect. 5.1, the pixels situated farther from the observer are more susceptible to the influence of haze compared to those closer. Consequently, the initial approximation of the image's depth map is formulated as

$$I_{\min}(p,q) = \min_{\tau \in \{\mathrm{R,G,B}\}} \left(I^{\tau}_{\mathrm{Hazy}}(p,q) \right) \tag{5.3}$$

where

– τ represents R, G, and B color channels in the observed hazy image $I^{\tau}_{\mathrm{Hazy}}(p,q)$.

As per the DCP theory [23], there is always at least one color channel that has zero pixel intensity in a haze-free image. It is known as a dark channel. Let us consider a local patch N_k of size $\sqrt{k} \times \sqrt{k}$ with k number of pixels, $\forall\ \sqrt{k} = 1, 3, 5, 7, 9, \ldots$, centered at location (p,q). The dark channel of an observed hazy image is

$$I_{\mathrm{dark}}(p,q) = \min_{\tau \in \{\mathrm{R,G,B}\}} \left(\min_{(i,j) \in N_k} \left(I^{\tau}_{\mathrm{Hazy}}(i,j) \right) \right) \tag{5.4}$$

where

– $\forall\ 1 \leq (i,j) \leq \sqrt{k}$.

The critical input parameter for estimating scene transmission in the context of the DCP method is the size of the local patch. As highlighted by He et al. in [23], the scene

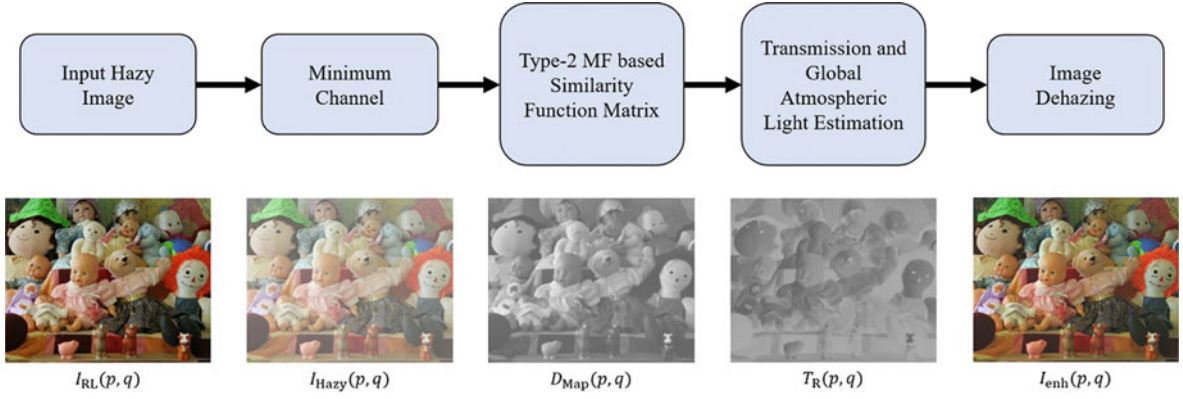


Fig. 5.1 Visual representation of Type-2 fuzzy approach based dehazing process of an image [52]. Here, $I_{RL}(p,q)$, $I_{Hazy}(p,q)$, $D_{Map}(p,q)$, $T_R(p,q)$, and $I_{enh}(p,q)$ are the reference haze-free image, observed hazy image, depth map, transmission map, and dehazed image, respectively. The observed hazy image is synthetically generated with haze concentration $H_C = 40\%$

transmission remains constant within a local patch, and the true scene transmission is determined by maximizing visibility. This leads to the following observations:

(i) A larger patch size results in a darker dark channel, which challenges the assumption of constant scene transmission within a local patch and introduces halo effects near edges.
(ii) Conversely, a smaller patch size leads to an over-saturation of the recovered scene image.

However, the Type-2 MF-based similarity function effectively mitigates halo artifacts near edges by incorporating neighborhood information with spatial and grayness ambiguities. In Fig. 5.2, a comparison of scene transmissions estimated using the DCP method and the presented approach for various patch sizes is presented. The figure illustrates that the scene transmission estimated using the DCP method undergoes changes near edges as the patch size increases. In contrast, the scene transmission estimated using the presented method maintains sharpness near edges, ensuring minimal halo effects. The next subsection provides an explanation of the Type-2 MF-based similarity function matrix.

5.2.2 Type-2 MF Based Similarity Function Matrix

When dealing with a set of elements, the conventional Type-1 MF assigns a single membership value to accommodate uncertainties associated with the elements. In contrast, a Type-2 MF, as introduced by Zadeh [27], addresses uncertainties in the membership values of the Type-1 fuzzy set. In other words, membership values within a Type-2 fuzzy set have a range of values to manage a more extensive range of uncertainty.

For an observed hazy image I_{Hazy}, if p_{ij} denotes the pixel intensity at location (i, j), the Type-1 fuzzy set is defined as

$$M_{ij} = \{\big(p_{ij}, \mu(p_{ij})\big) \,|\forall p_{ij} \in I_{\text{Hazy}}\} \tag{5.5}$$

where
– $0 \leq \mu_{ij} \leq 1$ denotes the membership values associated with pixel intensity value p_{ij}.

Thus, the Type-2 fuzzy set is defined by

$$\widetilde{M_{ij}} = \{\big(M_{ij}, \mu(M_{ij})\big)\}. \tag{5.6}$$

The formulation of the Type-2 MF-based similarity function draws inspiration from Singh et al. [28], where a similar approach is applied for image denoising. This design is rooted in fuzzy sets introduced by Zadeh [27]. The process begins by obtaining the minimum channel of the observed hazy image using (5.3). Subsequently, a

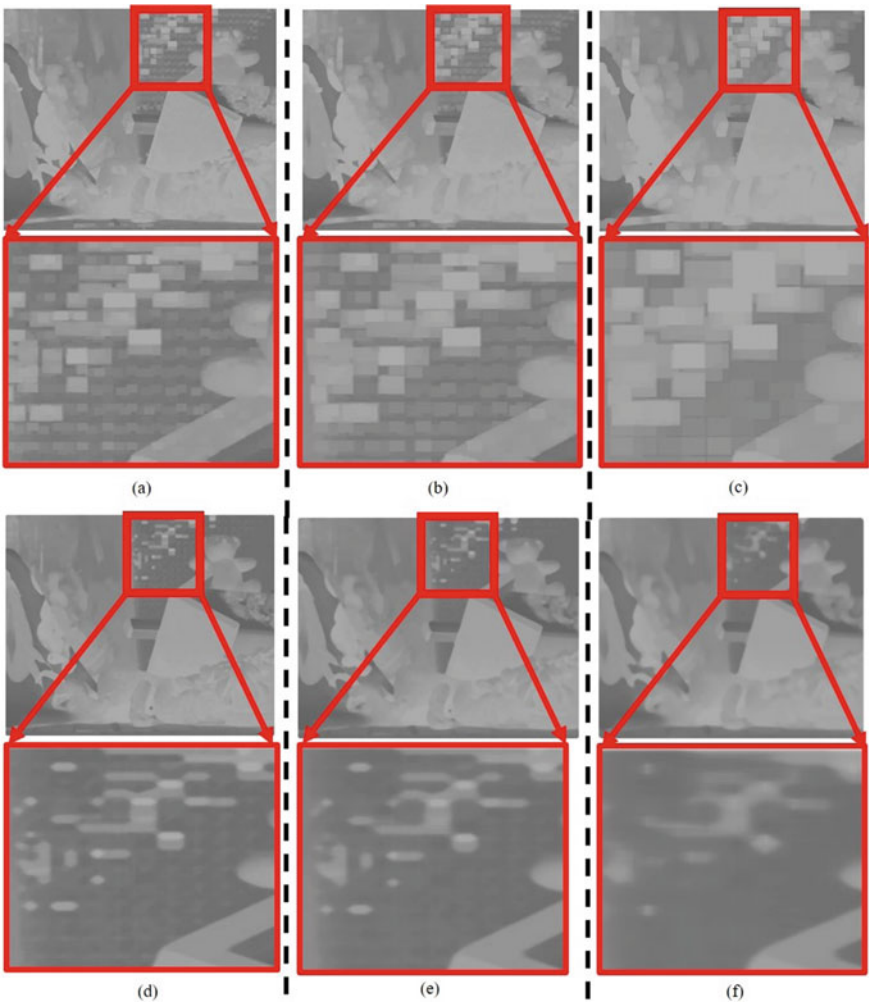


Fig. 5.2 Perceptual comparison of scene transmissions for analyzing the block artifacts [52]: Transmissions obtained using DCP method [23] for different sizes of patch, i.e., **a** 5×5, **b** 7×7, and **c** 11×11, Transmissions obtained using Type-2 fuzzy approach for different sizes of patch, i.e., **d** 5×5, **e** 7×7, and **f** 11×11. For each image, one is the transmission of the full scene view (top image) and another is the enlarged version of an image region marked in the full scene view (bottom image)

similarity function matrix is crafted for each pixel location of the minimum channel using Type-2 MF, taking into account the neighborhood pixels within a local patch.

A local patch N_k centered at (i, j) with dimensions $\sqrt{k} \times \sqrt{k}$ is extracted from the minimum channel as

$$N_k = \{I_{\min}(i+u, j+v)\} \; \forall \; (u, v) \in [-F_{HW}, F_{HW}] \tag{5.7}$$

where
– F_{HW} represents the half-filter width considered around the pixel. It is given by

$$F_{HW} = \frac{\sqrt{k}-1}{2}. \tag{5.8}$$

Remark 5.2 The value of patch size $\sqrt{k}$ may be $\{3, 5, 7, 9, ...\}$. If $k = 9$, the patch size will be $\sqrt{k} \times \sqrt{k} = 3 \times 3$. Hence, the value of half-filter width F_{HW} will be $\frac{3-1}{2} = 1$.

Hence, the similarity function for a local patch N_k is defined as

$$\eta_k^h = f(N_k, M^h) = \exp\left(-\frac{1}{2}\left(\frac{N_k - M^h}{\sigma}\right)^2\right) \tag{5.9}$$

where
– σ is the standard deviation and $M^h = \{l_1, l_2, ..., l_h\}$ is h-middle means for a local patch. h-middle means for a local patch with an odd number of pixels are calculated as

$$M^h = \frac{1}{2h-1} \sum_{i=n_m-h+1}^{n_m+h-1} N_i \tag{5.10}$$

where

$$n_m = \frac{k+1}{2} \tag{5.11}$$

and

$$h = 1, 2, 3, ..., n_m. \tag{5.12}$$

Remark 5.3 In (5.10), h is the number of means obtained for a local patch. If size of a local patch is $k = 9$ i.e., $\sqrt{k} \times \sqrt{k} = 3 \times 3$, the neighborhood pixels under consideration for this patch will be $\{N_1, N_2, N_3, N_4, N_5, N_6, N_7, N_8, N_9\}$. The value of n_m is $\frac{k+1}{2} = \frac{9+1}{2} = 5$. Hence, the number of means for this local patch will be $h = 1, 2, 3, 4, 5$. Using (5.10), the h-middle means will be calculated by

$$M^1 = N_5, \tag{5.13}$$

$$M^2 = \frac{1}{3}\left[N_4 + N_5 + N_6\right], \tag{5.14}$$

$$M^3 = \frac{1}{5}\left[N_3 + N_4 + N_5 + N_6 + N_7\right], \tag{5.15}$$

$$M^4 = \frac{1}{7}\left[N_2 + N_3 + N_4 + N_5 + N_6 + N_7 + N_8\right], \tag{5.16}$$

and

$$M^5 = \frac{1}{9}\left[N_1 + N_2 + N_3 + N_4 + N_5 + N_6 + N_7 + N_8 + N_9\right]. \tag{5.17}$$

The Type-2 MF-based similarity function matrix using (5.9) can also be expressed as

$$\eta_k^h = \begin{bmatrix} f(N_1, l_1) & f(N_2, l_1) & \dots & f(N_k, l_1) \\ f(N_1, l_2) & f(N_2, l_2) & \dots & f(N_k, l_2) \\ \vdots & \vdots & \ddots & \vdots \\ f(N_1, l_h) & f(N_2, l_h) & \dots & f(N_k, l_h) \end{bmatrix}_{h \times k}. \tag{5.18}$$

Equation (5.18) can be simply rewritten as

$$\eta_k^h = \begin{bmatrix} \eta_1^1 & \eta_2^1 & \dots & \eta_k^1 \\ \eta_1^2 & \eta_2^2 & \dots & \eta_k^2 \\ \vdots & \vdots & \ddots & \vdots \\ \eta_1^h & \eta_2^h & \dots & \eta_k^h \end{bmatrix}_{h \times k}. \tag{5.19}$$

Equation (5.19) undergoes additional processing to derive the estimates for the depth map and the global atmospheric light of the observed hazy image. The subsequent subsection provides an explanation of the process involved in estimating scene transmission using the depth map of the observed hazy image.

5.2.3 *Depth Map and Scene Transmission*

The MF value associated with each pixel in a local patch N_k is estimated using (5.19) as

$$\eta_k' = \left[\eta_1' \ \eta_2' \ \dots \ \eta_k'\right] \tag{5.20}$$

where η_k' is the mean of MF values obtained for h-middle means of kth pixel in the patch N_k.

Thus, the elements of (5.20) can be rewritten as

$$\begin{aligned}
\eta_1' &= \frac{\sum_{i=1}^{h} \eta_1^i}{h} = \eta_1^1 + \eta_1^2 + \eta_1^3 + \cdots + \eta_1^h \\
\eta_2' &= \frac{\sum_{i=1}^{h} \eta_2^i}{h} = \eta_2^1 + \eta_2^2 + \eta_2^3 + \cdots + \eta_2^h \\
&\vdots \quad \vdots \quad \vdots \quad \vdots \quad \vdots \\
\eta_k' &= \frac{\sum_{i=1}^{h} \eta_k^i}{h} = \eta_k^1 + \eta_k^2 + \eta_k^3 + \cdots + \eta_k^h.
\end{aligned} \tag{5.21}$$

The estimation of the depth map for the observed hazy image at pixel location (p, q) involves calculating the weighted average of the pixels within a local patch N_k surrounding the pixel of interest.

The associated membership function (MF) values η_k' are obtained using (5.21). This process is expressed as

$$D_{\text{Map}}(p, q) = \frac{\sum_{i=1}^{k} \eta_i'(p, q) N_i(p, q)}{\sum_{i=1}^{k} \eta_i'(p, q)}. \tag{5.22}$$

Therefore, the true scene transmission is estimated by

$$T_{\text{R}}(p, q) = \exp\left(-\beta \times D_{\text{Map}}(p, q)\right) \tag{5.23}$$

where β has been taken as 1 empirically. It represents the angular scattering coefficient of the atmosphere.

Remark 5.4 For a visible light spectrum, the relation between the scattering coefficient β, and the wavelength λ is given by the inverse power law as

$$\beta(\lambda) = \frac{\text{Constant}}{\lambda^{\gamma}} \tag{5.24}$$

where $\gamma \in [0, 4]$. In the context of fog and dense haze, it is commonly observed that $\gamma = 0$ [29, 30], indicating that the parameter β does not undergo significant changes with wavelength. Specifically, for the visible light range, where the spectral bandwidth of the camera is constrained, the scattering coefficient $\beta(\lambda)$ is assumed to remain constant over this bandwidth. Narasimhan and Nayar [21] conducted simulations and concluded that "the scattering coefficient does not vary with wavelength within the visible spectrum".

In this chapter, the optimal experimental value for β has been determined to be 1. Selecting $\beta = 1$ has been found to yield the best results in the conducted experiments. When β is less than 1, it tends to enhance brightness in the illuminated regions, while for β greater than 1, the darker regions become even darker. This imbalance can result in over-saturation in the enhanced image.

In the following subsection, the explanation of global atmospheric light estimation is provided, employing the MF-based similarity function matrix as expressed in (5.19).

5.2.4 Global Atmospheric Light

In a local patch, the h-middle means exhibit a monotonic increase and decrease in homogeneous regions and a non-monotonic variation in the presence of discontinuities [31, 32]. This characteristic variation in means also influences the structure of the similarity function matrix (5.19). Based on these observations, the atmospheric light is computed at each pixel location utilizing a Type-2 MF-based similarity function matrix, and the process is expressed as

$$L(p,q) = \min\left(\eta_k^h(p,q)\right). \tag{5.25}$$

Typically, the atmospheric light estimation involves selecting the most haze-opaque regions. However, the influence of the nature of means on the similarity function matrix leads to the exclusion of haze-opaque regions containing the brightest objects. This is because variations in means are more pronounced for bright objects, where smoothness is inherently lower compared to hazy regions. Consequently, the global atmospheric light is estimated through the following process,

$$A_{\mathrm{G}} = \mathrm{mean}\left(L(p,q)\right). \tag{5.26}$$

The global atmospheric light serves as a representation of the pixel intensity at the farthest hazy pixel location, predominantly found in sky regions. However, conflicts may arise in the presence of brighter objects within hazy scenes. The estimate of the global atmospheric light effectively mitigates the impact of brighter or white objects, thanks to the inherent nature of variation in means within the Type-2 MF-based similarity function.

Remark 5.5 In the scenario where a patch encompasses a bright or white object within a haze-free region, the similarity function matrix will exhibit h different means. The computation of the weighted average of associated MFs for these means, combined with the pixel intensity values of the patch, will be predominantly influenced by the darker pixel intensity values within the patch. Consequently, this process ensures that the global atmospheric light is determined while mitigating the impact of bright or white objects.

5.2.5 *Image Dehazing*

Finally, the dehazed output image using the observed hazy image $I_{\text{Hazy}}(p, q)$, estimated scene transmission $T_{\text{R}}(p, q)$ using (5.23), and global atmospheric light A_{G} using (5.26) is obtained by rearranging (1.2) as

$$I_{\text{enh}}(p, q) = \frac{I_{\text{RL}}(p, q) - A_{\text{G}}}{T_{\text{R}}(p, q)} + A_{\text{G}} \tag{5.27}$$

where $I_{\text{enh}}(p, q)$ is a dehazed image and tries to approach a clear or haze-free image $I_{\text{RL}}(p, q)$.

Algorithm 1 outlines the step-by-step implementation of the presented Type-2 MF-based single image dehazing method. This approach yields a dehazed image that closely resembles its haze-free counterpart, with the added benefit of enhancing illumination for visual appeal. Notably, it effectively addresses over-saturation issues in haze-free and dark regions, providing better visibility. The zoomed view of the "Books" image in Fig. 5.3 illustrates the changes in the enhanced output image using both the DCP method [23] and the Type-2 fuzzy approach. In this particular visualization, a patch size of 3×3 has been employed.

Algorithm 1: Type-2 fuzzy approach for single image dehazing

Patch Size (Input): $k = 9 \Rightarrow \sqrt{k} \times \sqrt{k} = 3 \times 3$
Min Channel: $I_{\text{min}}(p, q) = \min_{\tau \in \{\text{R,G,B}\}} I^{\tau}_{\text{Hazy}}(p, q)$
for $I_{\text{min}}(i, j)$ **do**
  $N_k(i, j) = \{I_{\text{min}}(i + u, j + v)\} \ \forall\, u, v \in [-F_{HW}, F_{HW}]$
  Similarity Function Matrix: $\eta^h_k(i, j) = f(N_k, M^h)$
  MF Value: $\eta'_k(i, j)$ by (5.20)
  Depth map: $D_{\text{Map}}(i, j)$ by (5.22)
  Scene transmission: $T_{\text{R}}(i, j)$ by (5.23)
end for
Global Atmospheric Light: A_{G} by (5.26)
Dehazed Image: $I_{\text{enh}}(p, q)$ by (5.27)

5.3 Results and Discussions

In this section, the experimental analysis of the Type-2 fuzzy approach has been presented and provides a comparison with other state-of-the-art dehazing methods. The experiments were conducted on a system running the Windows 10 64-bit Operating System with an x64-based Intel(R) Core(TM) i7-8550U CPU, using MATLAB2019a. The comparison includes well-known dehazing methods, namely,

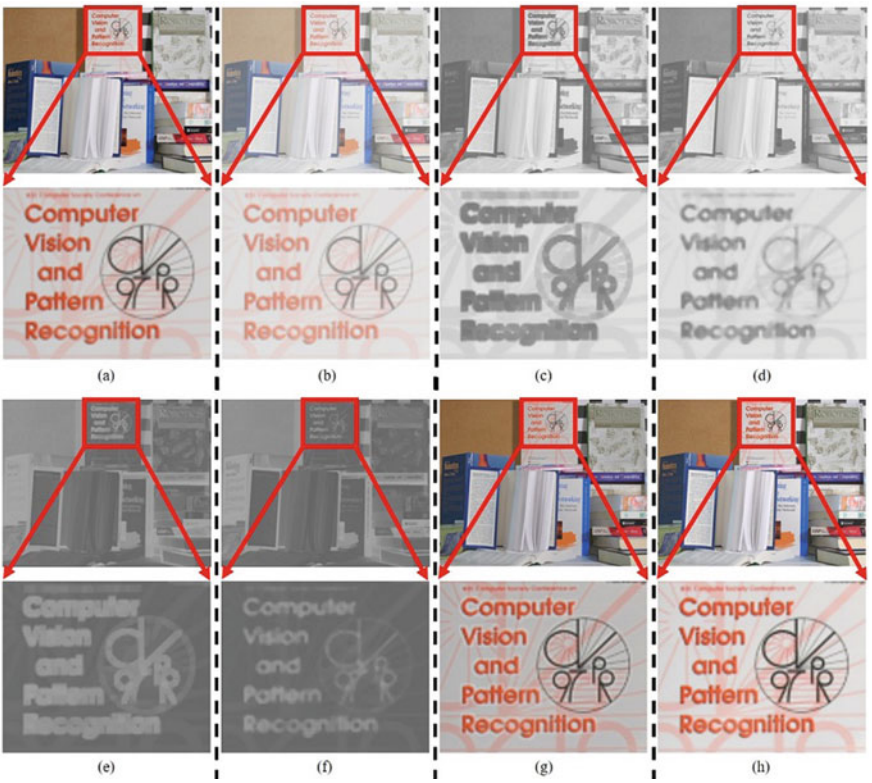


Fig. 5.3 Perceptual comparison of depth and transmissions maps of an image named "Books" to depict the step-by-step enhancement process [52]: **a** Image as reference or ground truth, **b** Image captured under haze, Depth estimation of the image shown in **b** using **c** DCP [23], and **d** Type-2 fuzzy approach, Transmission estimation of the image shown in **b** using **e** DCP [23], and **f** Type-2 fuzzy approach, and Dehazed version of the image shown in **b** using **g** DCP [23], and **h** Type-2 fuzzy approach. For each image, one is the full scene view (top image) and another is the enlarged version of an image region marked in the full scene view (bottom image)

CLAHE [33], DCP [23], BCCR [34], CAP [35], MSCNN [36], DehazeNet [37], AOD-Net [38], and GMAN [39].

In the following subsections, datasets used for experimental analysis, performance metrics, and results have been detailed.

5.3.1 Datasets

The experimentation utilized a diverse set of datasets to evaluate and validate the methods. The datasets can be categorized as follows:

(i) Natural Hazy Benchmark Images: A collection of natural hazy images widely referenced in the literature [23].
(ii) Synthetic Hazy Images from MiddleBury Stereo Dataset: 32 synthetic hazy images generated using the MiddleBury Stereo dataset [40–42].
(iii) Indoor Scene SOTS Images from RESIDE Dataset: 500 hazy images from the RESIDE dataset specifically focused on indoor scenes [43].
(iv) Outdoor Scene SOTS Images from RESIDE Dataset: 500 hazy images from the RESIDE dataset specifically focused on outdoor scenes [43].
(v) Real-World Synthetic Hazy Images of HSTS from RESIDE Dataset: 10 real-world synthetic hazy images from the RESIDE dataset, particularly from the HSTS subset [43] , and
(vi) Synthetic Hazy Images from Stanford ImageNet Dataset: 250 synthetic hazy images generated from the Stanford ImageNet dataset [44].

Furthermore, to assess the efficacy of the method in the context of vision-based applications, specifically the dehazing+detection pipeline, an analysis has been conducted to obtain the detection results using the YOLOv2 architecture [45]. This architecture was trained on the COCO dataset [46], encompassing 80 distinct classes. The evaluation of detection results was performed on a Real-world Task-driven Testing Set (RTTS) sourced from the RESIDE-β dataset [43].

The RTTS consists of 4,322 real-world hazy images, each annotated with object bounding boxes for five different classes: person, bicycle, car, bus, and motorbike. The detection results on dehazed images were scrutinized to validate the effectiveness of the presented approach in enhancing vision-based applications, particularly in scenarios involving object detection in hazy conditions.

In the presence of haze, a clear scene appears whitish due to obstruction in the scene transmission reaching the camera. The scene transmission $T_{\mathrm{R}}(p, q)$ is assumed to be 1 if the image is haze-free. Thus, a hazy image $I_{\mathrm{Hazy}}(p, q)$ with haze concentration H_C from haze-free image $I_{\mathrm{RL}}(p, q)$ is generated as

$$I_{\mathrm{Hazy}}(p, q) = I_{\mathrm{RL}}(p, q)\,(1 - H_C) + H_C. \tag{5.28}$$

Remark 5.6 The first term and second term in (5.28) represent attenuation and airlight, respectively. It shows that the scene transmission reduces with the increase in haze concentration; meanwhile, airlight increases.

5.3.2 Performance Metrics

The assessment of dehazed image quality is inherently subjective, subject to individual perceptions. To facilitate a meaningful comparison between dehazed image $I_{\mathrm{enh}}(p, q)$ and its haze-free counterpart $I_{\mathrm{RL}}(p, q)$, various performance measures have been employed in this chapter. These measures can be categorized as follows:

(i) Quantitative Measures for Dehazed Images: Objective metrics designed to quantitatively assess the quality of dehazed images.
(ii) Qualitative Measures for Dehazed Images: Subjective evaluations and visual assessments aimed at capturing the perceptual quality and fidelity of dehazed images, and
(iii) Receiver Operating Characteristic (ROC) Curve Analysis: An analysis involving ROC curves, as outlined in [47], specifically tailored for evaluating the performance of the dehazing+detection pipeline.

The performance metrics used for quantitative comparison are PSNR [48] and SSIM index [49]. PSNR is used for comparison of image quality.

In the evaluation of dehazing methods, the Peak Signal-to-Noise Ratio (PSNR) serves as an indicator of how closely the dehazed image matches its haze-free counterpart. A higher PSNR implies a better match.

The Structural SIMilarity (SSIM) index measures the perceptual difference between two images, considering luminance, contrast, and saturation. SSIM values range from 0 to 1, with a higher SSIM indicating minimal perceptual differences.

For qualitative assessment, two no-reference image quality metrics are employed:

Lightness Order Error (LOE) [50]: LOE evaluates the naturalness of enhanced images in comparison to their hazy counterparts. A lower LOE value suggests better perceptual quality.

Naturalness Image Quality Evaluator (NIQE) [51]: NIQE assesses the naturalness and quality of images without a reference image. Lower NIQE values indicate better image quality.

Additionally, the Receiver Operating Characteristic (ROC) curve is utilized to provide precision and recall values for estimating the F1-score and Area Under the Curve (AUC) measures for each class.

The higher the values of the F1-score and AUC, the better the model is capable of distinguishing the classes, which leads to the conclusion that the input dehazed image is better.

In the next subsection, the performance comparisons have been tabulated and discussed.

5.3.3 Quantitative Comparison

Tables 5.1, 5.2, 5.3, and 5.4 provide a comprehensive comparison between the Type-2 fuzzy approach and state-of-the-art methods, evaluating performance in terms of PSNR, SSIM, LOE, and NIQE for images depicted in Figs. 5.4, 5.5, 5.6, and 5.7. High PSNR and SSIM values indicate enhanced perceived quality of dehazed images, closely resembling their reference counterparts, while low LOE and NIQE values signify closer alignment with human perception in terms of naturalness.

To validate the superiority of the Type-2 fuzzy approach, experimental analyses were extended to RESIDE and RESIDE-β datasets [43]. Tables 5.5, 5.6, and 5.7

Table 5.1 Performance assessment of Type-2 fuzzy approach on an image named "Books" from the MiddleBury Stereo dataset [40–42] as shown in Fig. 5.4

Performance metric	CLAHE [33]	DCP [23]	BCCR [34]	CAP [35]	MSCNN [36]	DehazeNet [37]	AOD-Net [38]	GMAN [39]	Type-2 fuzzy approach [52]
PSNR [48]	17.6044	12.9717	13.7686	26.8754	24.2730	23.8629	28.0353	18.7716	**28.8661**
SSIM [49]	0.9096	0.8555	0.6964	0.9585	0.9422	0.9585	0.8707	0.8636	**0.9709**
LOE [50]	1.076	0.604	1.571	0.320	0.523	0.801	0.405	1.756	**0.203**
NIQE [51]	3.47	3.39	3.56	3.53	3.50	3.34	3.95	2.86	**2.66**

Table 5.2 Performance assessment of Type-2 fuzzy approach on an image named "Cloth" from the MiddleBury Stereo dataset [40–42] as shown in Fig. 5.5

Performance metric	CLAHE [33]	DCP [23]	BCCR [34]	CAP [35]	MSCNN [36]	DehazeNet [37]	AOD-Net [38]	GMAN [39]	Type-2 fuzzy approach [52]
PSNR [48]	16.8967	16.6430	16.0363	25.8608	19.3254	21.4286	28.0976	21.8120	**30.8010**
SSIM [49]	0.8770	0.8222	0.6876	0.9079	0.8454	0.9037	0.8394	0.9360	**0.9906**
LOE [50]	1.962	0.306	0.399	0.392	0.300	0.319	0.511	1.552	**0.260**
NIQE [51]	13.77	14.01	16.23	12.66	13.09	12.26	6.58	9.22	**2.93**

Table 5.3 Performance assessment of Type-2 fuzzy approach on an image named "Computer" from the MiddleBury Stereo dataset [40–42] as shown in Fig. 5.6

Performance metric	CLAHE [33]	DCP [23]	BCCR [34]	CAP [35]	MSCNN [36]	DehazeNet [37]	AOD-Net [38]	GMAN [39]	Type-2 fuzzy approach [52]
PSNR [48]	15.4275	17.5518	15.6319	22.4719	22.4988	19.0182	27.6855	16.4630	**27.7401**
SSIM [49]	0.8684	0.8650	0.5872	0.9441	0.7613	0.8400	0.9046	0.8724	**0.9764**
LOE [50]	0.732	0.798	1.165	0.342	0.649	0.511	0.352	1.621	**0.263**
NIQE [51]	5.32	5.42	4.93	5.50	5.21	5.87	4.84	5.54	**2.67**

Table 5.4 Performance assessment of Type-2 fuzzy approach on an image named "Cones" from the MiddleBury Stereo dataset [40–42] as shown in Fig. 5.7

Performance metric	CLAHE [33]	DCP [23]	BCCR [34]	CAP [35]	MSCNN [36]	DehazeNet [37]	AOD-Net [38]	GMAN [39]	Type-2 fuzzy approach [52]
PSNR [48]	15.9313	19.9807	22.8435	28.3163	15.9033	20.2857	27.6788	22.7143	**29.0336**
SSIM [49]	0.8529	0.9219	0.9265	0.9778	0.8047	0.9643	0.7851	0.9513	**0.9880**
LOE [50]	1.094	0.391	0.667	0.387	0.662	0.322	0.462	0.959	**0.298**
NIQE [51]	3.75	3.52	3.65	3.38	3.69	3.51	3.91	2.92	**2.74**

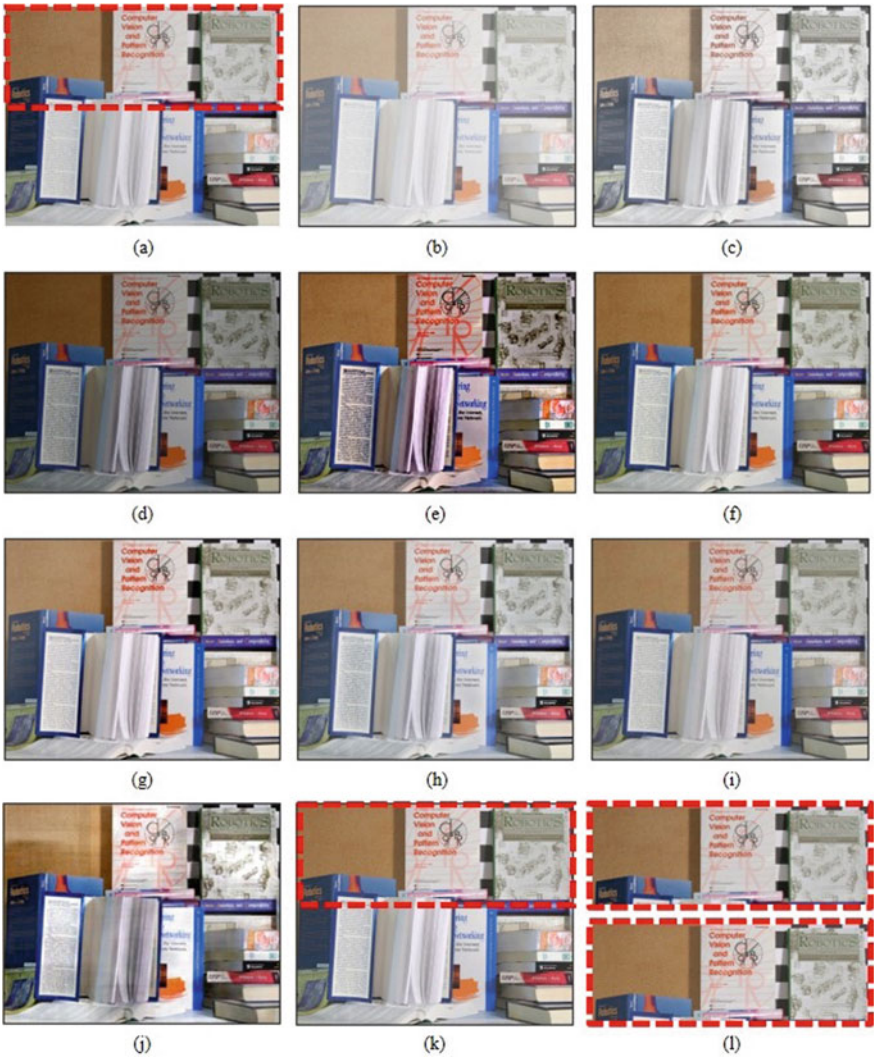


Fig. 5.4 Perceptual analysis of an image named "Books" from the MiddleBury Stereo dataset images [40–42]: **a** Image with no haze (Ground truth), **b** Image with some haze present, Dehazed version of the image shown in **b** using **c** CLAHE [33], **d** DCP [23], **e** BCCR [34], **f** CAP [35], **g** MSCNN [36], **h** DehazeNet [37], **i** AOD-Net [38], **j** GMAN [39], **k** Type-2 fuzzy approach [52], and **l** Enlarged versions of an image regions marked in **a** and **k**. (Please also see Figs. 5.5, 5.6, and 5.7 for more results)

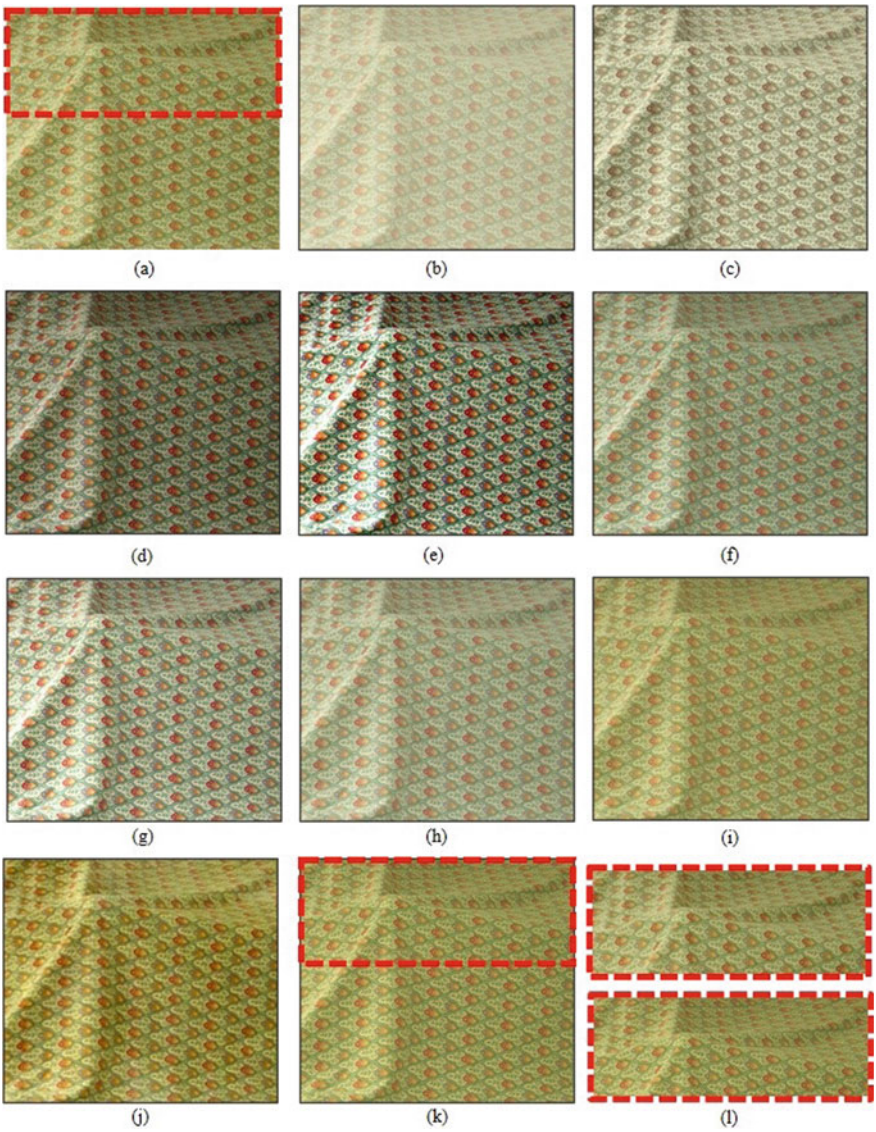


Fig. 5.5 Perceptual analysis of an image named "Cloth" from the MiddleBury Stereo dataset images [40–42]: **a** Image with no haze (Ground truth), **b** Image with some haze present, Dehazed version of the image shown in **b** using **c** CLAHE [33], **d** DCP [23], **e** BCCR [34], **f** CAP [35], **g** MSCNN [36], **h** DehazeNet [37], **i** AOD-Net [38], **j** GMAN [39], **k** Type-2 fuzzy approach [52], and **l** Enlarged versions of an image regions marked in (a) and (k). (Please also see Figs. 5.4, 5.6, and 5.7 for more results)

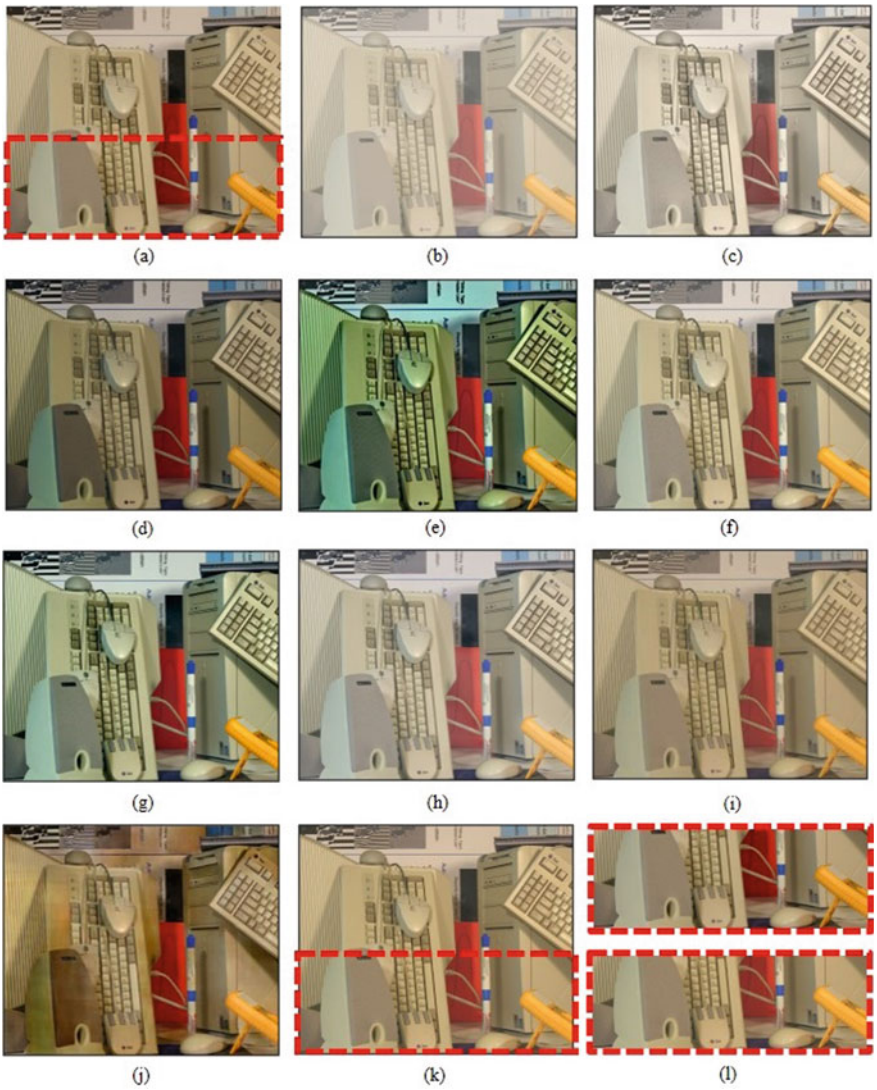


Fig. 5.6 Perceptual analysis of an image named "Computer" from the MiddleBury Stereo dataset images [40–42]: **a** Image with no haze (Ground truth), **b** Image with some haze present, Dehazed version of the image shown in **b** using **c** CLAHE [33], **d** DCP [23], **e** BCCR [34], **f** CAP [35], **g** MSCNN [36], **h** DehazeNet [37], **i** AOD-Net [38], **j** GMAN [39], **k** Type-2 fuzzy approach [52], and **l** Enlarged versions of an image regions marked in **a** and **k**. (Please also see Figs. 5.4, 5.5, and 5.7 for more results)

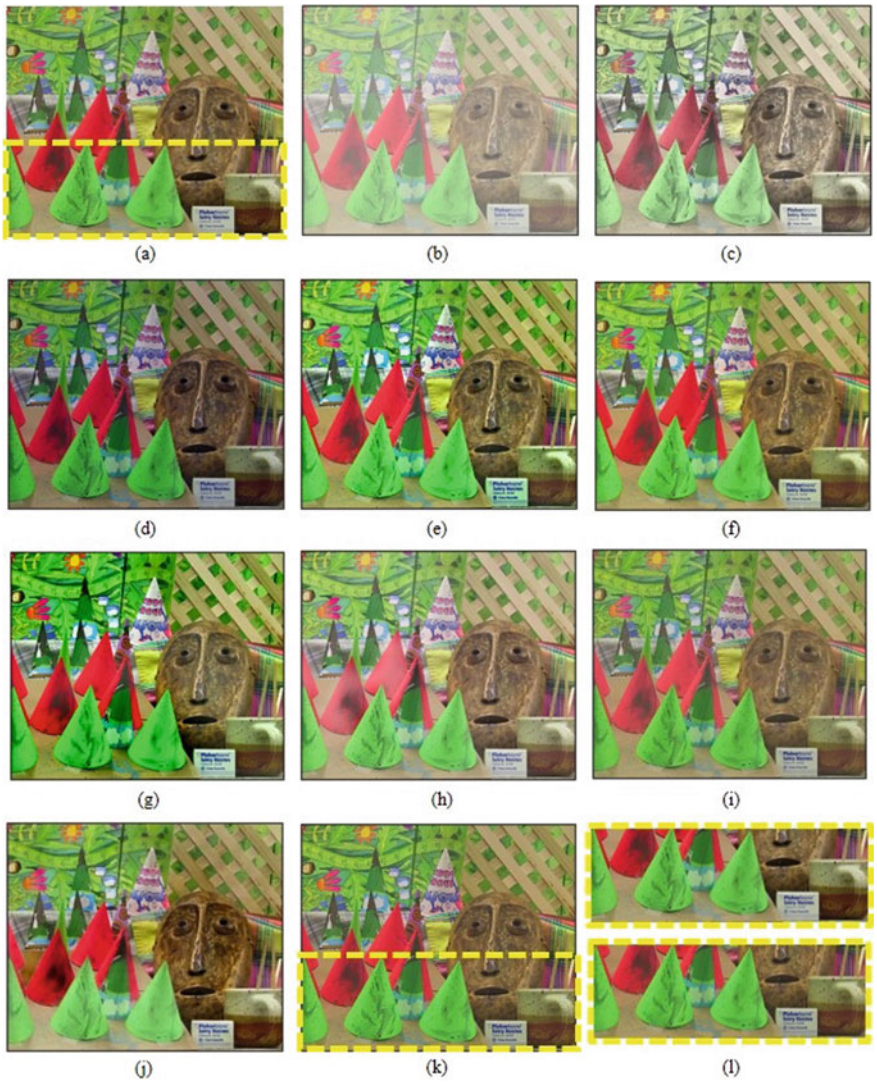


Fig. 5.7 Perceptual analysis of an image named "Cones" from the MiddleBury Stereo dataset images [40–42]: **a** Image with no haze (Ground truth), **b** Image with some haze present, Dehazed version of the image shown in (b) using **c** CLAHE [33], **d** DCP [23], **e** BCCR [34], **f** CAP [35], **g** MSCNN [36], **h** DehazeNet [37], **i** AOD-Net [38], **j** GMAN [39], **k** Type-2 fuzzy approach [52], and **l** Enlarged versions of an image regions marked in **a** and **k**. (Please also see Figs. 5.4, 5.5, and 5.6 for more results)

Table 5.5 Performance assessment of Type-2 fuzzy approach on images of indoor SOTS subset from the RESIDE dataset [43]

Performance metric	CLAHE [33]	DCP [23]	BCCR [34]	CAP [35]	MSCNN [36]	DehazeNet [37]	AOD-Net [38]	GMAN [39]	Type-2 fuzzy approach [52]
PSNR [48]	13.1359	11.8304	17.0468	18.9634	17.1292	21.1671	18.7337	28.0794	**29.7248**
SSIM [49]	0.7140	0.6346	0.7938	0.8524	0.8069	0.8741	0.8456	0.9664	**0.9702**
LOE [50]	1.326	1.485	1.071	0.941	0.982	0.839	1.012	0.947	**0.821**
NIQE [51]	3.128	2.947	3.013	3.159	3.178	3.12	3.224	3.753	**2.381**

Table 5.6 Performance assessment of Type-2 fuzzy approach on images of outdoor SOTS subset from the RESIDE dataset [43]

Performance metric	CLAHE [33]	DCP [23]	BCCR [34]	CAP [35]	MSCNN [36]	DehazeNet [37]	AOD-Net [38]	GMAN [39]	Type-2 fuzzy approach [52]
PSNR [48]	16.4549	15.0527	15.5109	18.1236	19.4735	22.2760	19.4997	18.8861	**23.1524**
SSIM [49]	0.7921	0.6993	0.7575	0.7622	0.8334	0.8570	0.8606	0.8548	**0.8696**
LOE [50]	0.872	1.039	0.858	0.549	0.730	0.547	0.607	0.939	**0.529**
NIQE [51]	2.799	3.097	2.799	2.826	2.745	2.746	2.837	2.903	**2.713**

Table 5.7 Performance assessment of Type-2 fuzzy approach on images of synthetic HSTS subset from the RESIDE dataset [43]

Performance metric	CLAHE [33]	DCP [23]	BCCR [34]	CAP [35]	MSCNN [36]	DehazeNet [37]	AOD-Net [38]	GMAN [39]	Type-2 fuzzy approach [52]
PSNR [48]	13.2242	18.9331	11.4926	11.8863	16.3276	14.2192	13.7651	10.0533	**22.4193**
SSIM [49]	0.7117	0.6917	0.5971	0.6126	0.8048	0.7334	0.8027	0.5353	**0.8699**
LOE [50]	0.905	1.151	0.945	0.487	0.696	0.495	0.375	0.932	**0.212**
NIQE [51]	2.75	3.28	2.80	2.73	2.73	2.76	3.43	2.67	**2.73**

Table 5.8 Performance assessment of Type-2 fuzzy approach on synthetic hazy images subset from the Stanford ImageNet dataset [44]

Performance metric	CLAHE [33]	DCP [23]	BCCR [34]	CAP [35]	MSCNN [36]	DehazeNet [37]	AOD-Net [38]	GMAN [39]	Type-2 fuzzy approach [52]
PSNR [48]	15.1316	19.6324	18.0292	22.2278	19.2127	24.7768	24.7332	24.8459	**28.4413**
SSIM [49]	0.7940	0.8472	0.7708	0.9011	0.8283	0.9412	0.9025	0.9023	**0.9431**

present comparative analyses of the Type-2 fuzzy approach against state-of-the-art methods on 500 indoor SOTS, 500 outdoor SOTS, and 10 real-world synthetic HSTS images from the RESIDE dataset, respectively. These tables contribute to the holistic assessment of the Type-2 fuzzy approach's performance across different datasets and scenarios.

In addition to the datasets discussed earlier, the experiments were expanded to include synthetic hazy images generated from 250 randomly selected images in the Stanford ImageNet dataset. A detailed comparative analysis of these results is presented in Table 5.8.

5.3.4 Qualitative Comparison

The figures presented in this section, namely, Figs. 5.4, 5.5, 5.6, and 5.7, illustrate the comparative analysis of experimental results obtained through the Type-2 fuzzy approach in contrast to state-of-the-art methods. These comparisons are conducted on images representing diverse scenes, namely, "Books", "Cloth", "Computer", and "Cones", sourced from the MiddleBury Stereo dataset images [40–42].

The hazy images showcased in these figures are synthetically generated with a haze concentration (H_C) of 40%. The Type-2 fuzzy approach, utilizing a patch size of 5×5, demonstrates remarkable performance by producing dehazed images that closely resemble clear counterparts. Notably, the Type-2 fuzzy approach outperforms other methods in terms of haze removal.

In contrast, the CLAHE method [33] leaves residual haze in the dehazed image, while DCP [23] results in over-saturation, leading to a loss of illumination. BCCR [34] introduces oversaturated colors and increased illumination, altering color information significantly, though the visual results remain acceptable.

Among the deep learning-based models, including CAP [35], MSCNN [36], DehazeNet [37], AOD-Net [38], and GMAN [39], CAP and AOD-Net exhibit better performance but introduce blurring in image textures. MSCNN modifies color information, while DehazeNet leaves some residual haze. Notably, the GMAN model [39], based on an encoder-decoder architecture, achieves visually appealing results, albeit at the expense of compromised quantitative performance due to over-saturation.

These comparative visualizations provide insights into the strengths and weaknesses of each method, aiding in a comprehensive understanding of their performance across different scenes and scenarios.

Figures 5.8 and 5.9 showcase the visual comparison of experimental results on benchmarked natural hazy images of "Girls" and "Canyon" [23], frequently referenced in literature for visual assessments. The displayed images underscore the visual superiority of the Type-2 fuzzy approach in generating dehazed images that are more aesthetically pleasing compared to other methods. The clarity and quality enhancement achieved by the Type-2 fuzzy approach in these benchmarked scenarios highlight its effectiveness in addressing challenges posed by natural hazy conditions.

To provide a visual representation of the comparative results, Fig. 5.10 illustrates the outcomes on an image of HSTS from the RESIDE dataset. The performance measures detailed in the tables consistently confirm the superior performance of the Type-2 fuzzy approach compared to other methods. This emphasizes the effectiveness of the presented approach not only across real-world datasets but also in handling synthetic hazy images, showcasing its robustness and versatility.



Fig. 5.8 Perceptual analysis of a naturally hazy image named as "Girls" [23]: **a** Image captured under haze, Dehazed version of image shown in **a** using **b** CLAHE [33], **c** DCP [23], **d** BCCR [34], **e** CAP [35], **f** MSCNN [36], **g** DehazeNet [37], **h** AOD-Net [38], **i** GMAN [39], and **j** Type-2 fuzzy approach [52]

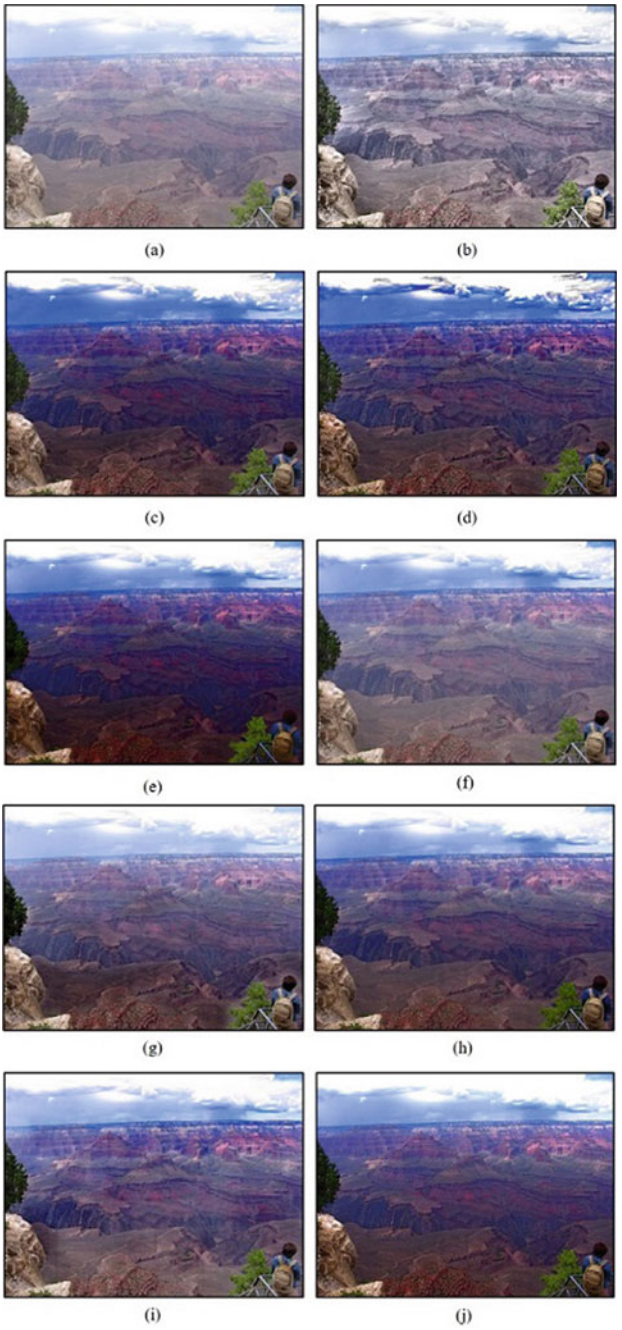


Fig. 5.9 Perceptual analysis of a naturally hazy image named as "Canyon" [23]: **a** Image captured under haze, Dehazed version of image shown in **a** using **b** CLAHE [33], **c** DCP [23], **d** BCCR [34], **e** CAP [35], **f** MSCNN [36], **g** DehazeNet [37], **h** AOD-Net [38], **i** GMAN [39], and **j** Type-2 fuzzy approach [52]

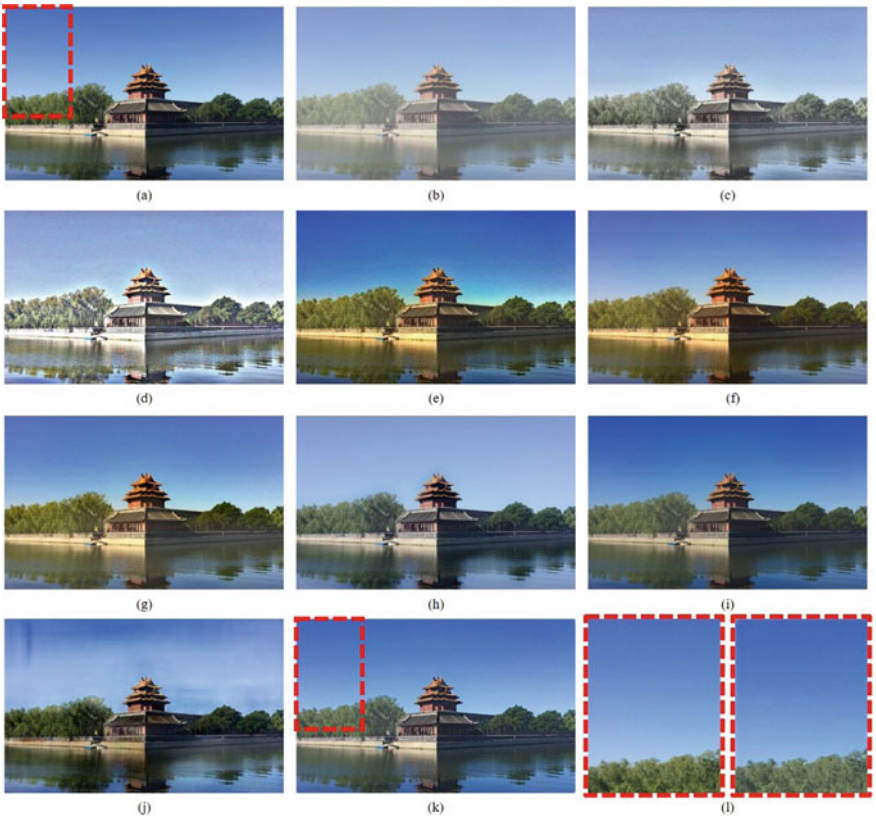


Fig. 5.10 Perceptual analysis of an image belongs to HSTS [43]: **a** Image with no haze (Ground truth), **b** Image captured under haze, Dehazed version of image shown in **b** using **c** CLAHE [33], **d** DCP [23], **e** BCCR [34], **f** CAP [35], **g** MSCNN [36], **h** DehazeNet [37], **i** AOD-Net [38], **j** GMAN [39], **k** Type-2 fuzzy approach, and **l** Enlarged versions of an image regions marked in **a** and **k** [52]

5.3.5 Detection Results

Furthermore, the dehazing+detection pipeline underwent experimentation on 4,322 images of RTTS from the RESIDE-β dataset. The chosen detection network is YOLOv2 [45], trained on 80 classes of the COCO dataset [46]. Evaluation of detection results was performed for five distinct classes: person, bicycle, car, bus, and motorbike. Figures 5.11, 5.12, 5.13, 5.14, and 5.15 illustrate the comparison of detection outcomes, presenting F1-score and AUC values obtained through ROC curve analysis for the classes “Person”, “Bus”, “Bicycle”, “Motorbike”, and “Car”, respectively.

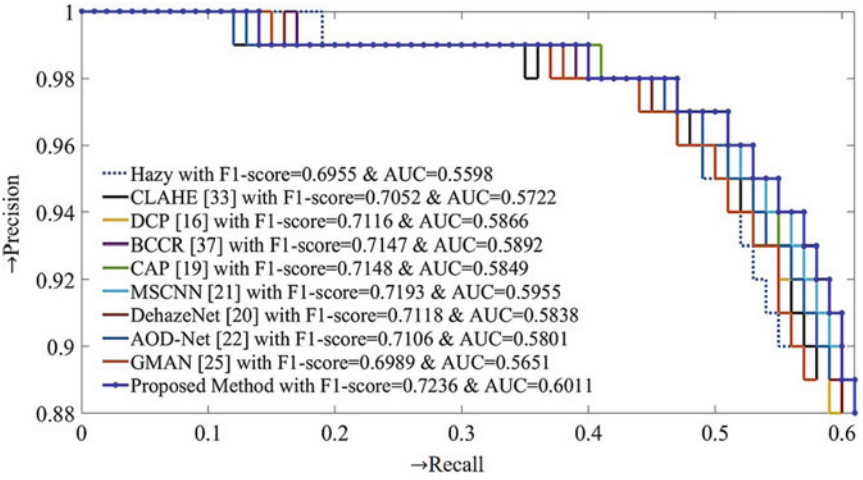


Fig. 5.11 A comparative analysis of dehazing+detection output on the RTTS subset from the RESIDE-β dataset [43] is assessed using F1-score and AUC values derived from the Precision-Recall curve for the "Person" class [52]

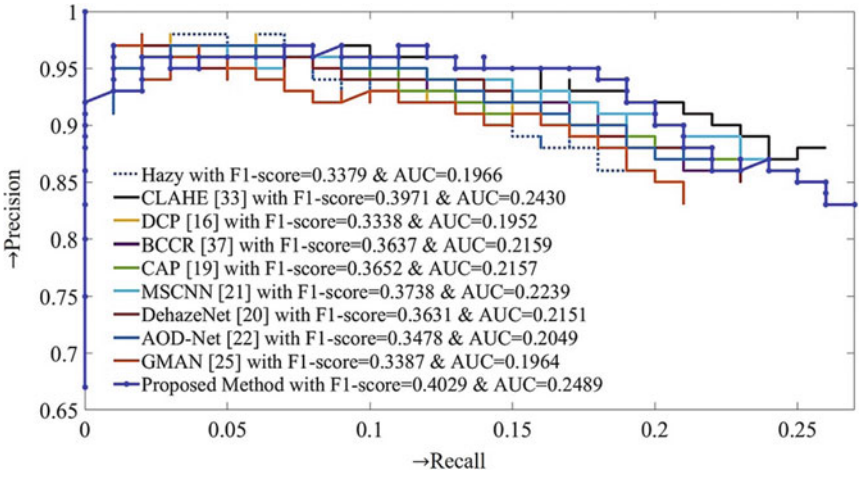


Fig. 5.12 A comparative analysis of dehazing+detection output on the RTTS subset from the RESIDE-β dataset [43] is assessed using F1-score and AUC values derived from the Precision-Recall curve for the "Bus" class [52]

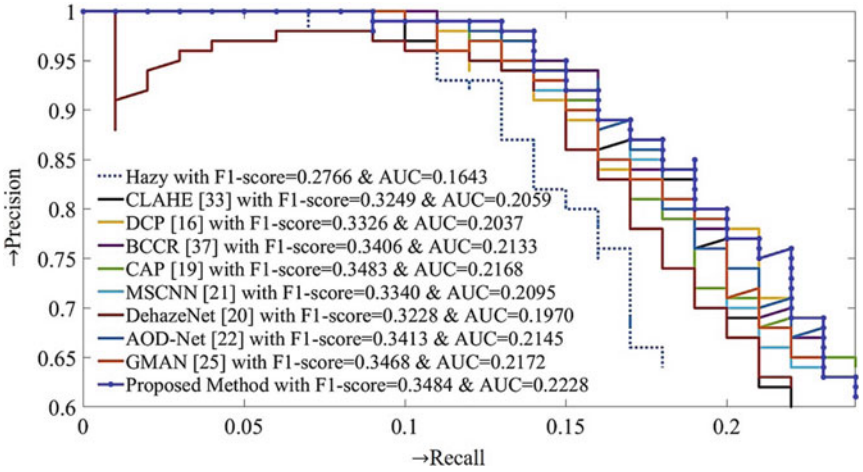


Fig. 5.13 A comparative analysis of dehazing+detection output on the RTTS subset from the RESIDE-β dataset [43] is assessed using F1-score and AUC values derived from the Precision-Recall curve for the "Bicycle" class [52]

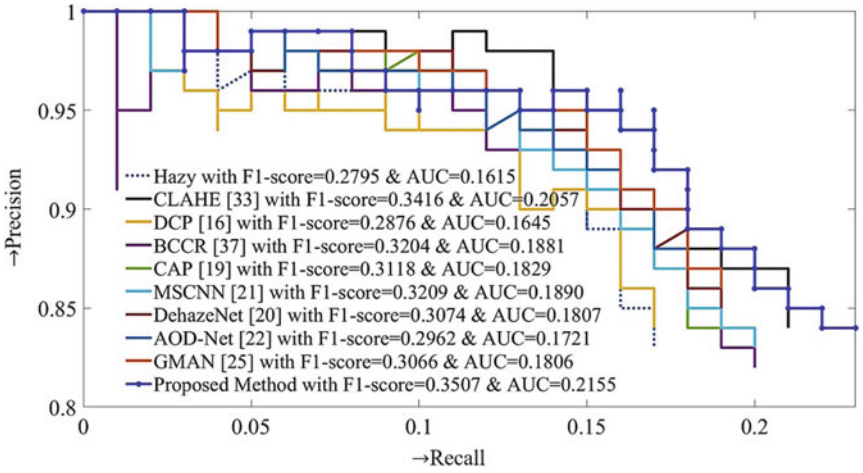


Fig. 5.14 A comparative analysis of dehazing+detection output on the RTTS subset from the RESIDE-β dataset [43] is assessed using F1-score and AUC values derived from the Precision-Recall curve for the "Motorbike" class [52]

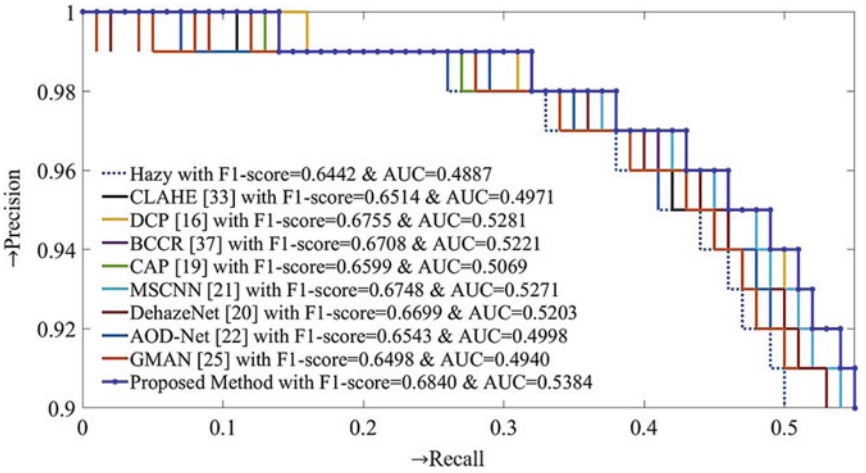


Fig. 5.15 A comparative analysis of dehazing+detection output on the RTTS subset from the RESIDE-β dataset [43] is assessed using F1-score and AUC values derived from the Precision-Recall curve for the "Car" class [52]

Remark 5.7 The F1-score serves as a crucial evaluation metric by harmonizing precision, which gauges the accuracy of detected relevant results, and recall, which measures the ability to capture the total relevant results in relation to the ground truth. By encapsulating both precision and recall into a single metric, the F1-score offers a comprehensive assessment of a model's performance. This metric is particularly advantageous in situations where achieving a balance between minimizing false positives and false negatives is essential, making it a valuable tool across diverse domains.

Table 5.9 tabulates the performance comparison of the Z-score approach (presented in the previous chapter) and the Type-2 fuzzy approach (presented in this chapter) in terms of PSNR, SSIM, LOE, and NIQE score for images from the RESIDE dataset. However, the computational burden of both is $O(R \times C)$ where R is the number of rows, and C is the number of columns in the input image being fed to the algorithm. Since there is no change in the computational complexity, the efforts remain the same in both algorithms, i.e., Z-score approach and Type-2 fuzzy approach for single image dehazing.

5.3.6 Variation in Patch Size

The Type-2 fuzzy approach employs the estimation of a similarity function matrix using Type-2 MF for a local patch centered at a pixel location under consideration. The patch size, which determines the extent of the considered neighborhood or

Table 5.9 Comparative performance assessment of methods, i.e., Z-Score and Type-2 fuzzy approaches. These methods are compared using the average values of PSNR, SSIM, LOE, and NIQE scores obtained on images belonging to the RESIDE dataset [43]

Datasets	Methods →	Z-score approach [1]	Type-2 fuzzy approach [52]
Indoor SOTS	**Average PSNR**	28.7952	**29.7248**
	Average SSIM	0.9695	**0.9702**
	Average LOE	0.822	**0.821**
	Average NIQE	**2.871**	2.381
Outdoor SOTS	**Average PSNR**	22.5978	**23.1524**
	Average SSIM	0.8679	**0.8696**
	Average LOE	0.530	**0.529**
	Average NIQE	2.601	**2.713**
HSTS	**Average PSNR**	20.1251	**22.4193**
	Average SSIM	0.8219	**0.8699**
	Average LOE	0.318	**0.212**
	Average NIQE	2.59	**2.73**

surrounding information, is an input parameter in this process. It's noteworthy that the depth map estimated using the DCP method [23] is also dependent on the patch size. The variation in patch size leads to changes in the obtained transmission maps, particularly near edges.

In Fig. 5.2, the transmission maps estimated using both DCP and the Type-2 fuzzy approach are presented for various patch sizes. Notably, the increase in patch size contributes to block effects near edges, potentially leading to halo effects in the dehazed images. The figure clearly illustrates that the block effects in transmission maps obtained using Type-2 MF near edges are minimal in comparison with those obtained using DCP.

Table 5.10 tabulates the average value of performance metrics obtained using the Type-2 fuzzy approach for MiddleBury Stereo dataset [40–42] images with various patch sizes and haze concentrations. The values of PSNR, SSIM, LOE, and NIQE show that the Type-2 fuzzy approach maintains its effectiveness with respect to the patch size variations.

Table 5.10 Comparative performance assessment of Type-2 fuzzy approach considering varying patch sizes and different levels of haze concentrations H_C on images belonging to the MiddleBury Stereo dataset [40–42]

Patch size →		5 × 5	7 × 7	9 × 9	11 × 11
H_C	Performance metric				
30%	PSNR [48]	25.8204	25.8299	25.9080	25.9094
	SSIM [49]	0.9447	0.9450	0.9464	0.9497
	LOE [50]	0.427	0.421	0.416	0.411
	NIQE [51]	2.95	2.90	2.89	2.88
40%	PSNR [48]	24.0475	24.0566	24.0626	24.0739
	SSIM [49]	0.9399	0.9400	0.9401	0.9405
	LOE [50]	0.355	0.349	0.344	0.339
	NIQE [51]	3.83	3.86	3.86	3.87
60%	PSNR [48]	21.4868	21.4863	21.4838	21.4782
	SSIM [49]	0.8999	0.8997	0.8995	0.8994
	LOE [50]	0.237	0.230	0.223	0.217
	NIQE [51]	3.82	3.82	3.81	3.75

5.4 Summary

This chapter introduces an approach using a Type-2 membership function based similarity function matrix for estimating the depth map and global atmospheric light in a single image dehazing. The presented method demonstrates qualitative and quantitative superiority over other techniques. The dehazed images produced maintain structural details near edges, preventing halo artifacts in the sky region and resulting in enhanced contrast with a natural appearance close to human perception.

A potential avenue for future improvement involves addressing the computational complexity associated with the estimation of the Type-2 membership function for each pixel. A speeded-up version of the method could be developed for more efficient single image dehazing. Additionally, addressing the issue of dark pixel intensity regions in input hazy scenes, which can sometimes appear visually oversaturated, could be achieved by refining the estimation of pixel-wise atmospheric light instead of relying on global atmospheric light for single image dehazing.

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Chapter 6
Adaptive Interval Type-2 Fuzzy Filter



6.1 Introduction

Artificial Intelligence provides a wide range of decision-making tools to address uncertainties in digital images. Among these, Fuzzy Set Theory (FST) [1] is capable of effectively tackling challenges related to grayness ambiguity in homogeneous regions and spatial ambiguity at edges within a digital image. It achieves this by considering the following factors:

(i) *Sparse constraint:* It emphasizes that information should exhibit smoothness in homogeneous regions.
(ii) *Boundary constraint:* It highlights the importance of preserving abrupt intensity changes to maintain the overall structure of the image.
(iii) *Range of illumination:* It ensures to preserve the image details such that the global ambiance remains unchanged, and there is no alteration in the direction of the light source.

In FST, Type-2 fuzzy logic was introduced by Karnik et al. [2]. In contrast to Type-1 fuzzy logic, it is more capable to model and minimize the effects of uncertainties for various real-time applications in image processing [3–26], control theory [27, 28], predictions and forecasting, [29], etc. Considering the complexity, Interval Type-2 Fuzzy Logic (IT2 FL) is preferred over General Type-2 Fuzzy Logic (GT2 FL) because secondary membership functions are non-uniformly weighted in GT2 FL and uniformly weighted in IT2 FL with membership values 1 [30].

In this chapter, an *Adaptive Interval Type-2 Fuzzy Filter (AIT2FF)* as an AI agent designed to address uncertainties in digital image enhancement has been introduced, aiming to preserve both image details and naturalness. The key takeaways of this chapter can be summarized as

(i) AIT2FF estimates the coarse illumination of non-uniformly illuminated images by incorporating neighborhood information while accounting for uncertainties. The estimated coarse illumination is then utilized to derive the true reflectance.

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(ii) The filter effectively manages grayness and spatial ambiguities in homogeneous regions and edges, respectively. This ensures the preservation of image details, contributing to visual realism and naturalness.
(iii) AIT2FF provides a luminance measure for each pixel location, enabling adaptive gamma correction to obtain refined coarse illumination for the final image composition.
(iv) Additionally, AIT2FF estimates the true transmission of a hazy input image, facilitating the generation of enhanced clear images. This capability makes the enhanced images obtained using AIT2FF valuable for tasks such as vehicle tracking in inclement weather conditions.

In the subsequent subsections, we delve into the detailed methodology of AIT2FF, followed by results, validations, and discussions.

6.2 Methodology

This section delineates the steps involved in the AIT2FF-based methodology, designed to eliminate uncertainties and preserve image naturalness during enhancement. The AIT2FF method consists of several stages: initial coarse illumination estimation (max channel), coarse illumination estimation, obtaining reflectance and refined coarse illumination, and, finally, image composition followed by denoising to produce the enhanced image. The block diagram of the AIT2FF-based methodology is depicted in Fig. 6.1.

The subsequent subsections provide a detailed explanation of each block in the process.

6.2.1 Initial Coarse Illumination

Let us consider an observed input RGB image $I^{\tau}_{\mathrm{NUI}}(p,q)$ of size $H \times W$ where H and W represent the number of rows (height) and columns (width), respectively of the image. According to the Retinex theory [31], the pixel intensity of the light channel reaching our eye is considered to be the maximum among the intensities of R, G, and B channels, which is termed as *Initial Coarse Illumination* and defined by

$$S_{\mathrm{m}}(p,q) = \max_{\tau \in \{\mathrm{R,G,B}\}} I^{\tau}_{\mathrm{NUI}}(p,q). \tag{6.1}$$

According to the literature [31], (6.1) lacks accuracy because it considers bright areas or bright objects as well. Further, it has been assumed that the illumination in a local patch is constant. Hence, for a k elements local patch N_k centered at (i, j) of size $\sqrt{k} \times \sqrt{k}$, the patch-based coarse illumination is defined by

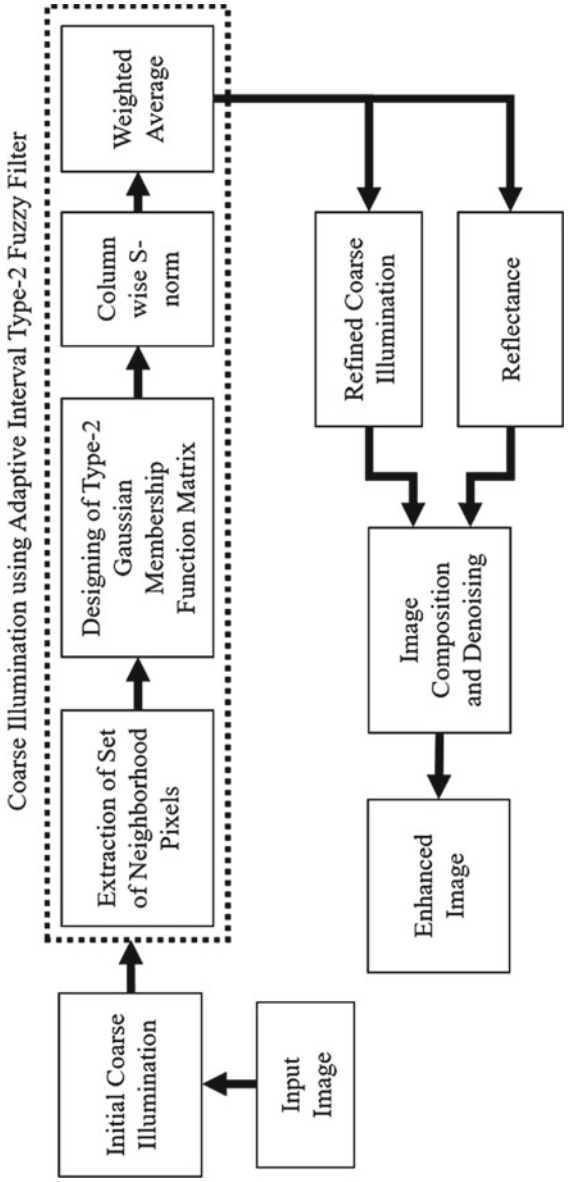


Fig. 6.1 Visual representation of AIT2FF-based enhancement process of an image [43]

$$S_{N_k}(p,q) = \max_{\tau\in\{\mathrm{R,G,B}\}} \left(\max_{(i,j)\in N_k} I^{\tau}_{\mathrm{NUI}}(i,j) \right);\ \forall 1 \le (i,j) \le \sqrt{k}. \tag{6.2}$$

where
– N_k is the set of neighborhood pixels.

The set of neighborhood pixels around a pixel intensity value P located at (i, j) for window size $\sqrt{k} \times \sqrt{k} = (2F+1) \times (2F+1)$ is extracted as

$$N_k = \{P(i+u_1, j+v_1)\}\ \forall\, u_1, v_1 \in [-F, F] \tag{6.3}$$

where
– F is the half-filter window size.

Since the patch-based illumination is not always constant, therefore (6.2) produces block effects near edges. To overcome this issue, the AIT2FF is discussed in the next subsection.

6.2.2 *Coarse Illumination Estimation Using Adaptive Interval Type-2 Fuzzy Filter*

Enhancement methods aimed at eliminating uncertainties in the pixel intensities of a digital image must possess specific properties to effectively address these uncertainties. The key properties include:

(i) Considering homogeneity and edginess to handle grayness and spatial ambiguities, respectively.
(ii) Incorporating neighborhood pixels with a weighting factor to maintain the global ambiance, i.e., the range of illumination.

AIT2FF is rooted in the FST introduced by Zadeh [1]. The methodology involves the extraction of a neighborhood pixel set, the design of a Type-2 Gaussian membership function matrix, processing it with column-wise S-norm, and finally, applying a weighted average to obtain the coarse illumination from the initial coarse illumination for each pixel location. These sub-steps are further detailed below.

6.2.2.1 Extraction of Set of Neighborhood Pixels

For a pixel centered at $(i,\ j)$ in $S_{\mathrm{m}}(p, q)$, the set of neighborhood pixels for a window size $(2F+1) \times (2F+1)$ is extracted as

$$N_k = \{S_{\mathrm{m}}(i+u_2, j+v_2)\}\ \forall\, u_2, v_2 \in [-F, F] \tag{6.4}$$

where
– k is the number of elements in the window.

6.2.2.2 Designing of Type-2 Gaussian Membership Function Matrix

FST-based image enhancement methods employ membership degrees for the belongingness of a pixel intensity among a set of pixel intensities. A Type-1 fuzzy set has crisp membership values, whereas a Type-2 fuzzy set has fuzzy membership values. Such a set is useful to handle more uncertainties where it is difficult to determine the exact membership function. Herein, the Gaussian Membership Function (GMF) has been chosen to assign the membership degrees. Hence, for neighborhood set N_k, the Type-2 GMF is defined as

$$\tilde{\rho}_k^h = \left\{ \left(N_k, \mu_{\rho_k^h}(N_k) \right), \mu_{\tilde{\rho}_k^h} \left(N_k, \mu_{\rho_k^h}(N_k) \right) \right\} \tag{6.5}$$

where
– $\mu_{\rho_k^h}(N_k) \in [0, 1]$ and

$$\mu_{\tilde{\rho}_k^h}(N_k) = f\left(N_k, M^h, \sigma\right) = \exp\left(-\frac{1}{2} \left(\frac{N_k - M^h}{\sigma} \right)^2 \right) \tag{6.6}$$

where
– M^h are h-middle means, and
– σ is the standard deviation.

The h-middle means for N_k are estimated by

$$M^h = \begin{cases} \dfrac{1}{2h-1} \displaystyle\sum_{i=n_m-h+1}^{n_m+h-1} N_i; \; n_m = \dfrac{k+1}{2}; & \text{for odd } k \\ \dfrac{1}{2h} \displaystyle\sum_{i=n_m-h+1}^{n_m+h} N_i; \quad n_m = \dfrac{k}{2}; & \text{for even } k \end{cases} \tag{6.7}$$

where
– n_m is the number of means, and
– $h = 1, 2, 3, \ldots, n_m$.

For h-middle means, the following statements can be made [3–5]:

(i) For a homogeneous region, M^h is monotonically increasing or decreasing function of h.
(ii) For an edge region, M^h is non-monotonic function of h.

This behavior of means within a window considers the homogeneity and edginess properties to preserve the naturalness. The standard deviation σ is estimated for the neighborhood window under consideration. It is defined as

$$\sigma = M^h\left(N_k'\right) \tag{6.8}$$

where

$$N'_k = \{|N_k - m_{\text{avg}}|\} \tag{6.9}$$

represents the deviation of pixel intensities for the window under consideration with an average of h-middle means obtained by

$$m_{\text{avg}} = \frac{1}{h}\sum_{i=1}^{h} M^h \tag{6.10}$$

Hence, the membership values of the Type-2 GMF matrix are written as

$$\mu_{\tilde{\rho}_k^h} = \begin{bmatrix} f(N_1, M^1, \sigma) & f(N_2, M^1, \sigma) & .. & f(N_k, M^1, \sigma) \\ f(N_1, M^2, \sigma) & f(N_2, M^2, \sigma) & .. & f(N_k, M^2, \sigma) \\ .. & .. & .. & .. \\ f(N_1, M^h, \sigma) & f(N_2, M^h, \sigma) & .. & f(N_k, M^h, \sigma) \end{bmatrix}_{h\times k}. \tag{6.11}$$

Remark 6.1 The Type-2 GMF is characterized by its Upper Membership Function (UMF) and Lower Membership Function (LMF). The UMF for Type-2 GMF is defined as

$$\overline{\mu}_{\tilde{\rho}_k^h}(N_k) = \begin{cases} U_1 & N_k < m_1 \\ 1 & m_1 \le N_k \le m_2 \\ U_2 & N_k > m_2 \end{cases} \tag{6.12}$$

where

$$U_1 = \exp\left(-\frac{1}{2}\left(\frac{N_k - m_1}{\sigma}\right)^2\right) \tag{6.13}$$

and

$$U_2 = \exp\left(-\frac{1}{2}\left(\frac{N_k - m_2}{\sigma}\right)^2\right) \tag{6.14}$$

for $m_1 = \min(M^h)$ and $m_2 = \max(M^h)$. The LMF for Type-2 GMF is defined as

$$\underline{\mu}_{\tilde{\rho}_k^h}(N_k) = \min(U_1, U_2), \tag{6.15}$$

Figure 6.2 shows UMF and LMF of a Type-2 GMF with uncertain means. For a homogeneous region, M^h is a monotonically increasing or decreasing function of h which results in the overlapped UMFs and LMFs, i.e., $\overline{\mu}_{\tilde{\rho}_k^h}(N_k) = \underline{\mu}_{\tilde{\rho}_k^h}(N_k)$. Similarly for an edge region, M^h is a non-monotonic function of h which results in $\overline{\mu}_{\tilde{\rho}_k^h}(N_k) > \underline{\mu}_{\tilde{\rho}_k^h}(N_k)$. Thus, the UMF and LMF using Eqs. (6.12) and (6.15) ensure that the UMF is always greater than or equal to the LMF for any region in a digital image.

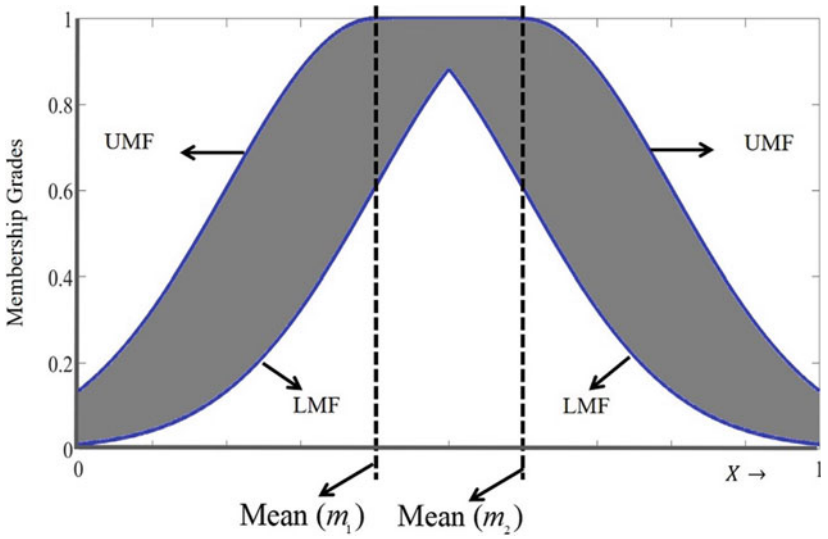


Fig. 6.2 Reconceptualizing the Uncertainty in Type-2 GMF with Uncertain Means [43]: In this context, the concept involves UMF and LMF denoting the upper and lower bounds for X belonging to N_k. Additionally, the interval $[m_1, m_2] \in M^h$ represents the uncertain means

Remark 6.2 The uncertainty present in a digital image arises in homogeneous regions and edges due to factors such as grayness and spatial ambiguities, respectively. AIT2FF effectively addresses these uncertainties by leveraging variations in means. Figure 6.3 illustrates the edge map of coarse illumination both before and after enhancement using a conventional Type-1 Fuzzy Filter (T1FF) and AIT2FF. The visual representation clearly indicates that the edge map becomes sharper near edges when processed with AIT2FF, and noise points in homogeneous regions are effectively filtered out. Additionally, AIT2FF preserves textural information in the edge map when compared to the edge map generated by the conventional T1FF method.

Remark 6.3 In homogeneous regions, characterized by regions with uniform pixel intensity values, AIT2FF simplifies to a T1FF. This simplification occurs because the h middle means denoted as M^h, become equal in such regions. Consequently, this equality leads to overlapped Gaussian MFs centered at a single mean. The collapse of AIT2FF to T1FF in homogeneous regions reflects the adaptation of the filtering approach based on the specific characteristics of the image data, streamlining the processing in scenarios where variations in means are minimal.

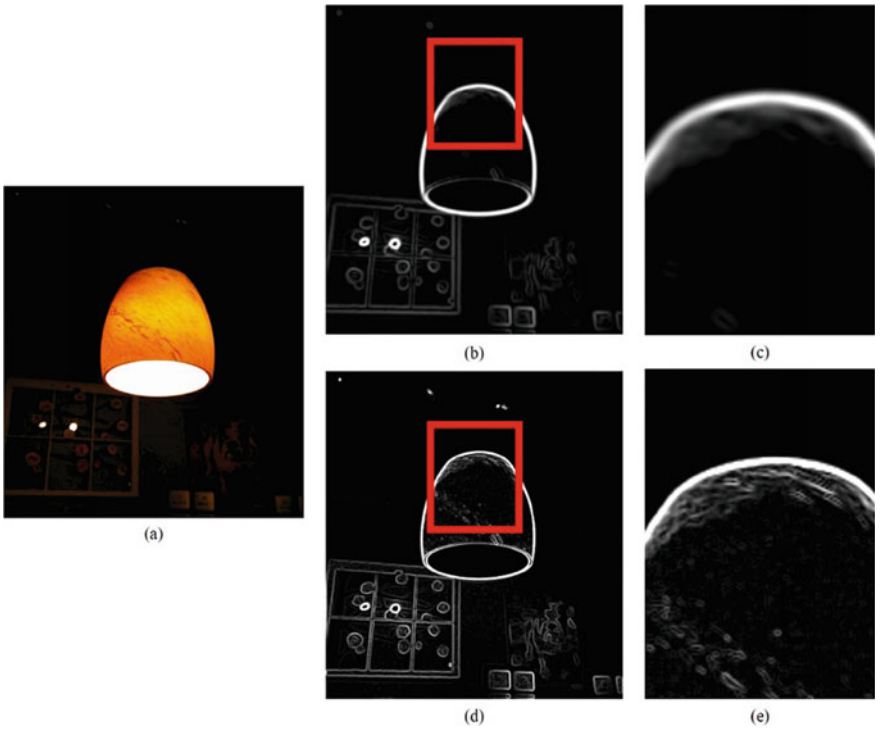


Fig. 6.3 Perceptual comparison of intensity variations, i.e., edge maps [43]: **a** Image captured under poor illumination condition, **b** Coarse illumination obtained using Type-1 fuzzy filter, **c** Enlarged version of an image region marked in (**b**), **d** Coarse illumination obtained using AIT2FF, and **e** Enlarged version of an image region marked in (**d**)

6.2.2.3 Column Wise S-norm

The Type-2 GMF matrix is processed with column-wise S-norm to obtain the associated membership degrees with each pixel intensity belonging to the window under consideration. The column-wise S-norm is obtained as

$$\eta_k = \left[\frac{\sum_{i=1}^{h} f\left(N_1, M^i, \sigma\right)}{h} \; \frac{\sum_{i=1}^{h} f\left(N_2, M^i, \sigma\right)}{h} \cdots \frac{\sum_{i=1}^{h} f\left(N_k, M^i, \sigma\right)}{h} \right]. \quad (6.16)$$

where

– η_k denotes the membership degree associated with kth pixel in the window N_k.

6.2.2.4 Weighted Average

Finally, the coarse illumination at each pixel location is estimated with the weighted average of the pixel intensities within the window around the pixel under consideration and the associated membership degrees. It can be written as

$$I_l(p,q) = \frac{\sum_{i=1}^{k} \eta_i(p,q) \times N_i(p,q)}{\sum_{i=1}^{k} \eta_i(p,q)}. \tag{6.17}$$

6.2.3 Reflectance

The reflectance at pixel location (p, q) is estimated by

$$R^{\tau}(p,q) = \frac{I^{\tau}_{\mathrm{NUI}}(p,q)}{I_l(p,q)} \tag{6.18}$$

where
– $I_l(p, q)$ is the coarse illumination estimated using AIT2FF explained in the previous subsection.

For darker regions, i.e., pixel intensities having "zero value", $I_l(p, q)$ results in an infinite value of reflectance as

$$R^{\tau}(p,q) = \frac{I^{\tau}_{\mathrm{NUI}}(p,q)}{I_l(p,q)} = \frac{I^{\tau}_{\mathrm{NUI}}(p,q)}{0} = \infty. \tag{6.19}$$

Since reflectance always lies in the range of [0, 1], therefore the condition occurring in (6.19) is impractical. This incurs inconsistency in the estimated reflectance. An example of this effect is shown with a red rectangle in Fig. 6.4. It is being treated by the region-growing method in the AIT2FF-based methodology. The region-growing method searches for a non-zero value in the neighborhood of the pixel location under consideration. The region-growing method is detailed in Algorithm 6.1. It is applied to the initial coarse illumination $S_{\mathrm{m}}(p, q)$. This results in a non-zero value of $I_l(p, q)$.

In the next subsection, image composition followed by denoising is explained for obtaining the enhanced output image.

6.2.4 Image Composition and Denoising

The estimated coarse illumination $I_l(p, q)$ is subjected to the adaptive gamma correction [32] with luminance measure for obtaining the refined coarse illumination. The

Algorithm 6.1: Region growing method

```
1: F ← 1
2: if S_m(i, j) = 0 then
3:    temp ← 0
4:    while temp = 0 do
5:       N_k = {S_m(i + u_3, j + v_3)}  ∀ u_3, v_3 ∈ [−F, F]
6:       temp1 = min(N'_k)  ∀ N'_k ∈ [N_k ≠ 0]
7:       if isempty(temp1) then
8:          F ← F + 1
9:       else
10:         temp = 1;
11:         S_m(i, j) = temp1;
12:      end if
13:   end while
14: end if
```



Fig. 6.4 Visual representation of region-growing helping to deal with darker regions [43]: **a** Image captured under poor illumination condition, **b** A segment which shows dark region for an image region marked in (**a**), **c** Initial coarse illumination, **d** Reflectance, and **e** Refined reflectance obtained using presented region-growing method

Type-2 GMF matrix (6.11) in the AIT2FF-based method also provides a luminance measure for each pixel location. It is given by

$$L(p,q) = \max_i \left(\min_j \left(\mu_{\tilde{\rho}_i^j} \right) \right) \ \forall 1 \le i \le k; 1 \le j \le h. \tag{6.20}$$

Remark 6.4 Equation (6.20) measures luminance using Zadeh's compositional rule of inference [1]. This can be interpreted using the following condition:

IF $N_k \in S_{\mathrm{m}}$ **THEN** the luminance measure L is $\max_i \left(\min_j \left(\mu_{\tilde{\rho}_i^j} \right) \right)$; where $i = 1, 2, \ldots k$ denotes ith pixel location in the neighborhood pixel set N_k and $j = 1, 2, \ldots h$ denotes jth mean obtained for N_k.

Hence, the refined coarse illumination is obtained by

$$I_l^{\mathrm{AGC}}(p,q) = I_l(p,q)^{\gamma(p,q)} \tag{6.21}$$

where

$$\gamma(p,q) = \frac{I_l(p,q) + \alpha}{(1+\alpha)^{\alpha}} \tag{6.22}$$

and

$$\alpha = 1 + \mathrm{mean}(L(p,q)). \tag{6.23}$$

Finally, the enhanced output image is obtained by

$$I_{\mathrm{enh}}^{\tau}(p,q) = R^{\tau}(p,q) \times I_l^{\mathrm{AGC}}(p,q). \tag{6.24}$$

For low-light input images, noises that are not visible due to darkness also get amplified during enhancement. Hence, image denoising is mandatory to hold the visual realism of enhanced images. The image denoising is performed using Block Matching and 3D (BM3D) filter [33]. The enhanced image $I_{\mathrm{enh}}^{\tau}(p,q)$ is first converted to YCbCr image domain as

$$\left(I_{\mathrm{enh}}^{\mathrm{Y}}, I_{\mathrm{enh}}^{\mathrm{Cb}}, I_{\mathrm{enh}}^{\mathrm{Cr}}\right) \xleftarrow{\text{RGB to YCbCr}} \left(I_{\mathrm{enh}}^{\mathrm{R}}, I_{\mathrm{enh}}^{\mathrm{G}}, I_{\mathrm{enh}}^{\mathrm{B}}\right). \tag{6.25}$$

The Y-channel of the enhanced image is subjected to a BM3D filter as

$$I_{\mathrm{ef}}^{\mathrm{Y}} \xleftarrow{\text{BM3D filter}} I_{\mathrm{enh}}^{\mathrm{Y}} \tag{6.26}$$

and then the updated Y-channel is converted back to the RGB domain as

$$\left(I_{\mathrm{ef}}^{\mathrm{R}}, I_{\mathrm{ef}}^{\mathrm{G}}, I_{\mathrm{ef}}^{\mathrm{B}}\right) \xleftarrow{\text{YCbCr to RGB}} \left(I_{\mathrm{ef}}^{\mathrm{Y}}, I_{\mathrm{enh}}^{\mathrm{Cb}}, I_{\mathrm{enh}}^{\mathrm{Cr}}\right). \tag{6.27}$$

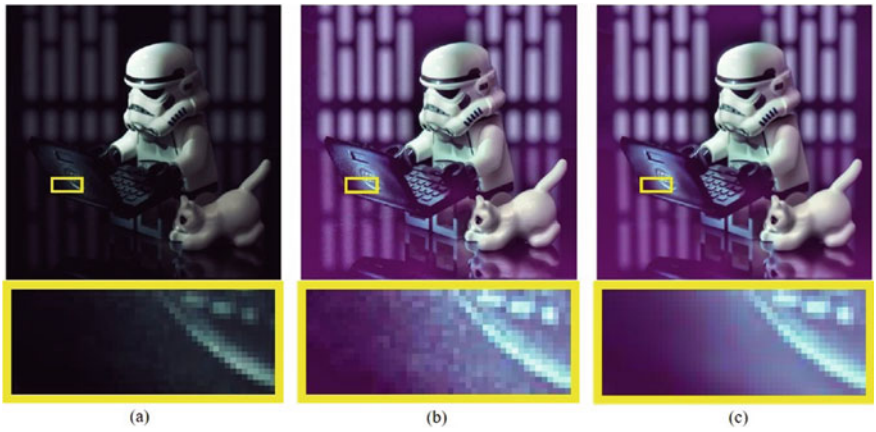


Fig. 6.5 Denoising of enhanced image: **a** Image captured under poor illumination condition, **b** Enhanced version of image shown in (**a**) using AIT2FF [43], and **c** Denoised version of image shown in (**b**) using BM3D filter [33]

Thus, $I_{\text{ef}}^{\tau}(p, q)$ is the final enhanced image after denoising. Figure 6.5 shows an example of noise suppression with zoomed views for an enhanced image. Algorithm 6.2 explains the step-by-step implementation of the AIT2FF-based method.

Algorithm 6.2: AIT2FF based methodology

Input: Observed input image $I_{\text{NUI}}^{\tau}(p, q)$
Define window size $k = (2F + 1) \times (2F + 1)$
Initial coarse illumination $S_{\text{m}}(p, q)$ using (6.1)
for all $P(i, j) \in S_{\text{m}}(p, q)$ **do**
    Extract set of neighborhood pixels N_k using (6.4)
    Obtain middle means M^h for N_k using (6.7)
    Estimate Type-2 GMF matrix using (6.11)
    Find column wise S-norm using (6.16)
    Coarse illumination $I_l(p, q)$ using (6.17)
end for
Estimate reflectance $R^{\tau}(p, q)$ using (6.18)
Luminance measure $L(p, q)$ using (6.20)
Apply adaptive gamma correction using (6.21)
Enhanced image $I_{\text{enh}}^{\tau}(p, q)$ using (6.24)
Perform Denoising using (6.25)–(6.27)
Output: Enhanced image $I_{\text{ef}}^{\tau}(p, q)$

6.3 Results and Discussions

This section provides a comprehensive evaluation, both quantitatively and qualitatively, comparing the outcomes achieved by the AIT2FF-based method to other existing methods. The comparison involves AHE [34], CVC [35], LDR [36], ENR [32], LIME [37], LECARM [38], and Zero-DCE [39]. The evaluations were conducted on a Windows 10 operating system, utilizing a system configuration with 8GB RAM and an Intel Core i7 processor. Both quantitative metrics and visual assessments contribute to the comprehensive comparison of the proposed AIT2FF-based method against the selected benchmark methods.

6.3.1 Datasets and Performance Metrics

For qualitative evaluation, the comparisons have been presented on standard images used in the literature, whereas for quantitative evaluation, 8 images from the High Dynamic Range (HDR) dataset [40] (as shown in Fig. 6.6) and 69 images captured with commercial DIgital CaMeras (DICM) [36] have been used. The quantitative performance has been measured in terms of LOE [41] and NIQE score [42]. LOE measures the performance by considering HDR tone-mapped result as the reference

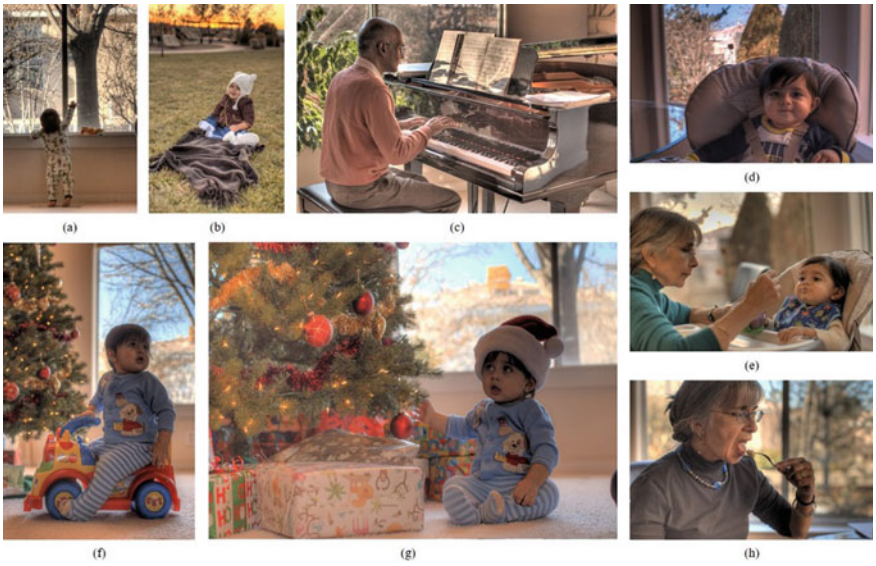


Fig. 6.6 Visualization of ground truth images from HDR dataset [40]: **a** BabyAtWindow, **b** BabyOnGrass, **c** PianoMan, **d** HighChair, **e** FeedingTime, **f** ChristmasRider, **g** SantasLittleHelper, and **h** LadyEating

for the HDR dataset and input low-light image as the reference for the DICM dataset. The LOE is defined as

$$\text{LOE} = \frac{1}{w}\sum_{x=1}^{w}\sum_{y=1}^{w}\left(U\left(\mathbf{Q}(x),\mathbf{Q}(y)\right)\oplus U\left(\mathbf{Q}_r(x),\mathbf{Q}_r(y)\right)\right) \tag{6.28}$$

where
– w is the pixel number,
– $U(s,t) = 1$ if $s \geq t$ else 0, $\mathbf{Q}(x)$ and
– $\mathbf{Q}_r(x)$ are the initial coarse illumination of enhanced and reference images, respectively, for pixel location x, and $\oplus$ is the exclusive-or operator.

NIQE serves as a metric to assess the perceptual quality of an enhanced image without relying on prior knowledge of human opinions or ground truths. Lower values for both LOE and NIQE indicate superior enhancement while preserving the natural characteristics of the image.

To streamline the computational complexity during LOE calculation, downsampling is employed with a resize factor of $\frac{100}{\min(H,W)}$, where H and W represent the height and width of the image, respectively.

Subsequent subsections will delve into various aspects, including quantitative and qualitative comparisons, the impact of varying window sizes, and the performance comparison between Type-1 and Type-2 fuzzy filters. These discussions aim to provide a comprehensive understanding of the presented method's efficacy in comparison to others and its behavior under different scenarios.

6.3.2 *Quantitative Comparison*

Figures 6.7, 6.8, 6.9 and 6.10 provide the visual comparison of results obtained using the AIT2FF-based method with state-of-the-art enhancement methods. The HDR dataset has eight groups of images, among which the visual comparison for four images is shown in the figures.

Table 6.1 shows the performance comparison for images from the HDR dataset in terms of LOE. From Table 6.1, it can be observed that the AIT2FF-based method outperforms others with an average LOE value of 1.929 for enhancement using the AIT2FF-based method rather than 2.573 for LIME. Table 6.2 tabulates the comparison among average LOE values for images from the DICM dataset. Moreover, Tables 6.3 and 6.4 show the comparison of NIQE values obtained for images from the HDR dataset and DICM dataset, respectively.

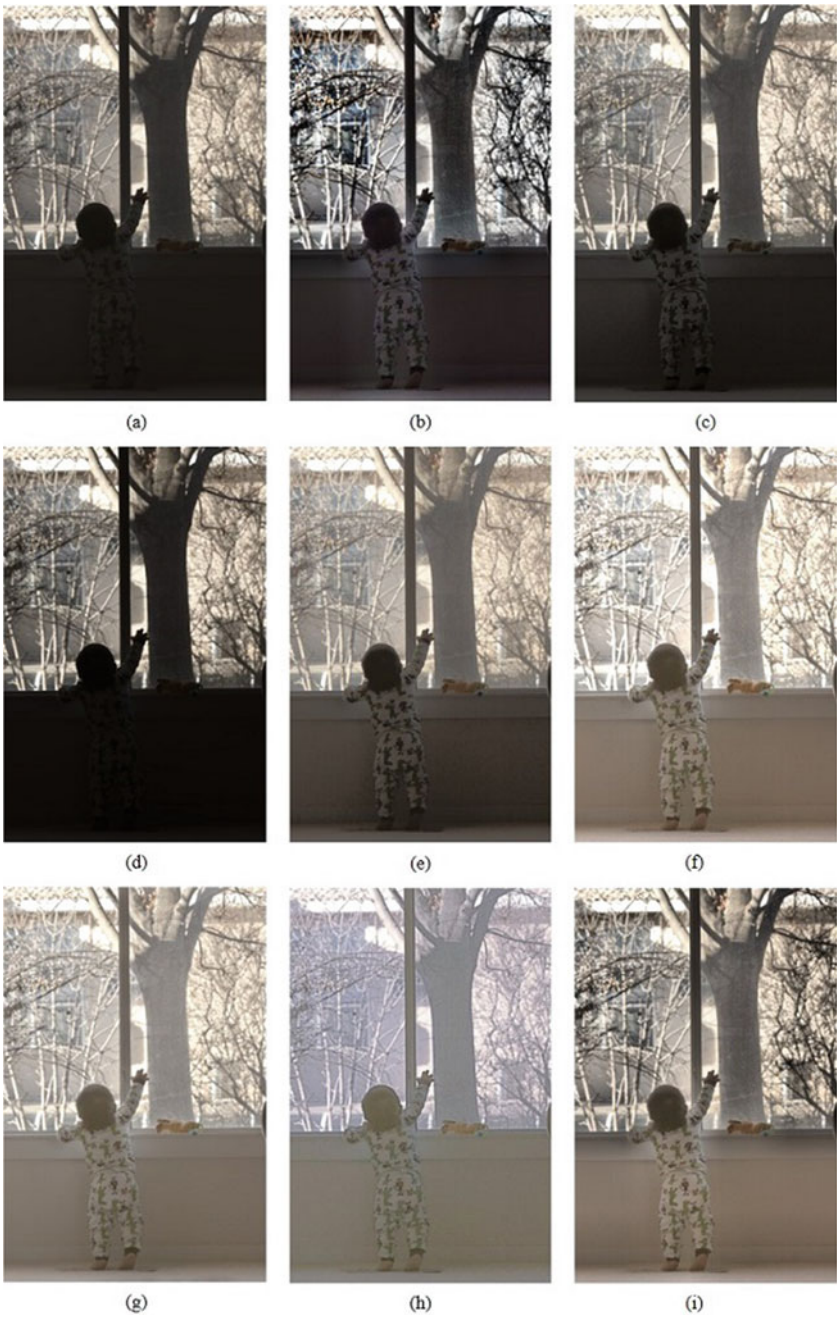


Fig. 6.7 Perceptual analysis of an image named "BabyAtWindow" from the HDR dataset [40]: **a** Image captured under poor illumination condition, Enhanced version of image shown in (**a**) using **b** AHE [34], **c** CVC [35], **d** LDR [36], **e** ENR [32], **f** LIME [37], **g** LECARM [38], **h** Zero-DCE [39], and **i** AIT2FF [43]



Fig. 6.8 Perceptual analysis of an image named "BabyOnGrass" from the HDR dataset [40]: **a** Image captured under poor illumination condition, Enhanced version of image shown in (**a**) using **b** AHE [34], **c** CVC [35], **d** LDR [36], **e** ENR [32], **f** LIME [37], **g** LECARM [38], **h** Zero-DCE [39], and **i** AIT2FF [43]



Fig. 6.9 Perceptual analysis of an image named "PianoMan" from the HDR dataset [40]: **a** Image captured under poor illumination condition, Enhanced version of image shown in (**a**) using **b** AHE [34], **c** CVC [35], **d** LDR [36], **e** ENR [32], **f** LIME [37], **g** LECARM [38], **h** Zero-DCE [39], and **i** AIT2FF [43]

6.3.3 Qualitative Comparison

The visual comparisons presented in Figs. 6.7, 6.8, 6.9 and 6.10 vividly illustrate the effectiveness of the AIT2FF-based method in preserving textural information within homogeneous regions and at edges. In contrast, the enhanced images produced using AHE exhibit halo artifacts in bright regions and introduce noise in darker regions. Notably, AHE, CVC, and LDR fall short of achieving luminance restoration, as evident in the images.

Comparatively, ENR, LIME, and LECARM successfully restore dark regions, but brighter regions remain unaddressed. On the other hand, Zero-DCE introduces illumination, resulting in an enhanced image that appears hazy. The AIT2FF-based method adeptly addresses both darker and brighter regions, effectively mitigating uncertainties during enhancement and avoiding over-enhancement. The ultimate goal of luminance restoration is accomplished with superior performance, as demonstrated in the visual comparisons.

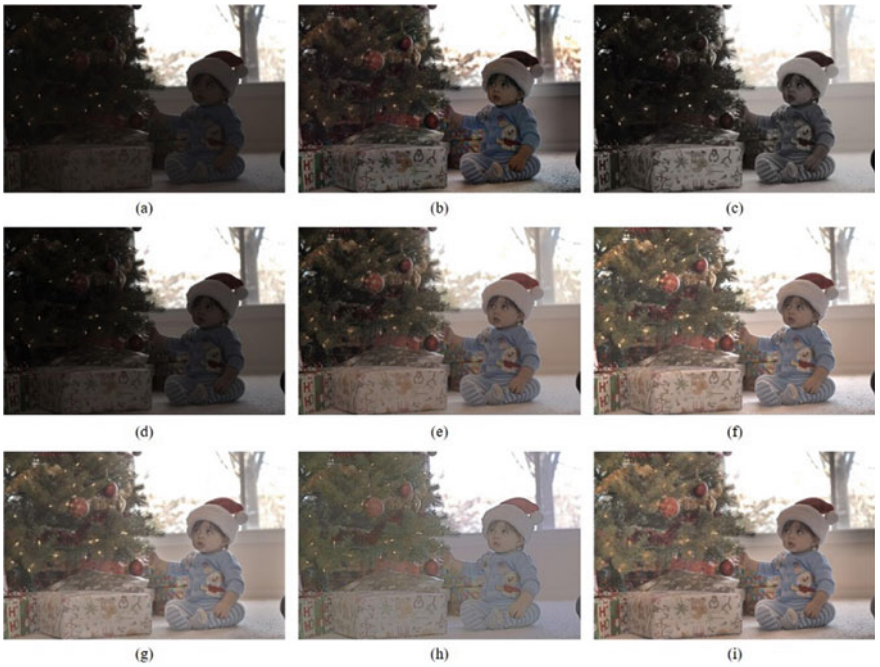


Fig. 6.10 Perceptual analysis of an image named "SantaLittleHelper" from the HDR dataset [40]: **a** Image captured under poor illumination condition, Enhanced version of image shown in (**a**) using **b** AHE [34], **c** CVC [35], **d** LDR [36], **e** ENR [32], **f** LIME [37], **g** LECARM [38], **h** Zero-DCE [39], and **i** AIT2FF [43]

Table 6.1 Performance assessment of AIT2FF-based method using the values of LOE ($\times 10^3$) estimated on image subset from the HDR dataset [40]

Methods → Image name ↓	AHE [34]	CVC [35]	LDR [36]	ENR [32]	LIME [37]	LECARM [38]	Zero-DCE [39]	AIT2FF [43]
BabyAtWindow [40]	3.541	4.540	4.505	4.518	3.543	4.108	2.326	**2.131**
BabyOnGrass [40]	2.483	4.472	4.504	4.505	2.854	3.273	2.997	**2.107**
ChristmasRider [40]	2.288	2.628	2.598	2.512	2.154	2.425	2.625	**2.002**
FeedingTime [40]	2.215	3.150	3.137	3.124	2.126	2.247	3.198	**1.843**
HighChair [40]	2.151	3.141	3.136	3.128	2.323	2.511	2.694	**2.003**
LadyEating [40]	2.624	3.413	3.416	3.405	2.429	3.115	2.361	**1.846**
PianoMan [40]	2.596	3.817	3.771	3.745	2.594	3.123	3.306	**1.563**
SantaLittleHelper [40]	2.993	3.717	3.702	3.672	2.565	3.409	2.266	**1.941**
Average LOE	2.611	3.609	3.596	3.576	2.573	3.026	2.723	**1.929**

Table 6.2 Performance assessment of AIT2FF-based method using the values of LOE ($\times 10^3$) estimated on image subset from the DICM dataset [36]

Methods →	AHE [34]	CVC [35]	LDR [36]	ENR [32]	LIME [37]	LECARM [38]	Zero-DCE [39]	AIT2FF [43]
Average LOE	3.4025	1.377	0.961	1.875	3.810	1.989	0.916	**0.777**

Table 6.3 Performance assessment of AIT2FF-based method using the values of NIQE scores estimated on image subset from the HDR dataset [40]

Methods →	AHE [34]	CVC [35]	LDR [36]	ENR [32]	LIME [37]	LECARM [38]	Zero-DCE [39]	AIT2FF [43]
Average NIQE	2.9859	3.1518	3.0575	2.8530	2.9719	2.8680	2.6035	**2.4713**

Table 6.4 Performance assessment of AIT2FF-based method using the values of NIQE scores estimated on image subset from the DICM dataset [36]

Methods →	AHE [34]	CVC [35]	LDR [36]	ENR [32]	LIME [37]	LECARM [38]	Zero-DCE [39]	AIT2FF [43]
Average NIQE	2.9582	2.8759	2.9080	3.5892	3.0569	3.7955	3.1546	**2.8231**

6.3.4 Variation in Patch Size

The computation of patch-based coarse illumination, as determined by (6.2), exhibits variation based on the chosen patch size. This variability introduces a block effect near edges due to spatial ambiguity. However, the AIT2FF mitigates such effects effectively by incorporating a Type-2 GMF and employing a weighting factor. Figure 6.11 visually demonstrates the impact of varying patch sizes on patch-based coarse illumination using (6.2) and the AIT2FF. Notably, a region rich in texture has been highlighted in each image for illustrative purposes. Throughout the performance analysis in this chapter, a patch size of $k = 9 = 3 \times 3$ has been adopted.

6.3.5 Type-1 Versus Type-2 Fuzzy Filter

The Type-2 Fuzzy Filter (T2FF) serves as an extension of the Type-1 fuzzy filter (T1FF), where the membership values of the ordinary fuzzy filter are inherently fuzzy. The AIT2FF is introduced to address uncertainties in situations where determining the exact MF using a standard T1FF is challenging. If T2FF is replaced with T1FF as an AI agent, AIT2FF simplifies into a conventional Gaussian T1FF. However, this simplification results in a blurred coarse illumination, leading to a loss

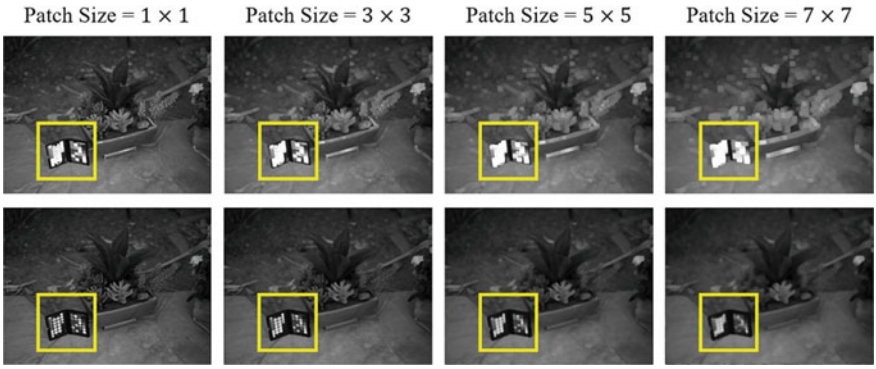


Fig. 6.11 **Top row:** Coarse illumination is obtained using different patch sizes with DCP as shown at the top of each image. **Bottom row:** Coarse illumination is obtained using different patch sizes with AIT2FF as shown at the top of each image [43]



Fig. 6.12 Perceptual comparison between T1FF and AIT2FF [43]: **a** Image captured under bad illumination condition, **b** Coarse illumination, **c** Reflectance, and **d** Enhanced version of image shown in (**a**) using T1FF, **e** Coarse illumination, **f** Reflectance, and **g** Enhanced version of image shown in (**a**) using AIT2FF, and **h** Enlarged version of image regions marked and shown in images of (**c**) and (**f**)

of textural information near edges. Figure 6.12 illustrates a comparison between the coarse illuminations and the corresponding enhanced versions obtained using T1FF and AIT2FF. The enlarged versions of reflectance clearly demonstrate that AIT2FF retains textural information near edges more effectively than the conventional T1FF.

Table 6.5 presents a performance comparison between the Z-score approach and AIT2FF, measured in terms of LOE and NIQE scores for images from HDR and

Table 6.5 Comparative performance assessment of methods, i.e., Z-Score and AIT2FF approaches. These methods are compared using the average values of LOE and NIQE scores obtained on images belonging to the HDR [40] and DICM [36] datasets

Datasets	Methods →	Z-Score approach [7]	AIT2FF [43]
HDR	Average LOE	2.095	**1.929**
	Average NIQE	**2.5506**	2.4713
DICM	**Average LOE**	0.892	**0.777**
	Average NIQE	2.8051	**2.8231**

DICM datasets. Despite the differences in the algorithms, both exhibit a computational burden of $O(R \times C)$, where R is the number of rows and C is the number of columns in the input image. Since there is no change in computational complexity, the efforts required for both algorithms, i.e., the Z-score approach and AIT2FF, remain consistent for non-uniform illumination enhancement.

6.4 Summary

In this chapter, an *Adaptive Interval Type-2 Fuzzy Filter (AIT2FF)* has been introduced as an AI agent using FST for vision-based applications. The AIT2FF efficiently addresses uncertainties in digital images to achieve human visual perception. It has been emphasized that preserving image naturalness is crucial for effective human vision. The filter adaptively extracts neighborhood information in homogeneous regions and at edges to handle grayness and spatial ambiguities, respectively, thereby preserving naturalness. The elimination of these ambiguities results in a decreased error value compared to alternative methods while maintaining the image structure. As a future direction, integrating the enhancement filter with a denoising filter could be explored for noise suppression, including Gaussian noise, impulse noise, etc.

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Chapter 7
Epilogue



The objective of vision-based applications is to automate human visual perception. This requires a pre-processed input image with fine details. The image details to achieve human vision suffer uncertainties in homogeneous regions and at edges. This results in an unnatural pre-processed image and degrades the performance. For example: Noisy images are not good for training deep learning models. Artificial intelligence offers one of the best tools, i.e., Fuzzy set theory, which is capable of handling uncertainties.

7.1 Summary

This book introduces artificial intelligent decision-making algorithms for the enhancement of digital images under hazy and non-uniform illumination conditions. The algorithms presented in this book are capable of boosting the performance of various vision-based applications like object detection, vehicle tracking, security, medical diagnostics, etc. The work presented in this book can be concluded in the following sub-points:

(i) A transmission map estimation function has been introduced to prevent the over-saturation of dehazed images.
(ii) A compact single image dehazing architecture trained with minimal resources and lesser run-time has been introduced.
(iii) A framework for single image dehazing and non-uniform illumination enhancement using Z-score weighting function map has been introduced.
(iv) A single image dehazing method using a Type-2 membership function based similarity function matrix has been introduced for the estimation of scene depth and global atmospheric light.

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(v) An adaptive interval Type-2 fuzzy filter as an artificially intelligent agent for preserving the naturalness of non-uniformly illuminated scene images has been introduced.

7.2 Directions for Future Work

The potential future scope of the work presented in the book can be summarized as follows:

(i) An artificial intelligent dehazing model using fuzzy set theory and biologically inspired models can be developed to combine the advantages of both in a single framework.
(ii) The development of an image enhancement method can be proposed to deal with the noise incurred during enhancement without compromising the textural information content of the image.